

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398183
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Varenne; Pierrick et al.

Wheat comprising male fertility restorer alleles

Abstract

A wheat transgenic plant carrying restorer of fertility genes specific to *T. timopheevii* CMS cytoplasm.

Inventors:	Varenne; Pierrick (Fontainebleau, FR), Comadran; Jordi (Riom, FR), Specel; Sébastien (Montpensier, FR), Murigneux; Alain (La Roche Blanche, FR), Melonek; Joanna (Crawley, AU), Small; Ian (Wattle Grove, AU), Perez; Pascual (Chanonat, FR), Duarte; Jorge (Clermont Ferrand, FR), Pichon; Jean-Philippe (Clermont Ferrand, FR), Levadoux; Sylvain (Clermont Ferrand, FR), Martin; Jérôme (Volvic, FR), Torney; François (Mahomet, IL)
Applicant:	VILMORIN & CIE (Paris, FR); LIMAGRAIN EUROPE (Saint Beuzire, FR)
Family ID:	1000008778292
Assignee:	VILMORIN & CIE (Paris, FR); LIMAGRAIN EUROPE (Saint Beuzire, FR)
Appl. No.:	16/760693
Filed (or PCT Filed):	October 31, 2018
PCT No.:	PCT/EP2018/079816
PCT Pub. No.:	WO2019/086510
PCT Pub. Date:	May 09, 2019

Prior Publication Data

Document Identifier	Publication Date
US 20200347104 A1	Nov. 05, 2020

Foreign Application Priority Data

EP	17306500	Oct. 31, 2017
EP	17306501	Oct. 31, 2017
EP	18305027	Jan. 12, 2018
EP	18306114	Aug. 14, 2018

Publication Classification

Int. Cl.: C12N15/82 (20060101); C07K14/415 (20060101)

U.S. Cl.:

CPC C07K14/415 (20130101); C12N15/8289 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11930756	12/2023	Varenne	N/A	A01H 5/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 2017/158128	12/2016	WO	A01H 1/04
WO-2017158126	12/2016	WO	A01H 1/02
WO-2017158128	12/2016	WO	A01H 1/023
2018/015403	12/2017	WO	N/A

OTHER PUBLICATIONS

Fuji, S. et al., PNAS vol. 108, No. 4 (Jan. 25, 2011) pp. 1723-1728. (Year: 2011). cited by examiner
Wurschum, T. et al. Genetic architecture of male fertility restoration of *Triticum timopheevii* cytoplasm and fine-mapping of the major restorer locus Rf3 on chromosome 1B; Theor Appl Genet (2017) 130:1253-1266. (Year: 2017). cited by examiner

Geyer, M. et al. Molecular Genetics and Genomics (2018) 293:451-462. (Year: 2018). cited by examiner

Wurschum, T. et al. Theor Appl Genet (2017) 130:1253-1266. (Year: 2017). cited by examiner

Kazama, T. et al. The Plant Journal (2008) 55, 619-628 (Year: 2008). cited by examiner

“Protein Rf1, mitochondrial-like (*Aegilops tauschii* subsp. *tauschii*);” Database Protein [Online]; XP-002781672; 2017; Database accession No. XP_020157157.1 sequence. cited by applicant

Geyer et al.; “Distribution of the fertility-restoring gene Rf3 in common and spelt wheat determined by an informative SNP marker;” Mol Breeding; 2016; pp. 1-11 with Electronic supplementary materials; vol. 36, No. 167. cited by applicant

Geyer et al.; “Exploring the genetics of fertility restoration controlled by Rf1 in common wheat

(*Triticum aestivum* L.) using high-density linkage maps;” Molecular Genetics and Genomics; 2018; pp. 451-462; vol. 293, No. 2. cited by applicant

Fujii et al.; “Selection patterns on restorer-like genes reveal a conflict between nuclear and mitochondrial genomes throughout angiosperm evolution;” PNAS; 2011; pp. 1723-1728; vol. 108, No. 4. cited by applicant

Lilienfeld; “H. Kihara: Genome-Analysis in Triticum and Aegilops. X. Concluding Review,” Cytologia; 1951; pp. 101-123; vol. 16. cited by applicant

Ahmed et al.; “QTL analysis of fertility-restoration against cytoplasmic male sterility in wheat;” Gene Genet. Syst.; 2001; pp. 33-38; vol. 76. cited by applicant

Zhou et al.; “SSR markers associated with fertility restoration genes against *Triticum timopheevii* cytoplasm in *Triticum aestivum*;” Euphytica; 2005; pp. 33-40; vol. 141. cited by applicant

Ma et al.; “Cell Biology & Molecular Genetics;” Crop Sci.; 1995; pp. 1137-1143; vol. 35. cited by applicant

Wilson, James A.; “Hybrid Wheat Breeding and Commercial Seed Development;” Plant Breeding Reviews; 1984; pp. 303-319. cited by applicant

Stojalowski et al.; “The importance of chromosomes from the sixth homeologous group in the restoration of male fertility in winter triticale with *Triticum timopheevii* cytoplasm;” J Appl Genetics; 2013; pp. 179-184; vol. 54. cited by applicant

Würschum et al.; Genetic architecture of male fertility restoration of *Triticum timopheevii* cytoplasm and fine-mapping of the major restorer locus Rf3 on chromosome 1B; Theor Appl Genet; 2017; pp. 1-23. cited by applicant

Brenchley et al.; “Analysis of the bread wheat genome using whole-genome shotgun sequencing;” Nature; 2012; pp. 705-710; vol. 491. cited by applicant

Wickersham et al.; “Male-Fertility Restoration in Crosses of R5 with Soft Red Winter Wheats;” Crop Science; 1980; pp. 100-102; vol. 20; No. 1. cited by applicant

Zhang et al.; “Location of the Fertility Restorer Gene for T-Type CMS Wheat by Microsatellite Marker,” Acta Genetica Sinica; 2003; pp. 459-464; vol. 30, No. 5. cited by applicant

Bahl et al.; “Chromosomal Location of Male Fertility Restoring Genes in Six Lines of Common Wheat;” Crop Science; 1973; pp. 317-320; vol. 13. cited by applicant

Ling et al.; “Draft genome of the wheat A-genome progenitor *Triticum urartu*;” Nature; 2013; pp. 87-90; vol. 496. cited by applicant

Dec. 19, 2018 Search Report issued in International Patent Application No. PCT/EP2018/079816. cited by applicant

Ma, Z.-Q et al., “Cell Biology & Molecular Genetics: Genetic Analysis of Fertility Restoration in Wheat Using Restriction Fragment Length Polymorphisms”. Crop Science, vol. 35, pp. 1137-1143, 1995. cited by applicant

Wurschum, Tobias et al., “Genetic architecture of male fertility restoration of *Triticum timopheevii* cytoplasm and fine-mapping of the major restorer locus Rf3 on chromosome 1B”. Theoretical and Applied Genetics, vol. 130, pp. 1253-1266, 2017. cited by applicant

Mar. 20, 2023 Office Action issued in U.S. Appl. No. 17/334,472. cited by applicant

Primary Examiner: Kallis; Russell

Attorney, Agent or Firm: Oliff PLC

Background/Summary

(1) The present application contains a Sequence Listing that has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. The ASCII copy, created on Dec. 8, 2020, is named Substitute Sequence Listing_ST25.txt and is 9,515,008 bytes in size.

(2) The invention is in the field of plant genetics and plant breeding. The invention more specifically relates to wheat plants carrying restorer of fertility genes specific to *T. timopheevii* CMS cytoplasm.

BACKGROUND

(3) Hybrid production is based on crossing two parental lines to increase heterosis and de facto, increase genetic variability to create new varieties or genotypes with higher yield and better adapted to environmental stresses. Even in a predominantly autogamous species like wheat, research studies have shown that hybrid lines exhibit improved quality and greater tolerance to environmental and biotic stresses.

(4) In order to promote commercially viable rates of hybrid production, self-fertilization must be avoided, i.e. fertilization of the female organ by the pollen of the same plant. It is desired that the female organ of the female parent is exclusively fertilized with the pollen of the male parent. In order to obtain a reliable and efficient system for producing seeds needed for hybrid production, one generally needs three essential elements: a means to induce male sterility, a means to propagate the sterility, and a means to restore fertility. For example a fully genetically based system is composed of a male-sterile line (female parent), a fertile maintainer line (male parent allowing propagation of the male-sterile line), and a fertility restorer line (male parent for hybrid production).

(5) Male sterility can be achieved by three different ways. Manual emasculation is the simplest one and is still used in some species where male and female flowers are separated, e.g. corn. However, it is impractical in species like wheat where flowers contain both female and male organs. Male sterility can also be induced by chemical hybridization agents (CHAs) with gametocidal effects. Currently, only a few commercial hybrid wheat cultivars are based on this technology as it can bear substantial financial risks.

(6) Finally, male sterility can also be induced by genetic means. There are many examples of hybrid systems in corn or sorghum based on male sterility induced by genetic means showing the preponderance of this technology compared to the two mentioned previously. However, in other species which are predominantly self-pollinated like wheat, hybrid production is still a challenge (Longin et al, 2012).

(7) The first case of male sterility in wheat was observed in 1951 (Kihara, 1951), where it was observed that sterility was caused by incompatibility between the cytoplasm of *Aegilops caudata* L. and the nucleus of *T. aestivum* var. *erythrospermum*. Subsequently research on *T. timopheevii* cytoplasm showed that this cytoplasm is able to induce sterility in bread wheat (*T. aestivum*) (Wilson and Ross, 1961, Crop Sci, 1:191-193). Orf256 was previously identified as a gene specific to the *T. timopheevii* mitochondrial genome (Rathburn and Hedgcoth, 1991; Song and Hedgcoth, 1994), however, it remains to be shown that orf256 is the genetic determinant of *T. timopheevii* CMS. It was expected that such a cytoplasm could be used in a hybrid production system. However, major limitations arose from the difficulty in finding a completely dominant and stable fertility restorer gene with no negative side effects (notably on yield).

(8) Fertility restoration of male sterile plants harboring *T. timopheevii* CMS cytoplasm (T-CMS cytoplasm) has been reported and eight major restorer loci (designated as Rf1 to Rf8) have been identified and located approximate within the wheat genome. One of the most effective restorer loci is Rf3 (Ma and Sorrells, 1995; Kojima et al, 1997; Ahmed et al 2001; Geyer et al 2016). Two SNP markers allowed the location of the Rf3 locus within a 2 cM fragment on chromosome 1B (Geyer et al, 2016). The author notes that these markers are not diagnostic markers.

(9) While it is understood that restoration to normal pollen fertility could require two or more Rf

loci, it is also well known that modifier loci exist that have either minor effects with low penetrance (Zhou et al 2005, Stojalowski et al 2013) or inhibitory effects on fertility, depending on environmental conditions (Wilson, 1984). It is not yet understood which combination of genes or loci is needed to complete a full restoration of T-CMS in different genetic backgrounds and environmental conditions.

(10) In this context, the development of technologies that enable a full restoration of pollen fertility is of major importance in wheat. It is therefore the object of this invention to propose suitable fertility restorer genes in wheat for the development of a hybrid production system useful for the seed industry.

SUMMARY

(11) A first object of the present disclosure relates to an isolated Rf1 nucleic acid encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to SEQ ID NO:361. An example of Rf1 nucleic acid comprises SEQ ID NO:3119.

(12) The disclosure also relates to a transgenic wheat plant comprising such Rf1 nucleic acid, and, optionally, one or more nucleic acids comprising Rf3, Rf4 and/or Rf7 restorer allele(s), as transgenic element(s).

(13) Another aspect relates to a genetically engineered wheat plant comprising such Rf1 nucleic acid, and, optionally, one or more nucleic acids comprising Rf3, Rf4 and/or Rf7 restorer allele(s), as genetically engineered element(s).

(14) in specific embodiments, said transgenic element(s) or genetically engineered element(s) express polypeptides which restore or improve male fertility to the plant as compared to the parent plant without such transgenic element(s) or genetically engineered element(s).

(15) Yet another aspect relates to a wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm comprising such Rf1 restorer allele, and at least two fertility restorer alleles within the restorer loci chosen amongst Rf3, Rf4 and Rf7, wherein, the Rf3 locus is located at most 10 cM from marker cfn1249269 of SEQ ID NO: 3205 or marker BS00090770 of SEQ ID NO:3228, the Rf7 locus is located at most 10 cM from marker cfn0919993 of SEQ ID NO: 3231, and, the Rf4 locus is located at most 10 cM from marker cfn0393953 of SEQ ID NO: 3233.

(16) The disclosure also provides methods for producing a transgenic wheat plant as described above, wherein the method comprises the steps of transforming a parent wheat plant with one or more Rf1 nucleic acids encoding protein restorer of *T. timopheevii* CMS cytoplasm, selecting a plant comprising said one or more nucleic acid(s) as transgene(s), regenerating and growing said wheat transgenic plant.

(17) Also part of the present disclosure is a method for producing a genetically modified wheat plant as described above, wherein the method comprises the steps of genetically modifying a parent wheat plant to obtain in their genome one or more nucleotide sequence encoding Rf1 protein restorer of *T. timopheevii* CMS cytoplasm, preferably by genome-editing, selecting a plant comprising said one or more nucleotide sequences as genetically engineered elements, regenerating and growing said wheat genetically engineered plant.

(18) The disclosure further relates to a method for producing a wheat plant by crossing, said method includes the following: providing a first wheat plant comprising one or two restorer allele selected among Rf1, Rf3 and Rf7 restorer alleles, crossing said first wheat plant with a second wheat plant comprising one or two restorer alleles selected among Rf1, Rf3 and Rf7 restorer alleles, wherein Rf1, Rf3 and Rf7 restorer alleles are represented at least once in the panel of restorer alleles provided by the first plant and the second plant, collecting the F1 hybrid seed, obtaining homozygous plants from the F1 plants, optionally detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles in the hybrid seed and/or at each generation.

(19) Preferably in such methods, the fertility score of the obtained wheat plant has a fertility score higher than the parent wheat plant.

(20) The disclosure also relates to a method for producing a transgenic or genetically engineered wheat plant, wherein the fertility level of said plant is modified, comprising the step of knocking-down Rf1 restorer allele expression, wherein said Rf1 restorer allele comprises a Rf1 nucleic acid.

(21) The disclosure also relates to the method for producing a wheat hybrid plant comprising the steps of: crossing a sterile female comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant as described above; collecting the hybrid seed; optionally detecting hybridity level of the hybrid seeds.

(22) The wheat hybrid plant as obtained by the above methods are also part of the present disclosure.

(23) The present disclosure also relates to a method of identifying a wheat plant as described above, wherein said wheat plant is identified by detecting the presence of at least one restorer allele Rf1 and, optionally, one or more further restorer alleles selected from the group consisting of Rf3, Rf4 and Rf7.

(24) Accordingly, nucleic acid probes or primers for the specific detection of the restorer allele Rf1 in a wheat plant, and, optionally, one or more of the Rf3, Rf4, and Rf7 restorer alleles, are also disclosed herein.

(25) Another aspect of the disclosure relates to a recombinant nucleic acid comprising a Rf1 nucleic acid encoding a Rf1 protein restorer of *T. timopheevii* CMS cytoplasm, operably linked to regulatory elements and the vectors for use in transformation of a wheat plant, comprising such recombinant nucleic acids.

DETAILED DESCRIPTION

(26) Nucleic Acids of the Present Disclosure

(27) An aspect of the present disclosure relates to the cloning and characterization of genes encoding restorer of fertility proteins that act on *T. timopheevii* CMS cytoplasm (hereafter referred as Rf genes or nucleic acids) in wheat plants and the use of the corresponding Rf nucleic acids for producing transgenic wheat plants, for modifying wheat plants by genome editing, and/or for detecting such Rf genes in wheat plants.

(28) Whenever reference to a “plant” or “plants” is made, it is understood that also plant parts (cells, tissues or organs, seed pods, seeds, severed parts such as roots, leaves, flowers, pollen, etc.), progeny of the plants which retain the distinguishing characteristics of the parents (especially, male fertility associated with the claimed Rf nucleic acids), such as seed obtained by selfing or crossing, e.g. hybrid seeds (obtained by crossing two inbred parent plants), hybrid plants and plant parts derived therefrom are encompassed herein, unless otherwise indicated.

(29) As used herein, the term “wheat plant” refers to species of the genus *Triticum* as for example, *T. aestivum*, *T. aethiopicum*, *T. araraticum*, *T. boeoticum*, *T. carthlicum*, *T. compactum*, *T. dicoccoides*, *T. dicoccon*, *T. durum*, *T. ispahanicum*, *T. karamyshevii*, *T. macha*, *T. militinae*, *T. monococcum*, *T. polonicum*, *T. spelta*, *T. sphaerococcum*, *T. timopheevii*, *T. turanicum*, *T. turgidum*, *T. urartu*, *T. vavilovii*, *T. zhukovskyi* Faegi. Wheat plant also refers to species of the genera *Aegilops* and *Triticale*.

(30) As used herein, the term “restorer of fertility of *T. timopheevii* CMS cytoplasm” refers to a protein whose expression in a wheat plant comprising *T. timopheevii* CMS cytoplasm contributes to the restoration of the production of pollen in the *Triticum timopheevii* CMS system.

(31) As used herein, the term “allele(s)” means any of one or more alternative forms of a gene at a particular locus. In a diploid, alleles of a given gene are located at a specific location or locus on a chromosome. One allele is present on each chromosome of the pair of homologous chromosomes. The same definition is used for plants bearing a higher level of ploidy like in *Triticum* gender wherein, for example, *T. aestivum* is an hexaploid plant.

(32) As used herein, the term “restorer allele of *T. timopheevii* CMS cytoplasm” refers to an allele which contributes to the restoration of the production of pollen in the CMS *Triticum timopheevii* system.

(33) The restoration of pollen fertility may be partial or complete. The pollen fertility can be evaluated by the pollen fertility tests as described in the Examples below. In particular, the fertility score of F1 wheat plants having CMS-*T. timopheevii* cytoplasm (from test restorer line with CMS hybrids) may be calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets and may be compared with the fertility scores of a panel of control fertile plants, for example elite inbred lines bearing a normal wheat cytoplasm, grown in the same area and under the same agro-environmental conditions. It is preferred that such panels of lines comprise a set of at least 5 elite inbred lines wherein these lines are representative of the area where the fertility test is achieved. Besides, it is preferred that at least 10 spikes from different individual F1 plants be assessed for a given experiment.

(34) If the fertility score is not null, then the plant has acquired partial or full restoration of fertility. For each fertility score, a statistical test is calculated to obtain a p-value. Examples of statistical tests are the Anova or mean comparison tests. A p-value below a 5% threshold will indicate that the two distributions are statistically different. Therefore, a significant decrease of the fertility score of the tested wheat plant as compared to the fertility score of the fully fertile control plant is indicative that the F1 plant has not acquired full restoration of fertility (i.e. partial restoration). A similar or higher fertility score is indicative that the F1 plant has acquired full restoration of fertility. In a preferred embodiment, the wheat plant, such as transgenic or genetically engineered wheat plant, according to the present disclosure, has acquired full restoration of fertility.

(35) The loci of the restorer alleles of *T. timopheevii* CMS cytoplasm within Rf1, Rf3, Rf4 and Rf7 have been mapped in the present disclosure. The corresponding restorer alleles are designated Rf1, Rf3, Rf4 and Rf7 restorer alleles and have been described in the art. In particular, a wheat plant source of the Rf3 restorer allele includes the commercial following lines: Allezy, Altigo, Altamira, see table 15. A wheat plant source of the Rf4 restorer allele includes the following lines: R113 or L13.

(36) In specific embodiments, representative alleles of Rf1, Rf3, Rf4 and Rf7 restorer alleles are provided by the seed sample chosen amongst: NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817.

(37) As used herein, the term “centimorgan” (“cM”) is a unit of measure of recombination frequency. One cM is equal to a 1% chance that a marker at one genetic locus will be separated from a marker at a second locus due to crossing over in a single generation.

(38) As used herein, the term “chromosomal interval” designates a contiguous linear span of genomic DNA that resides in planta on a single chromosome. The genetic elements or genes located on a single chromosomal interval are physically linked. The size of a chromosomal interval is not particularly limited. In some aspects, the genetic elements located within a single chromosomal interval are genetically linked, typically with a genetic recombination distance of, for example, less than or equal to 20 cM, or alternatively, less than or equal to 10 cM. That is, two genetic elements within a single chromosomal interval undergo recombination at a frequency of less than or equal to 20% or 10%.

(39) The present disclosure provides nucleic acids and their recombinant forms comprising the coding sequence of either Rf1, Rf3, Rf4, Rf7 or Rf-rye restorer of fertility proteins active in *T. timopheevii* CMS cytoplasm.

(40) As used herein, a “recombinant nucleic acid” is a nucleic acid molecule, preferably a DNA molecule, comprising a combination of nucleic acid molecules that would not naturally occur together and is the result of human intervention, e.g., a DNA molecule that is comprised of a combination of at least two DNA molecules heterologous to each other, and/or a DNA molecule that is artificially synthesized and comprises a polynucleotide sequence that deviates from the polynucleotide sequence that would normally exist in nature.

(41) Such nucleic acids encoding candidate restorer of fertility proteins of *T. timopheevii* CMS cytoplasm have been isolated as described in the Examples below. Accordingly, a first aspect of the

disclosure are nucleic acids encoding a protein restorer of fertility of *T. timopheevii* having an amino acid sequence at least 95% identical, typically at least 96% identical, to an amino acid sequence chosen amongst any one of SEQ ID NO:1 to SEQ ID NO: 1554.

(42) Percentage of sequence identity as used herein is determined by calculating the number of matched positions in aligned amino acid sequences, dividing the number of matched positions by the total number of aligned amino acids, and multiplying by 100. A matched position refers to a position in which identical amino acids occur at the same position in aligned amino acid sequences. For example, amino acid sequences may be aligned using the CD-hit (settings-c 0.96-n 5-G 0-d 0-AS 60-A 105-g 1, see <http://weizhongli-lab.org/cd-hit/>).

(43) The above candidate nucleic acids encoding any one of polypeptides SEQ ID NO: 1 to 1554, can further be assessed for their capacity to restore fertility of sterile wheat plant as described below.

(44) It is therefore disclosed herein a method for assessing the capacity of a nucleic acid to restore fertility, wherein the method comprises the steps of: a. introducing one or more candidate Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid encoding a putative amino acid sequence of at least 95% identical to any one of SEQ ID NO 1 to SEQ ID NO 1554 into a parent wheat sterile plant and *T. timopheevii* CMS cytoplasm, b. selecting the transgenic plant bearing one or more candidate nucleic acid as transgene(s), and c. evaluating the fertility of the transgenic plants as compared to the parent wheat sterile plant based on a fertility restoration assay, wherein an improvement in the fertility restoration is indicative that said nucleic acid has the capacity to restore fertility.

(45) In a specific embodiment, the parent wheat sterile plant is the Fielder line bearing the *T. timopheevii* CMS cytoplasm.

(46) In another specific embodiment of the above method, the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye candidate nucleic acid sequence is selected from those encoding an amino acid sequence having at least 95% identity, or at least 96% identity, for example 100% identity, to any one of SEQ ID NO 1 to SEQ ID NO 1554.

(47) Typically, the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye candidate nucleic acids are selected among the following nucleic acids of SEQ ID NO: 1555 to SEQ ID NO:3107 and 3133.

(48) In a further specific embodiment, where appropriate, the nucleic acid sequence may be optimized for increased expression in the transformed plant. There are a number of optimizations that can be performed at the DNA level, without changing the protein sequence, by conservative codon exchanges which replace one codon by another codon encoding the same amino acid. Besides, the nucleic acid sequence can be modified for cloning purpose. Like for optimization, such modification is achieved without changing the protein sequence.

(49) Rf1 Nucleic Acids

(50) In specific embodiment, the nucleic acid of the present disclosure is a Rf1 nucleic acid.

(51) As used herein, the term “Rf1 nucleic acid” refers to a nucleic acid comprising a gene encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs 1-2, SEQ ID NOs 288-290, SEQ ID NOs 293-296, SEQ ID NOs 343-346, SEQ ID NOs 349-354, SEQ ID NOs 359, 361 and 362, SEQ ID NOs 396 and 397, SEQ ID NOs 428-430, SEQ ID NO 517 and 519, SEQ ID NOs 752-754, SEQ ID NOs 1092, 1093 and 1095, typically, SEQ ID NOs 359, 361 and 362 and SEQ ID NO 428-430. In a particularly preferred embodiment, the Rf1 nucleic acid encodes an amino acid sequence having at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to SEQ ID NO:361. Examples of corresponding specific Rf1 nucleic acids are referred to in Table 7.

(52) In particular, and as shown in Example 6, the inventors have identified that RFL79 sequence of SEQ ID NO:361 (as depicted in Table 7) can restore male fertility of CMS-Fielder plants. Accordingly, in a preferred embodiment, examples of Rf1 nucleic acids comprises the disclosed

Rf1 nucleic acid sequences of SEQ ID NO:1913, SEQ ID NO: 1914, SEQ ID NO: 1915, SEQ ID NO: 1916 or SEQ ID NO:3119, preferably a Rf1 nucleic acid comprises SEQ ID NO:3119.

(53) Rf3 Nucleic Acids

(54) In specific embodiment, the nucleic acid of the present disclosure is a Rf3 nucleic acid.

(55) As used herein, the term “Rf3 nucleic acid” refers to a nucleic acid comprising a gene encoding a Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 124 and 125, SEQ ID NO:147, SEQ ID NO:150, SEQ ID NO: 156, SEQ ID NO: 158, SEQ ID NO:297, SEQ ID NO:299, SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 553 and 554, SEQ ID NOs: 557 and 558, SEQ ID NOs: 676 and 677, SEQ ID NOs: 684 and 685, SEQ ID NOs: 696 and 697, SEQ ID NOs: 938 and 939 and SEQ ID NOs: 1038 and 1039, typically, SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299. Preferred Rf3 nucleic acids encode a Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, with the corresponding amino acid sequence having at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 158, SEQ ID NO:676 and SEQ ID NO:684. Examples of corresponding specific Rf3 nucleic acids are referred to in Table 7 or further described in Example 12. Typically, examples of specific Rf3 nucleic acids comprises SEQ ID NO:1712, SEQ ID NO:2230, SEQ ID NO: 2238, SEQ ID NO:3146, SEQ ID NO:3147 or SEQ ID NO:3148.

(56) Rf4 Nucleic Acids

(57) In specific embodiment, the nucleic acid of the present disclosure is a Rf4 nucleic acid.

(58) As used herein, the term “Rf4 nucleic acid” refers to a nucleic acid comprising a gene encoding a Rf4 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO:477, and SEQ ID NOs 3135-3138, typically SEQ ID NO:477 and SEQ ID NOs 3136-3138. Examples of corresponding specific Rf4 nucleic acids are listed in Table 7 and further include any of SEQ ID NO: 2031, and SEQ ID NO:3140-3142.

(59) Rf7 Nucleic Acids

(60) In specific embodiment, the nucleic acid of the present disclosure is a Rf7 nucleic acid.

(61) As used herein, the term “Rf7 nucleic acid” refers to a nucleic acid comprising a gene encoding a Rf7 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 240-243, SEQ ID NOs 303-305, SEQ ID NO:363, SEQ ID NOs 375-377, SEQ ID NOs 497-499, SEQ ID NO:516, SEQ ID NOs 709-711, SEQ ID NO: 768, typically SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768. Examples of corresponding specific Rf7 nucleic acids are referred to in Table 7.

(62) Rf-Rye Nucleic Acids

(63) In specific embodiment, the nucleic acid of the present disclosure is a Rf-rye nucleic acid.

(64) As used herein, the term “Rf-rye nucleic acid” refers to a nucleic acid comprising a gene encoding a Rf-rye protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859. Examples of corresponding specific Rf-rye nucleic acids are referred to in Table 7.

(65) Rf Nucleic Acids as Transgene

(66) The present disclosure more specifically relates to DNA molecules including one or more of the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids. In particular, the disclosure relates to any DNA

molecule resulting from the insertion of a transgene in the wheat plant, said transgene including one or more of the above described Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids, and which insertion results in the expression of corresponding RNA and/or protein in the wheat plant.

(67) Also part of the present disclosure is a nucleic acid that has been extracted from cells, or tissues, or homogenate from a plant or seed or plant tissue; or can be produced as an amplicon from extracted DNA or RNA from cells, or tissues, or homogenate from a plant or seed or plant tissue, any of which is derived from such materials derived from a plant comprising such nucleic acid as disclosed above.

(68) As used herein, the term “transgene” or “transgenic element” refers to the nucleic acid (e.g. DNA molecule) incorporated into a host cell's genome. The term “transgene” or “transgenic element” refers in particular to a sequence that is not normally present in a given host genome in the genetic context in which the sequence is currently found. In this respect, the sequence may be native to the host genome, but be rearranged with respect to other genetic sequences within the host genomic sequence. For example, the transgene is rearranged at a different locus as compared to the native gene.

(69) Said one or more transgenic element(s) enables the expression of polypeptides which restore or improve male fertility to the plant having *T. timopheevii* CMS cytoplasm, as compared to the parent plant which do not comprise the transgenic element.

(70) A particular transgenic element is the recombinant nucleic acid as defined above, for example Rf1 nucleic acids as defined above. In specific embodiments, a transgenic element includes an Rf nucleic acid under the control of a constitutive promoter, such as the ZmUbi promoter.

(71) Recombinant Nucleic Acids for Use in Transforming Wheat Plants

(72) Such Rf nucleic acids as defined above are also useful to transform or genetically modify wheat plant, in particular wheat plant which does not have one or more of the fertility restoration alleles Rf1, Rf3, Rf4, Rf7 and Rf-rye.

(73) Another aspect of the present disclosure relates to a vector for use in transforming wheat plant, comprising one or more of Rf1, Rf3, Rf4, Rf7 and Rf-rye nucleic acids as described above.

(74) Vectors for use in transforming wheat plant includes at least the coding sequence of the corresponding protein restorer of fertility (either naturally occurring coding sequence, or improved sequence, such as codon optimized sequence), such coding sequence being operably linked to a regulatory element such as a promoter.

(75) The term “promoter” as used herein refers to a region of DNA upstream of the coding sequence (upstream of start codon) and including DNA regions for recognition and binding of RNA polymerase and other proteins to initiate transcription at the start codon. Examples of constitutive promoters useful for expression include the 35S promoter or the 19S promoter (Kay et al, 1987), the rice actin promoter (McElroy et al, 1990), the pCRV promoter (Depigny-This et al, 1992), the CsVMV promoter (Verdaguer et al. 1996), the ubiquitin 1 promoter of maize (Christensen and Quail, 1996), the regulatory sequences of the T-DNA of *Agrobacterium tumefaciens*, including mannopine synthase, nopaline synthase, octopine synthase.

(76) Promoters may be «tissue-preferred», i.e. initiating transcription in certain tissues or “tissue-specific”, i.e. initiating transcription only in certain tissues. Examples of such promoters are DHN12, LTR1, LTP1 specific of the embryo, SS1 specific of the phloem, OSG6B specific of the tapetum (Gotz et al 2011 and Jones 2015).

(77) Other suitable promoters could be used. It could be an inducible promoter, a developmentally regulated promoter. An “inducible” promoter initiates transcription under some environmental control or any stress-induced like for example the abiotic stress-induced RD29, COR14b (Gotz et al, 2011).

(78) Constitutive promoters may be used, such as the ZmUbi promoter, typically the ZmUbi promoter of SEQ ID NO:3134. Finally, promoter of SEQ ID NO:3114, SEQ ID NO:3123 and SEQ ID NO:3113 corresponding to pTaRFL46, 79 and 104 can also be used.

(79) In specific embodiments, the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids of the present disclosure are operably linked to heterologous promoters, i.e. a promoter which is not the natural promoter of the corresponding Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids as found in wheat. Typical recombinant constructs of Rf3 nucleic acids with heterologous promoters include any one of SEQ ID NO:3150-3153, and any one of SEQ ID NO: 3156-SEQ ID NO:3159. Typical recombinant constructs of Rf1 nucleic acids with heterologous promoters includes SEQ ID NO:3122, or a nucleic acid of SEQ ID NO: 3119 under the regulation of the promoter of SEQ ID NO:2123.

(80) The vector may further comprise additional elements including selection gene marker, operably linked to regulatory element, that allows to select the transformed plant cells containing the vector, comprising the nucleic acids of the present disclosure as transgene.

(81) The vector may further comprise additional elements including counter-selection gene marker, operably linked to regulatory element that allows to counter-select the transformed plant cells which do not have maintained the counter-selection gene marker in its genome.

(82) In a specific embodiment, the vector according to the present disclosure may be vector suitable for *Agrobacterium*-mediated transformation, in particular *Agrobacterium tumefaciens* or *Agrobacterium rhizogenes* mediated transformation, as described in the next section.

(83) Methods for Producing a Wheat Transgenic Plant

(84) Another aspect of the present disclosure relates to the use of the above-described nucleic acids for producing wheat transgenic plant expressing protein restorer of fertility.

(85) The term “transgenic plant” refers to a plant comprising such a transgene. A “transgenic plant” includes a plant, plant part, a plant cell or seed whose genome has been altered by the stable integration of recombinant DNA. A transgenic plant includes a plant regenerated from an originally-transformed plant cell and progeny transgenic plants from later generations or crosses of a transformed plant. As a result of such genomic alteration, the transgenic plant is distinctly different from the related wild type plant. An example of a transgenic plant is a plant described herein as comprising one or more of the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids, typically as transgenic elements. For example, the transgenic plant includes one or more Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids as transgene, inserted at loci different from the native locus of the corresponding Rf gene(s). Accordingly, it is herein disclosed a method for producing a wheat transgenic plant, wherein the method comprises the steps of (i) transforming a parent wheat plant with Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids, (ii) selecting a plant comprising said one or more nucleic acid(s) as transgene(s), (iii) regenerating and (iv) growing said wheat transgenic plant.

(86) For transformation methods within a plant cell, one can cite methods of direct transfer of genes such as direct micro-injection into plant embryos, vacuum infiltration or electroporation, direct precipitation by means of PEG or the bombardment by gun of particles covered with the plasmidic DNA of interest.

(87) It is preferred to transform the plant cell with a bacterial strain, in particular *Agrobacterium*, in particular *Agrobacterium tumefaciens*. In particular, it is possible to use the method described by Ishida et al. (Nature Biotechnology, 14, 745-750, 1996) for the transformation of monocotyledons.

(88) Descriptions of *Agrobacterium* vector systems and methods for *Agrobacterium*-mediated gene transfer are provided by Moloney et al., Plant Cell Reports 8:238 (1989). See also, U.S. Pat. No. 5,591,616 issued Jan. 7, 1997.

(89) Alternatively, direct gene transfer may be used. A generally applicable method of plant transformation is microprojectile-mediated transformation wherein DNA is carried on the surface of microprojectiles measuring 1 to 4 micron. The expression vector is introduced into plant tissues with a biolistic device that accelerates the microprojectiles to speeds of 300 to 600 m/s which is sufficient to penetrate plant cell walls and membranes. Sanford et al., Part. Sci. Technol. 5:27 (1987), Sanford, J. C., Trends Biotech. 6:299 (1988), Klein et al., BioTechnology 6:559-563

(1988), Sanford, J. C., *Physiol Plant* 7:206 (1990), Klein et al., *BioTechnology* 10:268 (1992). Several target tissues can be bombarded with DNA-coated microprojectiles in order to produce transgenic plants, including, for example, callus (Type I or Type II), immature embryos, and meristematic tissue.

(90) Following transformation of wheat target tissues, expression of the selectable marker genes allows for preferential selection of transformed cells, tissues and/or plants, using regeneration and selection methods now well known in the art.

(91) The foregoing methods for transformation would typically be used for producing a transgenic plant including one or more of Rf1, Rf3, Rf4, Rf7 or Rf Rye nucleic acids as transgenic element(s).

(92) The transgenic plant could then be crossed, with another (non-transformed or transformed) inbred line, in order to produce a new transgenic line. Alternatively, a genetic trait which has been engineered into a particular line using the foregoing transformation techniques could be moved into another line using traditional backcrossing techniques that are well known in the plant breeding arts. For example, a backcrossing approach could be used to move an engineered trait from a public, non-elite inbred line into an elite inbred line, or from an inbred line containing a foreign gene in its genome into an inbred line or lines which do not contain that gene. As used herein, "crossing" can refer to a simple X by Y cross, or the process of backcrossing, depending on the context.

(93) When the term transgenic wheat plant is used in the context of the present disclosure, this also includes any wheat plant including, as a transgenic element one or more of Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids and wherein one or more desired traits have further been introduced through backcrossing methods, whether such trait is a naturally occurring one or a transgenic one.

Backcrossing methods can be used with the present invention to improve or introduce one or more characteristic into the inbred. The term backcrossing as used herein refers to the repeated crossing of a hybrid progeny back to one of the parental wheat plants. The parental wheat plant which contributes the gene or the genes for the desired characteristic is termed the nonrecurrent or donor parent. This terminology refers to the fact that the nonrecurrent parent is used one time in the backcross protocol and therefore does not recur. The parental wheat plant to which the gene or genes from the nonrecurrent parent are transferred is known as the recurrent parent as it is used for several rounds in the backcrossing protocol (Fehr et al, 1987).

(94) In a typical backcross protocol, the recurrent parent is crossed to a second nonrecurrent parent that carries the gene or genes of interest to be transferred. The resulting progeny from this cross are then crossed again to the recurrent parent and the process is repeated until a wheat plant is obtained wherein all the desired morphological and physiological characteristics of the recurrent parent are recovered in the converted plant in addition to the gene or genes transferred from the nonrecurrent parent. It should be noted that some, one, two, three or more, self-pollination and growing of a population might be included between two successive backcrosses.

(95) Methods for Producing a Wheat Genetically Engineered Plant

(96) An aspect of the present disclosure relates to a DNA fragment of the corresponding protein restorer of fertility (either naturally occurring coding sequence, or improved sequence, such as codon optimized sequence) combined with genome editing tools (such TALENs, CRISPR-Cas, Cpf1 or zing finger nuclease tools) to target the corresponding Rf restorer alleles within the wheat plant genome by insertion at any locus in the genome or by partial or total allele replacement at the corresponding locus. In particular, the disclosure relates to a genetically modified (or engineered) wheat plant, wherein the method comprises the steps of genetically modifying a parent wheat plant to obtain in their genome one or more nucleotide sequence encoding protein restorer of *T.*

timopheevii CMS cytoplasm Rf1, Rf3, Rf4 or Rf7 as disclosed herein, preferably by genome-editing, selecting a plant comprising said one or more nucleotide sequences as genetically engineered elements, regenerating and growing said wheat genetically engineered plant.

(97) As used herein, the term "genetically engineered element" refers to a nucleic acid sequence

present in the genome of a plant and that has been modified by mutagenesis or by genome-editing tools, preferentially by genome-editing tools. In specific embodiments, a genetically engineered element refers to a nucleic acid sequence that is not normally present in a given host genome in the genetic context in which the sequence is currently found but is incorporated in the genome of plant by use of genome-editing tools. In this respect, the sequence may be native to the host genome, but be rearranged with respect to other genetic sequences within the host genomic sequence. For example, the genetically engineered element is a gene that is rearranged at a different locus as compared to a native gene. Alternatively, the sequence is a native coding sequence that has been placed under the control of heterologous regulatory sequences. Other specific examples are described hereafter. The term “genetically engineered plant” or “genetically modified plant” refers to a plant comprising such genetically engineered element. A “genetically engineered plant” includes a plant, plant part, a plant cell or seed whose genome has been altered by the stable integration of recombinant DNA. As used herein, the term “genetically engineered plant” further includes a plant, plant part, a plant cell or seed whose genome has been altered by genome editing techniques. A genetically engineered plant includes a plant regenerated from an originally-engineered plant cell and progeny of genetically engineered plants from later generations or crosses of a genetically engineered plant. As a result of such genomic alteration, the genetically engineered plant is distinctly different from the related wild type plant. An example of a genetically engineered plant is a plant described herein as comprising one or more of the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids. For example, the genetically engineered plant includes one or more Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids as genetically engineered elements, inserted at loci different from the native locus of the corresponding Rf gene(s).

(98) In specific embodiments, said genetically engineered plants do not include plants which could be obtained exclusively by means of an essentially biological process.

(99) Said one or more genetically engineered element(s) enables the expression of polypeptides which restore or improve male fertility to the plant having *T. timopheevii* CMS cytoplasm, as compared to the parent plant which do not comprise the genetically engineered element(s).

(100) A particular genetically engineered element is the Rf nucleic acid as defined above, for example Rf1 nucleic acids as defined above. In specific embodiments, a genetically engineered element includes an Rf nucleic acid under the control of expression elements as promoter and/or terminator. Suitable promoter can be a constitutive promoter, such as the ZmUbi promoter or an endogenous Rf promoter native or modified.

(101) Another aspect of the disclosure relates to a genetically engineered wheat plant, which comprises the modification by point mutation insertion or deletion of one or few nucleotides of an allele sequence rf or Rf, as genetically engineered element, into the respectively Rf or rf allele, by any of the genome editing tools including base-editing tool as described in WO2015089406 or by mutagenesis.

(102) The present disclosure further includes methods for modifying fertility level in a plant by genome editing, comprising providing a genome editing tool capable of replacing partially or totally a rf1, rf3, rf4, rf7 or rf-rye non-restorer allele sequence or form in a wheat plant by its corresponding Rf1, Rf3, Rf4, Rf7 or Rf-rye restorer allele sequence as disclosed herein.

(103) The term “rf non-restorer allele sequence or form” can be related to the presence in the genome of a rf non-restorer allele or to the absence in the genome of any rf or Rf allele. For example, in the rf3/Rf3 system, one wheat non-restorer line can be characterized by the presence of a rf3 allele sequence while another non-restorer line is characterized by the absence of any Rf3 or rf3 allelic sequence. The Rf3 restorer plant will be characterized by the presence of the Rf3 allele sequence in the genome. In the case of the rf4/Rf4 system, the non-restorer plant is characterized by the absence of any rf4 or Rf4 allelic forms while the Rf4 restorer plant will be characterized by the presence in the genome of the Rf4 gene sequence.

(104) In specific embodiments, methods for modifying fertility level in a plant by genome editing

comprises providing a genome editing tool capable of replacing or modifying a rf3 non-restorer allele to obtain a Rf3 restorer allele comprising SEQ ID NO:3146 (RFL29a). In other specific embodiments, rf3 non-restorer allele may comprise RFL29c characterized by a frameshift compared to RFL29a nucleotide sequence as shown in Example 22 and FIG. 12 and SEQ ID NO:3457.

(105) The disclosure further includes methods for modifying fertility level in a plant introducing the endogenous promoter of a restorer Rf gene in order to increase the expression of the corresponding endogenous Rf gene either by genome-editing or by mutagenesis.

(106) In a specific embodiment, the disclosure includes methods for modifying fertility level in a plant by genome editing of a weak fertile plant by modifying the 5'UTR sequence of the Rf3 "weak" RFL29b allele which 5'UTR region includes a 163 bp insertion to be deleted as shown in example 15.

(107) In a further specific embodiment, the invention includes a method for modifying fertility level in a plant by mutagenesis or by genome editing the endogenous promoter of a restorer Rf gene in order to increase the expression of the endogenous Rf gene. As an example, the sequence of proTaRFL79 depicted in SEQ ID N°3123 could be mutated or edited to increase RFL79 protein level.

(108) In another specific embodiment, whenever a stronger promoter is located upstream to the promoter of the Rf restorer gene, a deletion of that promoter and the region upstream can be achieved in order to juxtapose the stronger promoter to the Rf gene.

(109) In a specific embodiment, one rf non-restorer allele is replaced partially or totally by anyone of the Rf1, Rf3, Rf4, Rf7 or Rf-rye restorer allele. In such disclosure, a non-restorer rf1 allele could be replaced, for example, by a Rf3, or a Rf4, or a Rf7, or Rf-rye allele.

(110) In another aspect of the disclosure, at least one restorer allele from among Rf1, Rf3, Rf4, Rf7 and Rf-Rye can be integrated at one or more target sites in the wheat plant genome, typically in order to get an expression of said restorer alleles. Said expression can be achieved either by taking advantage of the presence at the targeting locus of a promoter, more specifically a strong promoter, and/or a terminator and by targeting with the Rf allele said expression elements. In a specific embodiment, the endogenous Rf allele is deleted from one first locus and further integrated downstream a suitable promoter at a second locus in the same plant genome.

(111) In a specific embodiment of the invention, the target sites can be located in the Rf1, Rf3, Rf4 and/or Rf7 locus as defined in example 17. In a more specific embodiment, the target sites can either be the Rf1, Rf3, Rf4, Rf7 or Rf-Rye endogenous gene sequence or any other target sites different from the Rf1, Rf3, Rf4, Rf7 or Rf-Rye endogenous gene sequences.

(112) In a preferred aspect of the disclosure, a wheat plant can comprise in its genome, at only one locus, the Rf1, Rf3 and Rf7 restorer alleles.

(113) Such genome editing tool includes without limitation targeted sequence modification provided by double-strand break technologies such as, but not limited to, meganucleases, ZFNs, TALENs (WO2011072246) or CRISPR CAS system (including CRISPR Cas9, WO2013181440), Cpf1 or their next generations based on double-strand break technologies using engineered nucleases.

(114) Method for Decreasing the Fertility Level in Wheat Plant:

(115) Alternatively, the present disclosure further includes methods for modifying fertility level in a plant by reverting a restorer line comprising a Rf allele to a maintainer line comprising a rf allele. Such method could be used for any plant comprising a Rf allele, including Rf1. It is of particular interest in the case of a hybrid production system based on Rf3 restorer allele, wherein for example, the Rf3 sequence is RFL29a and the rf3 sequence is RFL29c.

(116) The decrease in fertility could be obtained by knocking-down Rf gene or allele as described below.

(117) More specifically, the method can correspond first to an inhibition of the expression by RNAi

directed against the Rf allele of interest or by impairing the promoter function of the Rf allele either by mutagenesis or by genome editing.

(118) In another aspect, the fertility level decrease is obtained by mutagenesis, classically induced with mutagenic agents, or by genome editing technologies to impair the protein function, by deleting totally or partially the gene, or by modifying the gene sequence reading frame to impair protein translation.

(119) In a further aspect, the disclosure relates to the transgenic or genetically engineered wheat plant with decreased fertility level obtained by the methods described above.

(120) The Transgenic or Genetically Engineered Wheat Plant of the Present Disclosure

(121) Another aspect of the present disclosure relates to a transgenic or genetically engineered wheat plant comprising one or more Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acid(s) as described in the previous sections, as transgenic or genetically engineered elements respectively.

(122) Said transgenic or genetically engineered plant may be obtained by the methods described in the previous section.

(123) The transgenic or genetically engineered plants of the present disclosure may advantageously be used as parent plant in order to produce fertile wheat transgenic or genetically engineered plant restorer of *T. timopheevii* CMS cytoplasm. In particular, in specific embodiment, the wheat transgenic or genetically engineered plant is a fertile wheat transgenic or genetically engineered plant restorer of *T. timopheevii* CMS cytoplasm. Typically, a transgenic or genetically engineered wheat plant according to the present disclosure comprises a combination of at least two different transgenic or genetically engineered elements selected from the group consisting of Rf1, Rf3, Rf4, Rf7 and Rf-rye encoding nucleic acids.

(124) The transgenic or genetically engineered wheat plant as disclosed herein may express such protein restorer of fertility Rf1, Rf3, Rf4, Rf7 and/or Rf-Rye together as the result of the transgene's expression or expression of the genetically engineered element and may additionally express other protein restorer of fertility, as the result of naturally occurring alleles.

(125) In one embodiment, said combination of at least two, three or four of Rf1, Rf3, Rf4, Rf7 and Rf-rye nucleic is found in the same locus in the genome of the transgenic or genetically engineered plant. In other embodiments, the corresponding nucleic acids of the combination are located in distinct loci. In one specific embodiment, said combination may be obtained by crossing transgenic plants of the present disclosure each bearing one nucleic acid of the combination as a transgene at a distinct locus.

(126) Typically, said transgenic or genetically engineered plant includes the following combination of nucleic acids as transgenic or genetically engineered elements: a. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, b. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO: 516 and SEQ ID NO:768, c. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity, with any one of SEQ ID NO: 227, SEQ ID NO:378 and SEQ ID NO:859, d. a Rf3 nucleic acid, preferably encoding an amino acid

sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, e. a Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859, f. a Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with SEQ ID NO:363, SEQ ID NO: 516 and SEQ ID NO:768, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859, g. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO: 684, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768; h. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO: 684, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity, or at least 96% identity, with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859; i. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity, for example at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO: 361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf7 nucleic acid of claim 4, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity, with any one of SEQ ID NO: 227, SEQ ID NO:378 and SEQ ID NO:859; or, j. a Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid, preferably preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859.

(127) In other specific embodiments, said transgenic or genetically engineered plant further includes a Rf4 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 477, and SEQ ID NOs 3135-3138 in addition to any one of the above defined combination of Rf1, Rf3, Rf7 and/or Rf-Rye nucleic acids as defined in the previous paragraph, as transgenic elements.

(128) The disclosure also relates to hybrid wheat plants which can be produced by crossing a

transgenic or genetically engineered wheat plant restorer of fertility according to the present disclosure as described above with a second plant.

(129) In certain embodiments, the wheat plant according to the disclosure is alloplasmic and comprises the *T. timopheevii* cytoplasm.

(130) For example, a hybrid wheat plant may be obtained by crossing a wheat plant restorer of fertility according to the present disclosure as described above, preferably comprising Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids, and a wheat plant which does not express corresponding Rf1, Rf3, Rf4, Rf7 or Rf-rye protein restorer of fertility.

(131) It is also disclosed herein a method for producing a wheat hybrid transgenic or genetically engineered plant comprising the steps of: a. crossing a sterile female wheat plant comprising the *T. timopheevii* cytoplasm with a fertile male transgenic or genetically engineered wheat plant restorer of fertility of the present disclosure as described above; b. collecting the hybrid seed; c. optionally detecting the presence of *T. timopheevii* cytoplasm, and/or at least one or more of the Rf nucleic acids chosen amongst Rf1, Rf3, Rf4, Rf7 and Rf-rye in the hybrid seed, as transgenic elements or genetically engineered elements; and, d. optionally detecting hybridity level of the hybrid seed.

(132) Therefore, it is also disclosed herein the wheat transgenic or genetically engineered plants or lines according to the present disclosure developed to obtain such hybrid plants. Such transgenic or genetically engineered plants or lines typically comprise the cytoplasmic elements necessary for the implementation of the corresponding hybrid system. Preferably, the transgenic or genetically engineered plants or lines comprise a combination of at least two, three or four of Rf1, Rf3, Rf4, Rf7 and Rf-rye nucleic acids, and *T. timopheevii* cytoplasm.

(133) Alternatively, the detection of the presence of *T. timopheevii* cytoplasm and of at least one or more of the Rf nucleic acids chosen amongst Rf1, Rf3, Rf4, Rf7 and Rf-rye (step “c” of the method described above) can be performed on the parent lines in order to check their genotype before to start the cross (step “a”).

(134) In a certain embodiment of the disclosure, the male wheat plant is taller than the female wheat plant. This can be achieved by using male plant bearing Rht alleles that allows to obtain the size differences. Optionally, the disclosure further comprises the step of applying an herbicide to the fertile plants standing above the height of the shorter female plants, and further optionally, comprises the step of harvesting the seeds and selecting the seeds to remove undesirable self-fertilized male seeds, using a morphological character and/or a phenotypic character like size, shape, color etc. . . . An example of such method for hybrid production is described in WO2015135940.

(135) The T-CMS cytoplasm can be detected either phenotypically wherein a plant bearing rf genes and a T-CMS cytoplasm will be sterile or by molecular means able to detect the orf256 gene as described in Rathburn and Hedgcoth, 1991 and Song and Hedgcoth, 1994.

(136) The disclosure also relates to a method for improving the level of fertility restoration of a parent wheat plant bearing a fertility level lower than a full restoration level, comprising the steps of transforming said parent wheat plant with a vector comprising a Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid as described above, or genetically engineering said Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids in said wheat plant. The method further comprises the step of selecting a transgenic or genetically engineered wheat plant comprising said Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid(s) as a transgene or genetically engineered element, preferably Rf1, Rf3 and Rf4, regenerating and growing said transgenic or genetically engineered wheat plant, wherein said transgenic or genetically engineered wheat plant has an improved fertility restoration level as compared to the parent plant.

(137) The disclosure further provides a method for restoring fertility of a sterile wheat plant bearing the *T. timopheevii* CMS cytoplasm, comprising the step of transforming a parent sterile wheat plant bearing the *T. timopheevii* CMS cytoplasm, with a Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid as described above or genetically engineering said sterile plant to express a Rf1, Rf3,

Rf4, Rf7 and/or Rf-rye nucleic acids.

(138) Use of the Nucleic Acids of the Present Disclosure for Identifying Rf1, Rf3, Rf4, Rf7 and Rf-Rye Restorer Alleles or Transgenic Elements

(139) The present disclosure further provides methods of identifying the respective Rf1, Rf3, Rf4, Rf7 and/or Rf-rye restorer alleles and/or Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids as disclosed in the previous sections, and more generally methods of selecting or breeding wheat plants for the presence or absence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye fertility restorer alleles and/or corresponding Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

(140) Such methods of identifying, selecting or breeding wheat plants comprise obtaining one or more wheat plants and assessing their DNA to determine the presence or absence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye fertility restorer alleles and/or corresponding Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

(141) Such methods may be used, for example, to determine which progeny resulting from a cross have the required fertility restorer allele or Rf nucleic acids (or combination thereof) and accordingly to guide the preparation of plants having the required fertility restorer allele or Rf nucleic acids in combination with the presence or absence of other desirable traits.

(142) The method will consist of identifying the presence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye allele or nucleic acids in the fertile plant, said plant being either a fertile transgenic plant or a non-transgenic plant. Optionally, the method further consists of identifying the absence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye allele or nucleic acids in non-restorer plants and/or sterile plants.

(143) Accordingly, it is disclosed herein the means for specifically detecting the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids in a wheat plant.

(144) Such means include for example a pair of primers for the specific amplification of a fragment nucleotide sequence of Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids from plant wheat genomic DNA.

(145) As used herein, a primer encompasses any nucleic acid that is capable of priming the synthesis of a nascent nucleic acid in a template-dependent process, such as PCR. Typically, primers are oligonucleotides from 10 to 30 nucleotides, but longer sequences can be employed. Primers may be provided in double-stranded form though single-stranded form is preferred.

(146) Alternatively, nucleic acid probe can be used for the specific detection of any one of Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

(147) As used herein, a nucleic acid probe encompass any nucleic acid of at least 30 nucleotides and which can specifically hybridizes under standard stringent conditions with a defined nucleic acid. Standard stringent conditions as used herein refers to conditions for hybridization described for example in Sambrook et al 1989 which can comprise 1) immobilizing plant genomic DNA fragments or library DNA on a filter 2) prehybridizing the filter for 1 to 2 hours at 65° C. in 6×SSC 5×Denhardt's reagent, 0.5% SDS and 20 mg/ml denatured carrier DNA 3) adding the probe (labeled) 4) incubating for 16 to 24 hours 5) washing the filter once for 30 min at 68° C. in 6×SSC, 0.1% SDS 6) washing the filter three times (two times for 30 min in 30 ml and once for 10 min in 500 ml) at 68° C. in 2×SSC 0.1% SDS. The nucleic acid probe may further comprise labeling agent, such as fluorescent agents covalently attached to the nucleic acid part of the probe.

(148) Methods of Producing a Wheat Plant Carrying a Modified Rf3 Restorer of Fertility

(149) The inventors have also identified two types of Rf3 restorer of fertility, one with a strong fertility restoration, for example, capable of providing plants with a fertility score above 1.0, for example comprised between 1.0 and 2.0 and another with a weak fertility restoration, for example, capable of providing plants having a fertility score below 0.1, for example comprised between 0.5 and 1.0.

(150) Surprisingly, the strong fertility restoration correlates with the absence in the genome of the wheat plant carrying Rf3 of a 163 bp fragment of SEQ ID NO:3174, located in the 5'UTR of Rf3 coding sequence.

(151) Accordingly, the disclosure relates to a method for producing a wheat plant carrying a Rf3 restorer of fertility, said method comprising (i) providing a parent wheat plant comprising in its genome at least a 163 bp fragment of SEQ ID NO:3174, and (ii) deleting a fragment of at least 10 bp of said fragment of SEQ ID NO:3174, for example at least 20 bp, 30 bp, 40 bp, 50 bp, 60 bp, 70 bp, 80 bp, 90 bp, 100 bp, 110 bp, 120 bp, 130 bp, 140 bp, 150 bp, 160 bp or the whole fragment of SEQ ID NO:3174 in the genome of said wheat plant, thereby obtaining said wheat plant carrying a Rf3 restorer of fertility.

(152) Advantageously, the fertility score of the obtained wheat plant has a fertility score higher than the parent wheat plant, due to the deletion of said fragment depicted in SEQ ID NO:3174. The skilled person may select the deletion such as to obtain an increase in fertility restoration as compared to the parent wheat plant with the full fragment of SEQ ID NO:3174 in its genome.

(153) In specific embodiments, the parent wheat plant has a fertility score below 1, for example comprised between 0.5 and 1.0 and the obtained wheat plant has a fertility score above 1.0, for example comprised between 1.0 and 2.0.

(154) It is also possible to restore fertility from plants having rf3 non-restorer allele presenting a frameshift due to nucleotides deletion or insertion as compared to the Rf3 restorer allele RFL29a sequence of SEQ ID NO:3146 (see also Example 22).

(155) Such deletion of genomic fragment or correction of frameshift may be obtained by any suitable methods known by the skilled person in the art, including genome editing tools such as, but not limited to, meganucleases, ZFNs, TALENs (WO2011072246) or CRISPR CAS system (including CRISPR Cas9, WO2013181440) or their next generations based on double-strand break technologies using engineered nucleases. Examples of such methods are also described in Example 15 and Example 22.

(156) The wheat plants as obtained by the method described above are also part of the disclosure. Typically, such wheat plant as obtained by the above method, or obtainable by such method, carries a Rf3 restorer of fertility, wherein only a part but not the whole genomic fragment of SEQ ID NO:3174 is deleted in the genome of said wheat plant. Typically, a fragment between 10 bp and 162 bp of SEQ ID NO:3174 is deleted in the genome of said wheat plant. It is expected that the obtained wheat plant with the genome deletion has a fertility score higher than the fertility score as measured in a parent wheat plant with identical genome except for the full sequence of SEQ ID NO: 3174 in its genome. Alternatively, such wheat plant as obtained by the above method, or obtainable by such method, carries a Rf3 restorer of fertility, wherein nucleotides of RFL29c has been deleted or added to restore inframe translation.

(157) Methods for Assessing Fertility Restoration in a Wheat Plant

(158) The disclosure also includes a method for assessing fertility restoration in a wheat plant, said method comprising determining the presence or absence of a fragment of SEQ ID NO:3174 in the genome of said plant, wherein the presence of the whole fragment is indicative of a weak restoration of fertility and a deletion of at least a part of such fragment is indicative of a strong restoration of fertility. Typically said method is performed in a wheat plant which is susceptible to carry a Rf3 restorer of fertility.

(159) It is also disclosed herein nucleic acid probes for use in the above methods for assessing fertility restoration in wheat plant, wherein said nucleic acid probe consists of a nucleic acid of at least 10 nucleotides within SEQ ID NO:3174.

(160) Typically, said nucleic acid probe is a fragment of at least 20 bp, 30 bp, 40 bp, 50 bp, 60 bp, 70 bp, 80 bp, 90 bp, 100 bp, 110 bp, 120 bp, 130 bp, 140 bp, 150 bp, 160 bp or the whole fragment of SEQ ID NO: 3174.

(161) The Wheat Plant Restorer of Fertility of *T. timopheevii* CMS Cytoplasm with at Least Three Specific Fertility Restorer Alleles

(162) The inventors have shown that a combination of at least 3 specific fertility restorer alleles within the restorer loci Rf1, Rf3, Rf4 and Rf7 enable the obtention of plants with full restoration of

fertility of *T. timopheevii* CMS cytoplasm.

(163) Therefore, a first aspect of the present disclosure relates to a wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the plant comprises at least three fertility restorer alleles within the restorer loci chosen amongst Rf1, Rf3, Rf4 and Rf7.

(164) In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf1, Rf3, Rf4.

(165) In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf1, Rf4, Rf7.

(166) In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf1, Rf3, Rf7.

(167) In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf3, Rf4, Rf7.

(168) As used herein, the Rf1 locus refers to the locus of the Rf1 restorer allele, which locus is located at most 10 cM, preferably at most 7 cM, more preferably at most 2 cM, from marker cfn0522096 of SEQ ID NO:4 and/or from marker cfn05277067 of SEQ ID NO: 10. In a specific embodiment, the wheat plant restorer of fertility according to the present disclosure includes at least one Rf1 restorer allele, said Rf1 restorer allele being located within the chromosomal interval between SNP markers cfn0522096 of SEQ ID NO:3190 and cfn05277067 of SEQ ID NO:3196. In specific embodiments, the wheat plant restorer of fertility includes one Rf1 restorer allele at the Rf1 locus characterized by the presence of one or more of the SNP allele(s) as identified by Table 1.

(169) TABLE-US-00001 TABLE 1 SNP markers for mapping of Rf1 locus

Marker	SEQ ID	SNP#	Marker Name	NO: Restorer Allele
SNP1	cfn523072	3187	T	SNP2
SNP2	cfn0523109	3188	A	SNP3
SNP3	276I13_96B22_97797	3189	C	SNP4
SNP4	cfn0522096	3190	C	SNP5
SNP5	cfn0527763	3191	C	SNP6
SNP6	104A4_105172	3192	TG	SNP7
SNP7	104A4_105588	3193	A	SNP8
SNP8	cfn0373248	3194	T	SNP9
SNP9	cfn1097828	3195	C	SNP10
SNP10	cfn0527067	3196	A	SNP11
SNP11	cfn0528390	3197	G	SNP12
SNP12	BWS0267	3198	A	SNP13
SNP13	cfn0527718	3199	T	SNP14
SNP14	cfn0524469	3200	G	SNP15
SNP15	cfn0524921	3201	G	SNP16
SNP16	cfn1122326	3202	C	

(170) Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf1 restorer allele at the Rf1 locus characterized by the presence of the SNP3 and/or SNP7 restorer alleles(s) as described in Table 1. More preferably, the wheat plant restorer of fertility is characterized by the haplotypes of the SNP3 and SNP7 restorer alleles "C" and "A". In specific embodiments, the wheat plant restorer of fertility with Rf1 restorer allele comprises a Rf1 nucleic acid of the present disclosure as described above. Examples of Rf1 nucleic acids comprises the disclosed Rf1 nucleic acid sequences of SEQ ID NO: 1913, SEQ ID NO: 1914, SEQ ID NO: 1915, SEQ ID NO: 1916 or SEQ ID NO:3119, preferably a Rf1 nucleic acid comprises SEQ ID NO: 3119.

(171) As used herein, the Rf3 locus refers to the locus of the Rf3 restorer allele, which locus is at most 10 cM, preferably at most 7 cM, more preferably at most 2 cM, from marker cfn1249269 of SEQ ID NO:3205 and/or from marker BS00090770 of SEQ ID NO:3228. In a specific embodiment, the wheat plant restorer of fertility includes at least one Rf3 restorer allele within the Rf3 locus, said Rf3 restorer allele being located within the chromosomal fragment between SNP markers cfn1249269 and BS00090770. In specific embodiment, the wheat plant restorer of fertility includes one Rf3 restorer allele at the Rf3 locus characterized by the presence of one or more of the SNP alleles(s) as identified by Table 2.

(172) TABLE-US-00002 TABLE 2 SNP Markers for mapping of Rf3 locus

SNP#	Marker Name	Marker	SEQ ID	Restorer Allele
SNP17	cfn1252000	3203	A	SNP18
SNP18	IWB14060*	3204	G	SNP19
SNP19	cfn1249269	3205	G	SNP20
SNP20	219K1_166464	3206	T	SNP21
SNP21	219K1_158251	3207	G	SNP22
SNP22	219K1_111446	3208	A	SNP23
SNP23	219K1_110042	3209	T	SNP24
SNP24	219K1_110005	3210	C	SNP25
SNP25	219K1_107461	3211	A	SNP26
SNP26	219K1_99688	3212	T	SNP27
SNP27	219K1_37	3213	C	SNP28
SNP28	cfn1270524	3214	T	SNP29
SNP29	136H5_3M5_7601	3215	T	SNP30
SNP30	cfn1288811	3216	G	SNP31

136H5_3M5_89176 3217 A SNP32 136H5_3M5_89263 3218 T SNP33 136H5_3M5_138211 3219 T SNP34 cfn0556874 3220 C SNP35 136H5_3M5_64154 3221 C SNP36 136H5_3M5_68807 3222 G SNP37 136H5_3M5_77916 3223 A SNP38 cfn1246088 3224 A SNP39 cfn1287194 3225 G SNP40 cfn1258380 3226 A SNP41 IWB72107* 3227 A SNP42 BS00090770 3228 T SNP43 cfn1239345 3229 A

(173) Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf3 restorer allele at the Rf3 locus characterized by the presence of the SNP29 and/or SNP31 restorer alleles(s) as described in Table 2. More preferably, the wheat plant restorer of fertility is characterized by the haplotype of the SNP29 and SNP31 restorer alleles “T” and “A” respectively.

(174) In another particular embodiment, that may be combined with the previous embodiments, the wheat plant restorer of fertility according to the present disclosure includes one Rf3 restorer allele at the Rf3 locus characterized by the presence of the SNP38 and SNP41 restorer alleles “A” and “A” respectively.

(175) Preferably, the wheat plant restorer of fertility according to the present disclosure includes a Rf3 nucleic acid comprising SEQ ID NO:1712, SEQ ID NO:2230, SEQ ID NO: 2238, SEQ ID NO:3146, SEQ ID NO:3147 or SEQ ID NO:3148, preferably SEQ ID NO: 3146.

(176) As used herein, the Rf7 locus is located at most 10 cM from marker cfn0919993 of SEQ ID NO:3231. In specific embodiment, the wheat plant restorer of fertility includes one Rf7 restorer allele at the Rf7 locus characterized by the presence of one or more of the SNP alleles(s) as identified by Table 3:

(177) TABLE-US-00003 TABLE 3 SNP markers of Rf7 locus

SNP#	Marker Name	Marker	SEQ ID
SNP44	cfn0917304	3230 T	SNP45
cfn0919993	3231 G	SNP46	cfn0920459
3232 C	SNP49	cfn0915987	3445 G
SNP50	cfn0920253	3446 A	SNP51
cfn0448874	3447 T	SNP52	cfn0923814
3448 C	SNP53	cfn0924180	3449 G
SNP54	cfn0919484	3450 G	

(178) Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf7 restorer allele at the Rf7 locus characterized by the presence of the nine SNP restorer alleles SNP44-SNP46 and SNP49-54 of “restorer allele” haplotype, as described in Table 3.

(179) As used herein, the Rf4 locus is located at most 10 cM from marker cfn0393953 of SEQ ID NO:3233. In specific embodiment, the wheat plant restorer of fertility includes one Rf4 restorer allele at the Rf4 locus characterized by the presence of one or more of the SNP alleles(s) as identified by Table 4.

(180) TABLE-US-00004 TABLE 4 SNP markers of Rf4 locus

SNP#	Marker Name	Marker	SEQ ID
SNP47	cfn0393953	3233 C	SNP48
cfn0856945	3234 G		

(181) Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf4 restorer allele at the Rf4 locus characterized by the presence of the two SNP restorer alleles, SNP47, and SNP48, of the haplotype “C” and “G” respectively, as described in Table 4.

(182) In specific embodiments, the wheat plant restorer of fertility with Rf4 restorer allele comprises a Rf4 nucleic acid of the present disclosure as described above. Examples of Rf4 nucleic acids comprises the disclosed Rf4 nucleic acid sequences of SEQ ID NO: 2031, SEQ ID NO:3140 to 3142.

(183) In a particular embodiment, the wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm comprises one Rf3 restorer allele and two other fertility restorer alleles selected amongst Rf1, Rf4 and Rf7 restorer alleles. Preferably, the wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm according to the present disclosure comprises the Rf1, Rf3, and Rf7 restorer alleles.

(184) In particular, it is hereby included a wheat plant comprising Rf1, Rf3 and Rf7 restorer alleles as provided by the seed samples as deposited on 25 Sep. 2017 under deposit number NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817 at the NCIMB collection. The NCIMB is located at Ferguson Building, Craibstone Estate, Bucksburn, Aberdeen, Scotland, UK, AB21 9YA.

(185) The disclosure also relates to hybrid wheat plants which can be produced by crossing a wheat plant restorer of fertility according to the present disclosure as described above with a second plant.

(186) In certain embodiments, the wheat plant according to the disclosure is alloplasmic and comprises the *T. timopheevii* cytoplasm.

(187) For example, a hybrid wheat plant may be obtained by crossing a wheat plant restorer of fertility according to the present disclosure as described above, preferably comprising Rf1, Rf3 and Rf7 restorer alleles, and a wheat plant which does not have said fertility restorer alleles.

(188) It is also disclosed herein a method for producing a wheat hybrid plant comprising the steps of: a. crossing a sterile female wheat plant comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant of the present disclosure as described above; b. collecting the hybrid seed; c. optionally detecting the presence of *T. timopheevii* cytoplasm, and/or at least three of the Rf locus chosen amongst Rf1, Rf3, Rf4 and Rf7 in the hybrid seed; d. optionally detecting hybridity level of the hybrid seed.

(189) Therefore, it is also disclosed herein the wheat plants or lines according to the present disclosure developed to obtain such hybrid plants. Such plants or lines typically comprise the cytoplasmic elements necessary for the implementation of the corresponding hybrid system. Preferably, the plants or lines comprise the fertility restorer alleles Rf1, Rf3 and Rf7 and *T. timopheevii* cytoplasm. In specific embodiments, such plants or lines comprise a fertility restorer allele Rf1 comprising a Rf1 nucleic acid of the present disclosure as disclosed above.

(190) Alternatively, the detection of the presence of *T. timopheevii* cytoplasm and of at least three of the Rf locus chosen amongst Rf1, Rf3, Rf4 and Rf7 (step “c” of the method described above) can be performed on the parent lines in order to check their genotype before to start the cross (step “a”).

(191) The T-CMS cytoplasm can be detected either phenotypically wherein a plant bearing rf genes and a T-CMS cytoplasm will be sterile or by molecular means able to detect the orf256 gene as described in Rathburn and Hedgcoth, 1991 and Song and Hedgcoth, 1994.

(192) Method of Producing and Selecting a Wheat Plant of the Disclosure

(193) The present disclosure also relates to the methods to produce the wheat plant with the fertility restorer alleles as described in the previous section.

(194) In one embodiment, said method for producing the wheat plant includes the following step: a. providing a first wheat plant comprising one or two restorer allele selected among Rf1, Rf3 and Rf7 restorer alleles, b. crossing said first wheat plant with a second wheat plant comprising one or two restorer alleles selected among Rf1, Rf3 and Rf7 restorer alleles, wherein Rf1, Rf3 and Rf7 restorer alleles are represented at least once in the panel of restorer alleles provided by the first plant and the second plant, c. collecting the F1 hybrid seed, d. obtaining homozygous plants from the F1 plants, e. optionally detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles in the hybrid seed and/or at each generation.

(195) Preferentially, the female plant in step b) is bearing the T-CMS cytoplasm. In this case, the presence of the restorer alleles is assessed at every generation from step b) to step d) by using the markers and optionally by further by assessing the fertility level.

(196) Method to generate homozygous plants are generally well known from skilled person of the art. This could be either by repetitive backcross or by double haploid development or by Single Seeds Descent (SSD) methods.

(197) The applicant has deposited a sample of seeds of the disclosed wheat plant with said Rf1, Rf3 and Rf7 restorer alleles, on 25 Sep. 2017 under the Budapest treaty, at NCIMB collection under the number NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817. The NCIMB is located at Ferguson Building, Craibstone Estate, Bucksburn, Aberdeen, Scotland, UK, AB21 9YA.

(198) The present disclosure further includes and provides methods of identifying the respective Rf1, Rf3, Rf4 and/or Rf7 restorer alleles as disclosed in the previous sections, and more generally

methods of selecting or breeding wheat plants for the presence or absence of the Rf1, Rf3, Rf4 and/or Rf7 fertility restorer alleles. Such methods of identifying, selecting or breeding wheat plants comprise obtaining one or more wheat plants and assessing their DNA to determine the presence or absence of the Rf1, Rf3, Rf4 and/or Rf7 fertility restorer alleles contained in the respective locus. (199) Such methods may be used, for example, to determine which progeny resulting from a cross have the required combination of fertility restorer alleles and accordingly to guide the preparation of plants having the required combination in combination with the presence or absence of other desirable traits.

(200) Accordingly, plants can be identified or selected by assessing them for the presence of one or more individual SNPs appearing in the above Tables 1, 2, 3 and 4, as well as the SNPs in Table 19, for assessing the presence of restorer alleles Rf1, Rf3, Rf7 or Rf4 respectively.

(201) More generally, it is disclosed herein the specific means for detecting the restorer alleles in a wheat plant, more specifically Rf1, Rf3, Rf4 and Rf7 restorer alleles and their combinations.

(202) Said means thus include any means suitable for detecting the following SNP markers within one or more of the following markers: SEQ ID NOs 3187-3235.

(203) Any method known in the art may be used in the art to assess the presence or absence of a SNP. Some suitable methods include, but are not limited to, sequencing, hybridization assays, polymerase chain reaction (PCR), ligase chain reaction (LCR), and genotyping-by-sequence (GBS), or combinations thereof.

(204) Different PCR based methods are available to the person skilled of the art. One can use the RT-PCR method or the Kaspar method from KBioscience (LGC Group, Teddington, Middlesex, UK).

(205) The KASPTM genotyping system uses three target specific primers: two primers, each of them being specific of each allelic form of the SNP (Single Nucleotide Polymorphism) and one other primer to achieve reverse amplification, which is shared by both allelic form. Each target specific primer also presents a tail sequence that corresponds with one of two FRET probes: one label with FAM® dye and the other with HEX® dye.

(206) Successive PCR reactions are performed. The nature of the emitted fluorescence is used to identify the allelic form or forms present in the mix from the studied DNA.

(207) The primers identified in Table 5 are particularly suitable for use with the KASPTM genotyping system. Of course, the skilled person may use variant primers or nucleic acid probes of the primers as identified in Table 5, said variant primers or nucleic acid probes having at least 90%, and preferably 95% sequence identity with any one of the primers as identified in Table 5, or with the DNA genomic fragment amplified by the corresponding set of primers as identified in Table 5.

(208) Percentage of sequence identity as used herein is determined by calculating the number of matched positions in aligned nucleic acid sequences, dividing the number of matched positions by the total number of aligned nucleotides, and multiplying by 100. A matched position refers to a position in which identical nucleotides occur at the same position in aligned nucleic acid sequences. For example, nucleic acid sequences may be aligned using the BLAST 2 sequences (Bl2seq) using BLASTN algorithms (www.ncbi.nlm.nih.gov).

(209) As used herein, a primer encompasses any nucleic acid that is capable of priming the synthesis of a nascent nucleic acid in a template-dependent process, such as PCR. Typically, primers are oligonucleotides from 10 to 30 nucleotides, but longer sequences can be employed. Primers may be provided in double-stranded form though single-stranded form is preferred.

Alternatively, nucleic acid probe can be used. Nucleic acid probe encompass any nucleic acid of at least 30 nucleotides and which can specifically hybridizes under standard stringent conditions with a defined nucleic acid. Standard stringent conditions as used herein refers to conditions for hybridization described for example in Sambrook et al 1989 which can comprise 1) immobilizing plant genomic DNA fragments or library DNA on a filter 2) prehybridizing the filter for 1 to 2 hours at 65° C. in 6×SSC 5×Denhardt's reagent, 0.5% SDS and 20 mg/ml denatured carrier DNA 3)

adding the probe (labeled) 4) incubating for 16 to 24 hours 5) washing the filter once for 30 min at 68° C. in 6×SSC, 0.1% SDS 6) washing the filter three times (two times for 30 min in 30 ml and once for 10 min in 500 ml) at 68° C. in 2×SSC 0.1% SDS.

(210) In specific embodiments, said primers for detecting the SNP markers of the present disclosure (specific for each allele “X” or “Y” or common) are as listed in the following table 5:

(211) TABLE-US-00005 TABLE 5 Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name) SEQ ID NO ID

Sequence	3253	cfn0238384	AlleleX	GAAGGTGACCAAGTTCATGCTGTAAAAAGATGTCT
GTGTGTCTAGC	3254	cfn0523109	AlleleX	GAAGGTGACCAAGTTCATGCTGGTGAACAAAACA GGCCTACAATCA
	3255	cfn0560679	AlleleX	GAAGGTGACCAAGTTCATGCTAATGATGTTTAACA TTGGAACGGTCC
	3256	cfn0917304	AlleleX	GAAGGTGACCAAGTTCATGCTGTGGTGGCGCTCT ACCCG
	3257	cfn0919993	AlleleX	GAAGGTGACCAAGTTCATGCTAAGTCATCGACTTA
CATGCTTCTTTG	3258	cfn0920459	AlleleX	GAAGGTGACCAAGTTCATGCTAGCCAAGGAAGCC CAGATTTTC
	3259	cfn1087371	AlleleX	GAAGGTGACCAAGTTCATGCTAGGGGAACTTTGG GTATACACCA
	3260	cfn1252000	AlleleX	GAAGGTGACCAAGTTCATGCTGTTAATGCTGTAG
CCATTCTTGCAA	3261	BWS0267	AlleleX	GAAGGTGACCAAGTTCATGCTTCAGCTGCATAAA AAMCAGAATACCA
	3262	cfn0524469	AlleleX	GAAGGTGACCAAGTTCATGCTGCACGTAGTAAGT
ATTGATTTTTCTGTG	3263	cfn0527067	AlleleX	GAAGGTGACCAAGTTCATGCTCAAATTACTTTTGT TCTTTTATTTTTTTTCGAAT
	3264	cfn0527718	AlleleX	GAAGGTGACCAAGTTCATGCTAATTGTTTACAACA
TGGACATGAGAAC	3265	cfn1082074	AlleleX	GAAGGTGACCAAGTTCATGCTTACTGATAAAATCC GGTTCAAATATATAAC
	3266	cfn1239345	AlleleX	GAAGGTGACCAAGTTCATGCTGGCTTCTTTTTTCT
CCCTATAATATGGA	3267	cfn0554333	AlleleX	GAAGGTGACCAAGTTCATGCTGAGAGGCATCACA TAGGCATAG
	3268	cfn0436720	AlleleX	GAAGGTGACCAAGTTCATGCTATTCTTCATTCCTT ACAACAAATATACCAAATT
	3269	cfn0522096	AlleleX	GAAGGTGACCAAGTTCATGCTAGTAGAATACCAC
CCAATAAATCACTG	3270	cfn0523072	AlleleX	GAAGGTGACCAAGTTCATGCTCTAGCGCATGAGG TCTATCG
	3271	cfn0523990	AlleleX	GAAGGTGACCAAGTTCATGCTACATGAAGAGTGC AGGCACACG
	3272	cfn0524921	AlleleX	GAAGGTGACCAAGTTCATGCTATTGTTTCCATGTT AAGCTTATATTGTGCA
	3273	cfn0528390	AlleleX	GAAGGTGACCAAGTTCATGCTAAAAACATCTATTC
CAAGCAAGTATTAGTAAT	3274	cfn0530841	AlleleX	GAAGGTGACCAAGTTCATGCTTCTTGTTTATATAT TCTCTTATCAGAAGTC
	3275	cfn1122326	AlleleX	GAAGGTGACCAAGTTCATGCTGAATCTGATTAAGA
CGCTGGAGAAC	3276	cfn1249269	AlleleX	GAAGGTGACCAAGTTCATGCTGATTCAAAGAGGT GACAAATATGTGTACT
	3277	contig46312_253_ BS00090770	AlleleX	GAAGGTGACCAAGTTCATGCTGGTTCGTAGCACAT
AGCCGTTTAC	3278	219K1_110042	AlleleX	GAAGGTGACCAAGTTCATGCTACGGAATCGAGTC
	3279	cfn0373248	AlleleX	AACCAATTCCT
	3280	cfn0527763	AlleleX	GAAGGTGACCAAGTTCATGCTAACAACAATTAYGA GGATCAAATGGTCA
	3281	cfn0556874	AlleleX	GAAGGTGACCAAGTTCATGCTATCTAGCCACGCA AATGCCCGT
	3282	cfn1097828	AlleleX	GAAGGTGACCAAGTTCATGCTAAGAGCATGTCA
GACACAATGCAG	3283	cfn1246088	AlleleX	GAAGGTGACCAAGTTCATGCTGGTTCCTGAGAGA GCAACCA
	3284	cfn1258380	AlleleX	GAAGGTGACCAAGTTCATGCTGACATCTGATGAG CCAGCATACA
	3285		AlleleX	GAAGGTGACCAAGTTCATGCTATCTACTCATCTAT TGCAGATGCTCTT

cfn1270524 AlleleX GAAGGTGACCAAGTTCATGCTAAATGCCTAGTCTA
TACCTGATAAACTAAA 3286 cfn1287194 AlleleX
GAAGGTGACCAAGTTCATGCTACCTCCTCCGTAT CTGATGGC 3287 cfn1288811 AlleleX
GAAGGTGACCAAGTTCATGCTTAATTTGGTTAACC AAATCCTTTTTTGATTTTT 3288
cfn1291249 AlleleX GAAGGTGACCAAGTTCATGCTTCCCAGATTTAGC ATGTGCATT
3289 cfn0231871 AlleleX GAAGGTGACCAAGTTCATGCTACTGTATTAAATTA
GCTAGTGTGGCG 3290 cfn0393953 AlleleX
GAAGGTGACCAAGTTCATGCTAAAAACAAGTTGTC ACCCAGATGAATC 3291
cfn0867742 AlleleX GAAGGTGACCAAGTTCATGCTGCATCCTCGACAA TGATTTCATCG
3292 cfn3126082 AlleleX GAAGGTGACCAAGTTCATGCTAGATTTTAGCACCT
AACGCCGCAAA 3293 104A4_105172 GAAGGTGACCAAGTTCATGCTGTCGMACCCAATG
AlleleX AATAATGTTT 3294 104A4_105588
GAAGGTGACCAAGTTCATGCTGTTCTTGTGACAT AlleleX GTACTCATAA 3295
136H5_3M5_138211 GAAGGTGACCAAGTTCATGCTACTGGGTGCAAAG AlleleX
CCAAGATGATT 3296 136H5_3M5_64154
GAAGGTGACCAAGTTCATGCTGGCGAAACTTCGC AlleleX CGCGATAAAT 3297
136H5_3M5_68807 GAAGGTGACCAAGTTCATGCTCAAGTTGCTCTTAA AlleleX
TTATCTGTGCGTA 3298 136H5_3M5_7601
GAAGGTGACCAAGTTCATGCTCGTCCCCCATGGC AlleleX ACCTGT 3299
136H5_3M5_77916 GAAGGTGACCAAGTTCATGCTATAGCAAGTAGAG AlleleX
TTAACTTATCAAGTTATTA 3300 136H5_3M5_89176
GAAGGTGACCAAGTTCATGCTGGATTTTCTCACC AlleleX GGCATCTCCA 3301
136H5_3M5_89263 GAAGGTGACCAAGTTCATGCTTCCCATGTTCTTTT AlleleX
TTTGCTCAAAC 3302 219K1_107461 GAAGGTGACCAAGTTCATGCTATATTGTTTGTATT
AlleleX AAAAAGTTGTGTGTTTTGA 3303 219K1_110005
GAAGGTGACCAAGTTCATGCTGCCTTTTCTTCTTC AlleleX CAGCATCTAC 3304
219K1_111446 GAAGGTGACCAAGTTCATGCTAGAATCGTTCTTC AlleleX
GAGAAGCACTCA 3305 219K1_158251
GAAGGTGACCAAGTTCATGCTCCTGGAGATGGAT AlleleX CCGGTCAG 3306
219K1_166464 GAAGGTGACCAAGTTCATGCTCCTGAGCTGGGCT AlleleX GCACC 3307
219K1_37 GAAGGTGACCAAGTTCATGCTAAAGGGCTATCCT GGTGAACAAC 3308
219K1_99688 GAAGGTGACCAAGTTCATGCTGTTGCCCTGCGCA AlleleX AAATCAAACCTT
3309 276113_96B22_97797 GAAGGTGACCAAGTTCATGCTGTACTATGGCTAT AlleleX
GTCTCTGAATGC 3310 CAP7_c3847_204
GAAGGTGACCAAGTTCATGCTCATTGACGCGTC AlleleX TTCCGCAATA 3311
Tdurum_contig50667_ GAAGGTGACCAAGTTCATGCTGATGACATGGAGG 306 AlleleX
ATTATATCGACGA 3312 cfn0856945 AlleleX
GAAGGTGACCAAGTTCATGCTGCACATGCTTTATT ACTGATCTGATTTG 3313
S100067637 GAAGGTGACCAAGTTCATGCTCCAAATGTCCGAA AlleleX TTCAGAGCAG
3404 S100069923 GAAGGTGACCAAGTTCATGCTACATATACGCGAG AlleleX CGCTCCTG
3315 S3045171 AlleleX GAAGGTGACCAAGTTCATGCTGGTTCTTGGCACA CTCCCCAG
3316 S3045222 AlleleX GAAGGTGACCAAGTTCATGCTAACCTAAGTAGTAA
GCTTGCTGGGT 3317 cfn0238384 Allele
GAAGGTGCGGAGTCAACGGATTGTAAAAAGATGTC Y TGTGTGTCTAGG 3318
cfn0523109 Allele GAAGGTGCGGAGTCAACGGATTGTGAACAAAACAG Y
GCCTACAATCC 3319 cfn0560679 Allele
GAAGGTGCGGAGTCAACGGATTCAATGATGTTTAA Y CATTGGAACGGTCT 3320
cfn0917304 Allele GAAGGTGCGGAGTCAACGGATTGGTGGTGGCGCT Y CTACCCT 3321
cfn0919993 Allele GAAGGTGCGGAGTCAACGGATTCAAGTCATCGACT Y
TACATGCTTCTTTT 3322 cfn0920459 Allele

GAAGGTCGGAGTCAACGGAGGAAAGC Y CCAGATTTTG 3323 cfn1087371
Allele GAAGGTCGGAGTCAACGGATTGGGGAACTTTGG Y GTATACACCG 3324
cfn1252000 Allele GAAGGTCGGAGTCAACGGATTGTAAATGCTGTAG Y
CCATTCTTGCAG 3325 BWS0267 Allele Y
GAAGGTCGGAGTCAACGGATTCAGCTGCATAAAA AMCAGAATACCG 3326
cfn0524469 Allele GAAGGTCGGAGTCAACGGATTGCACGTAGTAAGT Y
ATTGATTTTTCTGTT 3327 cfn0527067 Allele
GAAGGTCGGAGTCAACGGATTCAAATTACTTTTGT Y TCTTTTATTTTTTTTCGAAC 3328
cfn0527718 Allele GAAGGTCGGAGTCAACGGATTATAAATTGTTTCA Y
ACATGGACATGAGAAT 3329 cfn1082074 Allele
GAAGGTCGGAGTCAACGGATTCTTACTGATAAAAT Y CCGGTTCAAATATATAAT 3330
cfn1239345 Allele GAAGGTCGGAGTCAACGGATTGCTTCTTTTTTCTC Y
CCTATAATATGGG 3331 cfn0554333 Allele
GAAGGTCGGAGTCAACGGATTGAGAGGCATCACA Y TAGGCATAC 3332 cfn0436720
Allele GAAGGTCGGAGTCAACGGATTCTTCATTCTTACA Y ACAAATATACCAAATC 3333
cfn0522096 Allele GAAGGTCGGAGTCAACGGATTAGTAGAATAACCAC Y
CCAATAAATCACTC 3334 cfn0523072 Allele
GAAGGTCGGAGTCAACGGATTAACCTCTAGCGCAT Y GAGGTCTATCA 3335 cfn0523990
Allele GAAGGTCGGAGTCAACGGATTATACATGAAGAGT Y GCAGGCACACT 3336
cfn0524921 Allele GAAGGTCGGAGTCAACGGATTGTTTCCATGTTAA Y
GCTTATATTGTGCG 3337 cfn0528390 Allele
GAAGGTCGGAGTCAACGGATTAAACATCTATTCC Y AAGCAAGTATTAGTAAC 3338
cfn0530841 Allele GAAGGTCGGAGTCAACGGATTCTTCTTGTTTATAT Y
ATTCTCTTATCAGAAGTT 3339 cfn1122326 Allele
GAAGGTCGGAGTCAACGGATTGGAATCTGATTAA Y GACGCTGGAGAAT 3340
cfn1249269 Allele GAAGGTCGGAGTCAACGGATTCAAAGAGGTGACA Y
AATATGTGTACC 3341 contig46312 Allele
GAAGGTCGGAGTCAACGGATTAGGTCGTAGCACA Y_253_BS00090770 TAGCCGTTTAT
3342 219K1_110042 GAAGGTCGGAGTCAACGGATTTCGGAATCGAGTCA Allele Y
ACCAATTCCC 3343 cfn0373248 Allele
GAAGGTCGGAGTCAACGGATTAAACAACAATTAYG Y AGGATCAAATGGTCT 3344
cfn0527763 Allele GAAGGTCGGAGTCAACGGATTCTAGCCACGCAA Y TGCCCGC 3345
cfn0556874 Allele GAAGGTCGGAGTCAACGGATTGAAAGAGCATGTC Y
AGACACAATGCAA 3346 cfn1097828 Allele
GAAGGTCGGAGTCAACGGATTGGTTCCTGAGAGA Y GCAACCG 3347 cfn1246088
Allele GAAGGTCGGAGTCAACGGATTGACATCTGATGAG Y CCAGCATACC 3348
cfn1258380 Allele GAAGGTCGGAGTCAACGGATTCTACTCATCTATT Y
GCAGATGCTCTG 3349 cfn1270524 Allele
GAAGGTCGGAGTCAACGGATTAAATGCCTAGTCT Y ATACCTGATAAACTAAT 3350
cfn1287194 Allele GAAGGTCGGAGTCAACGGATTACCTCCTCCGTA Y TCTGATGGT
3351 cfn1288811 Allele GAAGGTCGGAGTCAACGGATTAAATTTGGTTAACC Y
AAATCCTTTTTTGATTTTG 3352 cfn1291249 Allele
GAAGGTCGGAGTCAACGGATTCTTCCCAGATTTA Y GCATGTGCATG 3353 cfn0231871
Allele GAAGGTCGGAGTCAACGGATTCTACTGTATTAAAT Y TAGCTAGTGTGGCT 3354
cfn0393953 Allele GAAGGTCGGAGTCAACGGATTAAAAAAACAAGTT Y
GTCACCCAGATGAATT 3355 cfn0867742 Allele
GAAGGTCGGAGTCAACGGATTGGCATCCTCGACA Y ATGATTTTCATCT 3356
cfn3126082 Allele GAAGGTCGGAGTCAACGGATTTTAGCACCTAACG Y CCGCAAC 3357
104A4_105172 GAAGGTCGGAGTCAACGGATTCTGTTCGMACCCAA Allele Y
TGAATAATGTTC 3358 104A4_105588 GAAGGTCGGAGTCAACGGATTGTTTCCTTGTGACA

Allele Y TGTACTCATAC 3359 136H5_3M5_1382
GAAGGTCGGAGTCAACGGATTACTGGGTGCAAAG 11 Allele Y CCAAGATGATA 3360
136H5_3M5_64154 GAAGGTCGGAGTCAACGGATTGCGAACTTCGCC Allele Y
GCGATAAAC 3361 136H5_3M5_68807 GAAGGTCGGAGTCAACGGATTAAAGTTGCTCTTAA
Allele Y TTATCTGTGCGTG 3362 136H5_3M5_7601
GAAGGTCGGAGTCAACGGATTGTCCCCCATGGCA Allele Y CCTGC 3363
136H5_3M5_77916 GAAGGTCGGAGTCAACGGATTAGCAAGTAGAGTT Allele Y
AACTTATCAAGTTATTG 3364 136H5_3M5_89176
GAAGGTCGGAGTCAACGGATTTTCTCACCGGCAT Allele Y CTCCG 3365
136H5_3M5_89263 GAAGGTCGGAGTCAACGGATTCTTCCCATGTTCT Allele Y
TTTTTTGCTCAAAT 3366 219K1_107461
GAAGGTCGGAGTCAACGGATTATATTGTTTGTATT Allele Y
AAAAAGTTGTGTGTTTTGC 3367 219K1_110005
GAAGGTCGGAGTCAACGGATTCGCCTTTTCTTCTT Allele Y CCAGCATCTAT 3368
219K1_111446 GAAGGTCGGAGTCAACGGATTAAATCGTTCTTCGA Allele Y
GAAGCACTCC 3369 219K1_158251 GAAGGTCGGAGTCAACGGATTCCTGGAGATGGAT
Allele Y CCGGTCAA 3370 219K1_166464
GAAGGTCGGAGTCAACGGATTGCCTGAGCTGGG Allele Y CTGCACT 3371 219K1_37
Allele Y GAAGGTCGGAGTCAACGGATTACAAAGGGCTATC CTGGTGAACAAT 3372
219K1_99688 GAAGGTCGGAGTCAACGGATTGCCCTGCGCAAAA Allele Y TCAAACCTC
3373 276I13_96B22_97797 GAAGGTCGGAGTCAACGGATTAAAGTACTATGGCT Allele Y
ATGTCTCTGAATGT 3374 CAP7_c3847_204
GAAGGTCGGAGTCAACGGATTCGACGCGTCTTCC Allele Y GCAATG 3375
Tdurum_contig50667_ GAAGGTCGGAGTCAACGGATTATGACATGGAGGA 306 Allele Y
TTATATCGACGG 3376 cfn0856945 Allele
GAAGGTCGGAGTCAACGGATTGGCACATGCTTTA Y TTAAGTATCTGATTTT 3377
S100067637 Allele GAAGGTCGGAGTCAACGGATTCCAAATGTCCGAA Y
TTCAGAGCAC 3378 S100069923 Allele
GAAGGTCGGAGTCAACGGATTGTACATATACGCG Y AGCGCTCCTA 3379 S3045171
Allele Y GAAGGTCGGAGTCAACGGATTGGTTCTTGGCACA CTCCCCAA 3380
S3045222 Allele Y GAAGGTCGGAGTCAACGGATTCCTAAGTAGTAAG CTTGCTGGGC
3381 cfn0238384 AGGGGGGGCGTACGGGGTGA Common 3382 cfn0523109
GTGTGTGCTAATGTGGATATACGTAAGTT Common 3383 cfn0560679
GACGTTGAAGGGGGCATAGATCAAA Common 3384 cfn0917304
CAACTGCTTGGAGAAAGGCAACACAA Common 3385 cfn0919993
CCATTAACAAGTACTGCATAGGTGCATAT Common 3386 cfn0920459
CCTCCTCCTAATTAAGCTCCTATAGATA Common 3387 cfn1087371
CCCCCTTCTTCTTCACTAGGGTAA Common 3388 cfn1252000
GTGCCCATAGACGACTGGGACAA Common 3389 BWS0267
CTGCGTTAAGGTTTCAGGCAACTGAT Common 3390 cfn0524469
GCCAATTTTCAAATCTAAGTCCACAGAGA Common 3391 cfn0527067
ATATGATTCACCCTAGATCCTTCACCTTA Common 3392 cfn0527718
GTTTCCTCCAATGTTCTTCCC Common 3393 cfn1082074
TGTCTCGCTCGCTCTGGTTAATTT Common 3394 cfn1239345
ACCCTCGCTGCAGTTCCTTCTTAAA Common 3395 cfn0554333
AAATTCACACCATCATTGATCTGGGGTAT Common 3396 cfn0436720
GTCCACTGAGAATTAAGGATGCATTCTTT Common 3397 cfn0522096
AAGTAGTACTCGTAGAGAGTTAACACAGA Common 3398 cfn0523072
GCTTGACAATGATAATGCCCCCGAA Common 3399 cfn0523990
AATAACTCTTGTACTTCAGGATGAACGTTT Common 3400 cfn0524921

GCCCTTTGTTAATTCAATCTTTT Common 3401 cfn0528390
GATGAGGAAGGTCTTCATGTTGGGT Common 3402 cfn0530841
GAGCAGCACATCGTTAGCTGTTCTA Common 3403 cfn1122326
CAGATGGCCTAGTCGTGACATATCTT Common 3404 cfn1249269
TAAAAGAACACAAATGTGGCCCTAGTGAT Common 3405 contig46312_
GAAACATTCTTCGGACAACATATGCATTA 253_BS00090770 Common 3406 219K1_110042
GCATCTTCAAGGGAGCCACTCAAAA Common 3407 cfn0373248
ATCATTGCCACGRAAAAAATCTCACAAGAT Common 3408 cfn0527763
CCTTGTCACCGAGACATGTACAAA Common 3409 cfn0556874
CCTGCTGGAAATGGGATTTCTTGTTTATT Common 3410 cfn1097828
GCTTCCTCTCGGTAGCGATGGAT Common 3411 cfn1246088
GGGACGTGGAATTTGGAAAGACACAT Common 3412 cfn1258380
TATAGGAGTGATAGCACCACACAATTCAT Common 3413 cfn1270524
TGTAACCGAACTCAACCAAATGACCATTT Common 3414 cfn1287194
CAGAAGGCACTGGGAGGGGATT Common 3415 cfn1288811
GCACAATGTTTGACATTTCGGTTTTCTAGTT Common 3416 cfn1291249
CTGACTGTCGTATCTTCAACATACTGATT Common 3417 cfn0231871
CCAAGGTATATGTGCCATTATCCTCAAA Common 3418 cfn0393953
CACTCACCGTCGACATTGACATAGTT Common 3419 cfn0867742
AGCCTCCGCGTCGTGATGGAAT Common 3420 cfn3126082
AAAGGGACAGCGATTTGATCTGG Common 3421 104A4_105172
GCCATCCTCTCGGAGCCAGAA Common 3422 104A4_105588
CAAGGATGGGGAGTATATGGCTCTT Common 3423 136H5_3M5_138211
CCTCCCAACGGCCATCAATCAATTT Common 3424 136H5_3M5_64154
GATCATCGGGGAACCTGATGATAGTT Common Common 3425 136H5_3M5_68807
TTGGTTGGTTACGTCAGGTTAAGACTTA Common 3426 136H5_3M5_7601
CTTCTCTGTGGCCGAAAACCTCTT Common 3427 136H5_3M5_77916
GCTKTAGACTCTAAGTACCACAGAAGAA Common 3428 136H5_3M5_89176
CCTACCATCCTTAAATACTCTTGCTCAAA Common 3429 136H5_3M5_89263
AAGCAACTAGAAAAATATTTGGACTAGCAT Common 3430 219K1_107461
GTTGATGCGAATTTGAAAATGACATAATAA Common 3431 219K1_110005
TTGACTCGATTCCGTGTGAGGCTAA Common 3432 219K1_111446
AATATGATACAGACCCAAGACAAACCATTT Common 3433 219K1_158251
TCCTCACAAATCACGGGCCCCT Common 3434 219K1_166464
GACCGTGGTATATGCCACCACGTT Common 3435 219K1_37
GGCTTCATTATCAAATTCTGACCCATCTT Common 3436 219K1_99688
GGGCGGGACCTGACTTGATGAT Common 3437 276I13_96B22_97797
ACGACAATATAGACAAATAAAACCAAACAA Common 3438 CAP7_c3847_204
CCGCGGCCGAAGCAGGCAA Common 3439 Tdurum_contig50667_
ATACATGTCGGCGTCCCAGTCC 306 Common 3440 cfn0856945
GGTGTAGGCAAACCTAAAATAAACAGTCAA Common 3441 S100067637
CAACGCCAAACGCCAACGCCAT Common 3442 S100069923
GCCTTGTACTGCAGTGAAGTGTGAT Common 3443 S3045171
TGACGGCTGCGAGGACGAGAAT Common 3444 S3045222
AGTCCAGAGTTACAGGACATGGCTA Common

Use of the Wheat Plants of the Disclosure

(212) The plant according to the disclosure can be crossed, with any another inbred line, in order to produce a new line comprising either an increase or a decrease in the fertility level. Alternatively, a genetic trait which has been engineered into a particular line using the foregoing techniques could be moved into another line using traditional backcrossing techniques that are well known in the

plant breeding arts. For example, a backcrossing approach could be used to move an engineered trait from a public, non-elite inbred line into an elite inbred line, or from an inbred line containing a foreign gene in its genome into an inbred line or lines which do not contain that gene. As used herein, “crossing” can refer to a simple X by Y cross, or the process of backcrossing, depending on the context.

(213) The wheat plant of the disclosure is also a wheat plant wherein one or more desired traits have further been introduced through backcrossing methods, whether such trait is a naturally occurring one or not.

(214) The disclosure also relates to the use of the wheat plant as described above or its seeds, for food applications, preferably for flour production and for feed applications, or for breeding applications, for example for use as a parent plant in breeding for improving agronomical value of a wheat plant, line, hybrid or variety.

(215) As used herein, breeding applications encompass pedigree breeding to improve the agronomical value of a plant, line, hybrid, or variety.

(216) The wheat plants disclosed herein are further useful, for example, for producing flour or for feed applications.

(217) Seeds harvested from plants described herein can be used to make flour by any available techniques in the art. The wheat plants or their flour are also useful as food compositions, for human or animal.

(218) The Examples below are given for illustration purposes only.

Specific Embodiments

(219) 1. An isolated nucleic acid encoding a protein restorer of fertility of *T. timopheevii*, wherein the corresponding amino acid sequence has at least 95% identity to an amino acid sequence chosen amongst any one of SEQ ID NO:1 to SEQ ID NO: 1554. 2. The nucleic acid of Embodiment 1, encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs 1-2, SEQ ID NOs 288-290, SEQ ID NOs 293-296, SEQ ID NOs 343-346, SEQ ID NOs 349-354, SEQ ID NOs 359, 361 and 362, SEQ ID NOs 396 and 397, SEQ ID NOs 428-430, SEQ ID NO 517 and 519, SEQ ID NOs 752-754, SEQ ID NOs 1092, 1093 and 1095. 3. The nucleic acid of Embodiment 1, encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to SEQ ID NO:361. 4. The nucleic acid of claim 1, encoding a Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NOs: 124 and 125, SEQ ID NO:147, SEQ ID NO: 150, SEQ ID NO: 156, SEQ ID NO:158, SEQ ID NO:297, SEQ ID NO:299, SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 553 and 554, SEQ ID NOs: 557 and 558, SEQ ID NOs: 676 and 677, SEQ ID NOs: 684 and 685, SEQ ID NOs: 696 and 697, SEQ ID NOs: 938 and 939 and SEQ ID NOs: 1038 and 1039. 5. The nucleic acid of Embodiment 4, encoding Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO:684. 6. The nucleic acid of Embodiment 1, encoding a Rf4 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NO: 477 and SEQ ID NOs 3135-3138. 7. The nucleic acid of Embodiment 1, encoding a Rf7 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence

selected from the group consisting of SEQ ID NOs: 240-243, SEQ ID NOs 303-305, SEQ ID NO:363, SEQ ID NOs 375-377, SEQ ID NOs 497-499, SEQ ID NO:516, SEQ ID NOs 709-711, SEQ ID NO:768. 8. The nucleic acid of Embodiment 1 encoding for a Rf-rye protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859. 9. A recombinant nucleic acid comprising a nucleic acid encoding a protein restorer of *T. timopheevii* CMS cytoplasm of any one of Embodiments 1 to 8, operably linked to regulatory elements. 10. A vector for use in transformation of a wheat plant, comprising the recombinant nucleic acid of Embodiment 9. 11. A wheat transgenic plant comprising one or more nucleic acid(s) of any one of Embodiments 1-9, as transgenic element(s). 12. The wheat transgenic plant of Embodiment 11, which is a fertile wheat plant restorer of *T. timopheevii* CMS cytoplasm and comprising a combination of at least two different transgenic elements, selected from the group consisting of Rf1, Rf3, Rf7 and Rf-rye nucleic acids of any one of Embodiments 1-9. 13. The transgenic wheat plant of Embodiment 11 or 12, wherein said transgenic plant includes the following combination of nucleic acids as transgenic elements: a. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, b. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:363, SEQ ID NO: 516 and SEQ ID NO:768, c. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO: 378 and SEQ ID NO: 859, d. a Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to SEQ ID NO: 363, SEQ ID NO:516 and SEQ ID NO:768, e. a Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859, f. a Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859, g. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO: 768; h. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO: 859; i. a Rf1

nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf7 nucleic acid of Embodiment 4 encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO: 363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859; or, j. a Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO: 859. 14. The transgenic wheat plant of Embodiment 13, which further contains a Rf4 nucleic acid encoding a Rf4 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, in combination with one, two, three or four of any of the restorer nucleic acid encoding Rf1, Rf3, Rf7 or Rf-rye protein, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NO:477 and SEQ ID NOs 3135-3138. 15. The transgenic wheat plant of any one of Embodiments 11-14, wherein said one or more transgenic element(s) express polypeptides which restore or improve male fertility to the plant as compared to the parent plant without such transgenic element(s). 16. Method for producing a wheat transgenic plant of any one of Embodiments 11-15, wherein the method comprises the steps of transforming a parent wheat plant with one or more nucleic acids encoding protein restorer of *T. timopheevii* CMS cytoplasm according to any one of Embodiments 1-9, selecting a plant comprising said one or more nucleic acid(s) as transgene(s), regenerating and growing said wheat transgenic plant. 17. A method for producing a wheat plant carrying a Rf3 restorer of fertility, said method comprising (i) providing a parent wheat plant comprising in its genome at least a 163 bp fragment of SEQ ID NO:3174, and (ii) deleting a region of at least 10 bp in said fragment of SEQ ID NO:3174, for example at least 20 bp, 30 bp, 40 bp, 50 bp, 60 bp, 70 bp, 80 bp, 90 bp, 100 bp, 110 bp, 120 bp, 130 bp, 140 bp, 150 bp, 160 bp or the whole fragment of SEQ ID NO:3174 in the genome of said wheat plant, thereby obtaining said wheat plant carrying a Rf3 restorer of fertility. 18. The method of Embodiment 17, wherein the fertility score of the obtained wheat plant has a fertility score higher than the parent wheat plant. 19. The method of Embodiment 17 or 18, wherein the parent wheat plant has a fertility score below 1, for example comprised between 0.5 and 1.0 and the obtained wheat plant has a fertility score above 1.0, for example comprised between 1.0 and 2.0. 20. A wheat plant carrying Rf3 restorer of fertility, as obtained by the method of any one of Embodiments 17-19, wherein only a part but not the whole genomic fragment of SEQ ID NO:3174 is deleted in the genome of said wheat plant. 21. A method for assessing fertility restoration in a wheat plant, said method comprising determining the presence or absence of a fragment of SEQ ID NO: 3174 in the genome of said plant, wherein the presence of the whole fragment is indicative of a weak restoration of fertility and a deletion of at least a part of such fragment or the whole fragment of SEQ ID NO:3174 is indicative of a strong restoration of fertility. 22. A nucleic acid probe for use in a method of any one of Embodiments 17-21, characterized in that it consists of a nucleic acid of at least 10 nucleotides within SEQ ID NO: 3174. 23. A wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the plant comprises at least three fertility restorer alleles within the restorer loci chosen amongst Rf1, Rf3, Rf4 and Rf7 wherein, a. the Rf1 locus is located at most 10 cM from marker cfn0522096 of SEQ ID NO: 3190 or marker cfn05277067 of SEQ ID NO: 3196, b. the Rf3 locus is located at most 10 cM from marker cfn1249269 of SEQ ID NO: 3205 or marker BS00090770 of SEQ ID NO:3228, c. the Rf7 locus is located at most 10 cM from marker cfn0919993 of SEQ ID NO: 3231, and, d. the Rf4 locus is located at most 10 cM from marker cfn0393953 of SEQ ID NO: 3233. 24. The wheat plant of Embodiment 23, wherein the plant comprises the Rf1, Rf3 and Rf7 restorer alleles. 25. The

wheat plant of any one of Embodiments 23 to 24, characterized in that it includes at least one Rf1 restorer allele within the Rf1 locus, said Rf1 restorer allele being located within the chromosomal interval between SNP markers cfn0522096 of SEQ ID NO:3190 and cfn05277067 of SEQ ID NO:3196. 26. The wheat plant of Embodiment 25, wherein said Rf1 locus is characterized by the presence of one or more of the following SNP allele(s):

(220) TABLE-US-00006 Marker SEQ ID SNP# Marker Name NO: Restorer Allele SNP1 cfn523072 3187 T SNP2 cfn0523109 3188 A SNP3 276113_96B22_97797 3189 C SNP4 cfn0522096 3190 C SNP5 cfn0527763 3191 C SNP6 104A4_105172 3192 TG SNP7 104A4_105588 3193 A SNP8 cfn0373248 3194 T SNP9 cfn1097828 3195 C SNP10 cfn0527067 3196 A SNP11 cfn0528390 3197 G SNP12 BWS0267 3198 A SNP13 cfn0527718 3199 T SNP14 cfn0524469 3200 G SNP15 cfn0524921 3201 G SNP16 cfn1122326 3202 C 27. The wheat plant of Embodiment 26, wherein the Rf1 locus is characterized by the haplotype “C” and “A” of the SNP3 and SNP7 restorer alleles as described in the table of Embodiment 5. 28. The wheat plant of any one of Embodiments 23 to 27, characterized in that it includes at least one Rf3 restorer allele within the Rf3 locus, said Rf3 restorer allele being located within the chromosomal fragment between SNP markers cfn1249269 and BS00090770. 29. The wheat plant of Embodiment 28, wherein said Rf3 locus is characterized by the presence of one or more of the following SNP allele(s):

(221) TABLE-US-00007 SNP# Marker Name Marker SEQ ID Restorer Allele SNP17 cfn1252000 3203 A SNP18 IWB14060* 3204 G SNP19 cfn1249269 3205 G SNP20 219K1_166464 3206 T SNP21 219K1_158251 3207 G SNP22 219K1_111446 3208 A SNP23 219K1_110042 3209 T SNP24 219K1_110005 3210 C SNP25 219K1_107461 3211 A SNP26 219K1_99688 3212 T SNP27 219K1_37 3213 C SNP28 cfn1270524 3214 T SNP29 136H5_3M5_7601 3215 T SNP30 cfn1288811 3216 G SNP31 136H5_3M5_89176 3217 A SNP32 136H5_3M5_89263 3218 T SNP33 136H5_3M5_138211 3219 T SNP34 cfn0556874 3220 C SNP35 136H5_3M5_64154 3221 C SNP36 136H5_3M5_68807 3222 G SNP37 136H5_3M5_77916 3223 A SNP38 cfn1246088 3224 A SNP39 cfn1287194 3225 G SNP40 cfn1258380 3226 A SNP41 IWB72107* 3227 A SNP42 BS00090770 3228 T SNP43 cfn1239345 3229 A 30. The wheat plant of Embodiment 29, wherein the Rf3 locus is characterized by the haplotype “T” and “A” of the SNP29 and SNP31 restorer alleles as described in the table of Embodiment 7.

(222) 31. The wheat plant of any one of Embodiments 23 to 30, wherein the Rf7 locus is characterized by the presence of one or more of the following restorer SNP allele(s):

(223) TABLE-US-00008 SNP# Marker Name Marker SEQ ID Restorer Allele SNP44 cfn0917304 3230 T SNP45 cfn0919993 3231 G SNP46 cfn0920459 3232 C SNP49 cfn0915987 3445 G SNP50 cfn0920253 3446 A SNP51 cfn0448874 3447 T SNP52 cfn0923814 3448 C SNP53 cfn0924180 3449 G SNP54 cfn0919484 3450 G 32. The wheat plant of any one of Embodiments 23 to 31, wherein the Rf4 locus is characterized by the presence of one or more of the following SNP allele(s), preferably by the haplotype “C” and “G” of the SNP47 and SNP48 restorer alleles:

(224) TABLE-US-00009 SNP# Marker Name Marker SEQ ID Restorer Allele SNP47 cfn0393953 3233 C SNP48 cfn0856945 3234 G 33. The wheat plant of any one of Embodiments 23-32, wherein representative alleles of Rf1, Rf3, Rf4 and Rf7 restorer alleles are provided by the seed sample chosen amongst: NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817. 34. The wheat plant according to any one of the Embodiments 23 to 33, wherein said wheat plant is alloplasmic and comprises the *T. timopheevii* cytoplasm. 35. A method of identifying a wheat plant according to any one of Embodiments 23 to 34, wherein said wheat plant is identified by detecting the presence of at least one restorer allele genetically associated with the restorer loci chosen amongst Rf1, Rf3, Rf4 and Rf7 loci. 36. Means for detecting one or more of SNPs of SEQ ID NOs 3187-3235. 37. The means according to Embodiment 36 consisting of one or more primers including any one of the following: SEQ ID NOs 3253-3444. 38. Method of production of a wheat hybrid plant comprising the steps of: a. crossing a sterile female wheat plant comprising the *T. timopheevii* cytoplasm with a fertile male

wheat plant according to any one of Embodiments 23 to 34; b. collecting the hybrid seed; c. optionally detecting the presence of *T. timopheevii* cytoplasm, and/or at least three of the Rf locus chosen amongst Rf1, Rf3, Rf4 and Rf7 in the hybrid seed; and, d. optionally detecting hybridity level of the hybrid seed.

Description

LEGENDS OF THE FIGURES

(1) FIGS. 1A and 1B is a table showing a summary of plant genomes used in the study and number of identified RFLs. In total, the analyses encompassed 16 genome data sets from Triticeae and 13 from Oryzae, respectively, as well as single data sets from Brachypodium distachyon, tef (*Eragrostis tef*), rye (*Secale cereale*), foxtail millet (*Setaria italica*), sorghum (*Sorghum bicolor*) and maize (*Zea mays*). *Lolium perenne* and *Triticum turgidum* transcriptome data sets were used as well.

(2) FIG. 2: Processing of orf256 in T-CMS wheat mitochondria (A) Structure of orf256 identified in the *T. timopheevii* mitochondrial genome. The binding site of the WORF256 probe (Song and Hedgoth 1994) used in the Northern blot analysis is indicated. (B) Differential processing of the orf256 in wheat lines with different restoring capabilities. No orf256 transcript was detected in *T. aestivum*, Primepii, Anapurna and Wheat-Rye-6R (WR_6R) lines. An additional, third band detected in R197 and R0934F accessions is indicated by asterisks. As a control for gel loading the picture of ethidium bromide (EtBr) stained agarose gel is shown.

(3) FIG. 3: FIG. 3 shows the list of RFL groups potentially corresponding to the Rf4 gene.

(4) FIGS. 4a and 4b: FIGS. 4A and 4B show respectively the alignment between nucleotide sequences of RFL120-spelt (Subject) (SEQ ID NO: 3136) with RFL120-timo (Query) (SEQ ID NO: 477) and amino acid sequences RFL120-17F3R-0377_1 (SEQ ID NO: 3135), RFL120-L13_1 (SEQ ID NO: 3137), RFL120-R113_1 (SEQ ID NO: 3138), RFL120-timo_1 (SEQ ID NO: 477), and RFL120-GSTR435_1 (SEQ ID NO: 3136).

(5) FIG. 5A: FIG. 5A shows the protein sequence alignment of RFL29a (SEQ ID NO: 3146), RFL29b (SEQ ID NO: 3149), RFL29c_1 (SEQ ID NO: 3458) and RFL29c_2 (SEQ ID NO: 3459).

(6) FIGS. 5B and 5C: FIGS. 5B and 5C respectively, show the protein sequence alignments of RFL164a (SEQ ID NO: 3147) and RFL164b (SEQ ID NO: 3144), and RFL166a (SEQ ID NO: 3148) and RFL166b (SEQ ID NO: 3145).

(7) FIG. 6: FIG. 6 shows an alignment of the 5'UTR regions as identified in the RFL29a (SEQ ID NO: 3460) and RFL29b genes (SEQ ID NO: 3461).

(8) FIG. 7: FIG. 7 shows the position of the different target sequences around and within the 163 bp region (SEQ ID NO: 3155) identified for different endonucleases.

(9) FIG. 8: The FIG. 8 shows the relative position of the Rf1 mapping intervals identified on our internal consensus genetic map.

(10) FIG. 9: The FIG. 9 shows the relative position of the Rf3 mapping intervals identified on our internal consensus genetic map.

(11) FIG. 10: The FIG. 10A shows position of the markers within the chromosomal interval of Rf1 locus. Left and right refer to the marker positions relative to the interval defined by cfn0522096 and cfn0527067 SNP markers. Interval refers to markers located within the mapping interval. The physical positions correspond to LG internal ordering of the scaffolds of the IWGSC Whole genome assembly, 'IWGSC WGA'.

(12) The FIGS. 10B, 10C, 10D show a subset of the diversity panel showing the haplotypes at the Rf1 locus for the restorer lines used in the genetic mapping (R197, R204, R0932E), derived lines LGWR16-0016 and LGWR16-0026 and a collection of maintainer lines. "-": correspond to dominant markers with no amplification in several maintainer lines. "H": means heterozygote

status wherein two alleles are detected.

(13) FIG. 11: The FIG. 11A shows the position of the markers within the chromosomal interval of Rf3 locus. Left and right refer to the marker positions relative to the interval defined by cfn1249269 and BS00090770 markers. Interval refers to markers located within the mapping interval. The physical positions correspond to LG internal ordering of the scaffolds of the IWGSC Whole genome assembly, 'IWGSC WGA'. *IWB14060 and IWB72107 are described in Geyer and al, 2016.

(14) The FIGS. 11B, 11C, 11D show a subset of the diversity panel showing the haplotypes at the Rf3 locus for the restorer lines LGWR16-0016 and LGWR16-0026, the TJB155 line used as restorer parental line in Rf3 QTL mapping and a series of maintainer lines.

(15) “-” corresponds to dominant markers with no amplification in several maintainer lines. “H” means heterozygote status wherein two alleles are detected.

(16) FIG. 12: The FIG. 12 shows nucleotide sequence alignment between RFL29a (SEQ ID NO: 3146) and RFL29c (SEQ ID NO: 3157) fragment sequences. The PAM motif and target sequence for CRISPR edition are respectively is in bold and underlined.

EXAMPLES

Example 1: Identification of 1188 RFL-PPR Sequences in Cereals

(17) 32 genomic and two transcriptome data sets from 27 cereal plant species and their wild relatives were downloaded from the public sequence depositories and analysed. A complete list of files and databases from which they were downloaded is presented in FIG. 1.

(18) The DNA sequences were screened for open reading frames (ORFs) in six-frame translations with the getorf program of the EMBOSS 6.6.0 package (Rice et al., 2000). Predicted ORFs longer than 92 codons were screened for the presence of P-and PLS-class pentatricopeptide repeat (PPR) motifs using hmmsearch from the HMMER 3.1b package (hmmer.org) and hidden Markov models defined by hmmbuild (Cheng et al., 2016). The post-processing of hmmsearch results was carried out according to rules described previously (Cheng et al., 2016). Sequences containing 10 or more P-class PPR motifs were retained for further analysis, as a previous study has shown that Restorer-to-Fertility-Like (RFL) genes are primarily comprised of tandem arrays of 15 to 20 PPR motifs (Fujii et al., 2011).

(19) For identification of RFL sequences among the P-class PPRs, the OrthoMCL algorithm (Li et al., 2003) was used via the OrthoMCL-DB website to cluster P-class PPR proteins from each data set (<http://www.orthomcl.org/orthomcl/>). The resulting output files were screened for groups containing reference RFLs (Fujii et al., 2011).

(20) In total, 633 RFLs were identified in the 34 cereal data sets by OrthoMCI analysis (see Table of FIGS. 1A and 1B). In addition, WGS data sets of 44 sorghum accessions including landraces and wild relatives (Mace et al., 2013) were analyzed and 517 additional RFL sequences were identified and included in the study (see Table of FIGS. 1A and B).

Example 2: Identification of Full Length RFL PPR Genes Potentially Involved in Fertility Restoration of *T. timopheevii* CMS in Wheat by Targeted Capture of RFL Genes

(21) A. Selection of Germplasm Accessions:

(22) Six wheat accessions were identified as potential restorer lines of *Timopheevii*-type CMS (T-CMS) derived from the interspecific cross between *Triticum timopheevii* and *Triticum aestivum*.

(23) The first accession is a wheat-rye addition line, “Wheat-Rye-6R”, wherein the Rf gene was mapped on the additional long arm of 6R chromosome of rye *Secale cereale* (Curtis and Lukaszewski, 1993). Four other wheat accessions are characterized by the presence of at least one of the mapped restorer genes in wheat: Rf1, Rf3 and Rf7. The commercial variety, Primepii, carries the Rf3 gene, and three Limagrain lines R197, R0934F and R0932E respectively carry both Rf1 and Rf7 genes, Rf3 or Rf1.

(24) The sixth accession, named Anapurna, is a maintainer line not able to restore T-CMS and carries no known Rf genes. Anapurna is considered as the negative control in this experiment. In

addition, a *T. timopheevii* line was included in the study as it is a fertile line expected to harbor more than one Rf gene able to restore T-type CMS (Wilson and Ross, 1962). To some extent, this line is considered as a positive control in this experiment.

(25) All six accessions were verified in regard to their restorer status by genetic analysis.

(26) Northern blot analysis was performed with restorer and sterile accessions using an orf256-specific probe (FIG. 2A). Orf256 was previously identified as a gene specific to the *T. timopheevii* mitochondrial genome (Rathburn and Hedgcoth, 1991; Song and Hedgcoth, 1994). The sequence from -228 to +33 of orf256 (numbering relative to the start codon) is identical to the homologous region of the *coxI* gene (encoding subunit 1 of mitochondrial complex IV) in *T. aestivum*, whereas the rest of orf256, including the 3' flanking region, is unrelated to *coxI* (FIG. 2A). Different patterns of orf256 transcript processing in the fertile *T. timopheevii* line and fertile restorer lines carrying the *T. timopheevii* cytoplasm were observed when compared to the sterile CMS line pattern which is coherent with the genetic analysis (FIG. 2B). The different processing patterns are consistent with (but not conclusive proof that) orf256 is involved in causing CMS.

(27) B. Bait Design and RFL-Capture from Different Wheat Genotypes:

(28) The 1188 RFL PPR sequences identified by our bioinformatics analysis underwent a pre-treatment process that included masking of the target sequences against wheat mitochondrial and chloroplastic genome sequences (accessions NC_007579.1 and AB042240) as well as repeated elements of wheat genome. The masked target sequences were used for capture probe design. Briefly, probes were designed to cover the target sequences with a frequency masking algorithm intended to rule out probes that match with high copy number sequences in the targeted genome(s). The final probes were synthesized as a probe pool.

(29) Seeds of each accession were sown and plantlets were grown in etiolated conditions. After DNA extraction, Illumina libraries (referred to as NGS libraries) were prepared from DNA fragments around 600 bp with the KAPA Biosystems chemistry according to the manufacturer's recommendations.

(30) The NGS libraries were then specifically enriched in RFL sequences using the probe pool and a capture protocol. The efficiency of the capture was confirmed by a specific qPCR assay and ultimately libraries were pooled and sequenced in paired-end mode with 300 nt read length on a MiSeq platform.

(31) C. Assembly of Full-Length Gene Sequences Encoding RFL Proteins and Identification of Putative Orthologous Groups:

(32) Sequence reads from the RFL capture experiment were assembled into full-length contigs spanning one or more sequences encoding RFL proteins as described below. Overlapping paired reads were merged into a single sequence using bbmerge from the bbmap package

(<https://sourceforge.net/projects/bbmap/>) with the parameters qtrim2=t trimq=10,15,20 minq=12 mininsert=150. Read pairs that could not be merged were discarded. The merged reads were downsampled to 300,000 reads using reformat.sh in the bbmap package

(samplereadtarget=300000). The merged and downsampled reads were assembled with Geneious 8 (set to Medium Sensitivity/Fast) (<http://www.geneious.com/>). Finally, contigs composed of more than 100 merged reads were retained for further analysis, with most of these composed of over 1000 reads. In this way, a total of 1457 contigs were generated (Table 6).

(33) Approximately 220 contigs were obtained from each accession, except for *Triticum timopheevii* for which only 138 contigs were assembled. This is consistent with the tetraploid nature of the *Triticum timopheevii* genome. The consistency of the results indicates that the RFL-capture experiment was, a priori, comprehensive.

(34) TABLE-US-00010 TABLE 6 Number of RFL contigs and ORFs identified per accession and number of orthologous groups to which the ORFs were assigned with CD-hit. Number of assembled Number of Number of orthologous Number of RFL ORFs >350 Accession contigs composed identified RFL groups with at least one aa assigned to name of >100 reads ORFs >210

aa* RFL from the accession orthologous groups Anapurna 211 221 202 156 Primepii 226 234 215 162 R197 219 241 219 174 R0932E 221 245 221 183 R0934F 223 237 215 174 *Triticum* 138 143 129 114 *timopheevi* Wheat-Rye-6R 219 233 212 163 TOTAL 1457 1554 397 (non-redundant) 1254

(35) Sequences encoding RFL proteins were identified within these contigs as follows.

(36) Open reading frames (ORFs) within the contigs were identified with getorf from the EMBOSS package (Rice et al. 2000) using the parameters-minsize 630-find 0-reverse true. Thus only ORFs longer than 210 amino acids were used for further analysis. In total, 1554 ORFs were identified across the seven accessions (Table 6). The number of ORFs per accession ranged from 143 ORFs in *T. timopheevi* to 245 in R0934F (Table 6). The ORFs were further analyzed using hmmsearch (with parameters -E 0.1 --domE 100) from the HMMER package (v3.1b1) to detect PPR motifs using the hidden Markov models developed by Cheng et al. (2016) and the post-processing steps described in the same paper. Finally, to identify putatively orthologous RFL sequences across all seven accessions, the 1554 RFL ORFs were clustered using CD-hit (settings -c 0.96 -n 5 -G 0 -d 0 -AS 60 -A 105 -g 1). Across all accessions, 397 non-redundant RFL clusters representing putatively orthologous groups were obtained (Table 6). We define an orthologous RFL group as a set of at least one RFL ORF from at least one accession and wherein, if at least two sequences are present, these sequences share at least 96% sequence identity over the alignment length. Some highly conserved RFL genes are present in all seven accessions, others are found in only a subset of the accessions, or in a single accession. In most cases, each orthologous RFL group contains only one RFL protein from each accession.

(37) Table 7 below is showing the different RFL groups, the corresponding ORF names and the corresponding protein sequence number and DNA encoding protein sequence number.

(38) However, we found that genes encoding RFL-PPR proteins are often inactivated by indels creating frameshifts that break the contiguity of the ORFs, resulting in two shorter ORFs corresponding to a single longer ORF in another accession. In these cases, both shorter ORFs could be within the same orthologous group. In our analysis, only ORF encoding more than 350 amino acids were considered as possibly functional. This threshold was used as it corresponds to 10 PPR motifs (each of 35 amino acids), and all known active Rf proteins contain at least this number of motifs (usually 15-20).

(39) Finally, a set of 397 orthologous RFL groups were identified and numbered from 1 to 397 (see Table 7 below).

(40) TABLE-US-00011 TABLE 7 RFL-Name Name SEQID-PRT1 SEQID-DNA RFL 1
R0932E.300k_Assembly_Contig_60_2 1 1555 RFL 1 R197.300k_Assembly_Contig_73_2 2 1556
RFL 2 R0934F.300k_Assembly_Contig_11_2 3 1557 RFL 2
Anapurna.300k_Assembly_Contig_47_1 4 1558 RFL 2 Primepii.300k_Assembly_Contig_16_2 5
1559 RFL 2 R0932E.300k_Assembly_Contig_39_1 6 1560 RFL 2
R197.300k_Assembly_Contig_31_1 7 1561 RFL 2 Wheat-Rye-6R.300k_Assembly_Contig_13_1 8
1562 RFL 3 R0934F.300k_Assembly_Contig_34_2 9 1563 RFL 3
R0932E.300k_Assembly_Contig_41_1 10 1564 RFL 3 Wheat-Rye-
6R.300k_Assembly_Contig_44_1 11 1565 RFL 3 Primepii.300k_Assembly_Contig_27_2 12 1566
RFL 3 R197.300k_Assembly_Contig_34_1 13 1567 RFL 3
Anapurna.300k_Assembly_Contig_21_1 14 1568 RFL 4 R197.300k_Assembly_Contig_38_2 15
1569 RFL 4 Anapurna.300k_Assembly_Contig_48_2 16 1570 RFL 4
Primepii.300k_Assembly_Contig_39_2 17 1571 RFL 4 R0932E.300k_Assembly_Contig_47_1 18
1572 RFL 4 R0934F.300k_Assembly_Contig_44_1 19 1573 RFL 4 Wheat-Rye-
6R.300k_Assembly_Contig_32_2 20 1574 RFL 4 *Triticum*- 21 1575
timopheevi.300k_Assembly_Contig_19_2 RFL 5 Primepii.300k_Assembly_Contig_9_3 22 1576
RFL 5 R0934F.300k_Assembly_Contig_7_3 23 1577 RFL 5
Anapurna.300k_Assembly_Contig_15_3 24 1578 RFL 5 R197.300k_Assembly_Contig_12_3 25
1579 RFL 5 R0932E.300k_Assembly_Contig_11_2 26 1580 RFL 5 Wheat-Rye-

6R.300k_Assembly_Contig_9_2 27 1581 RFL 5 Triticum- 28 1582
timopheevii.300k_Assembly_Contig_9_2 RFL 6 Wheat-Rye-6R.300k_Assembly_Contig_99_1 29
1583 RFL 6 Primepii.300k_Assembly_Contig_109_1 30 1584 RFL 6
Anapurna.300k_Assembly_Contig_99_2 31 1585 RFL 6 Anapurna.300k_Assembly_Contig_99_1
32 1586 RFL 7 Anapurna.300k_Assembly_Contig_82_1 33 1587 RFL 7
Primepii.300k_Assembly_Contig_25_1 34 1588 RFL 7 R0934F.300k_Assembly_Contig_86_1 35
1589 RFL 7 R197.300k_Assembly_Contig_11_1 36 1590 RFL 7
R0932E.300k_Assembly_Contig_61_1 37 1591 RFL 7 Wheat-Rye-
6R.300k_Assembly_Contig_34_1 38 1592 RFL 8 R0932E.300k_Assembly_Contig_85_1 39 1593
RFL 8 Wheat-Rye-6R.300k_Assembly_Contig_79_2 40 1594 RFL 8
R0934F.300k_Assembly_Contig_87_2 41 1595 RFL 8 Primepii.300k_Assembly_Contig_64_1 42
1596 RFL 8 R197.300k_Assembly_Contig_71_2 43 1597 RFL 8
Anapurna.300k_Assembly_Contig_54_2 44 1598 RFL 9 R0932E.300k_Assembly_Contig_66_1
45 1599 RFL 9 Wheat-Rye-6R.300k_Assembly_Contig_69_1 46 1600 RFL 9
Primepii.300k_Assembly_Contig_59_1 47 1601 RFL 9 R197.300k_Assembly_Contig_51_1 48
1602 RFL 9 R0934F.300k_Assembly_Contig_89_1 49 1603 RFL 9
Anapurna.300k_Assembly_Contig_66_1 50 1604 RFL 10 Primepii.300k_Assembly_Contig_62_2
51 1605 RFL 10 Anapurna.300k_Assembly_Contig_179_1 52 1606 RFL 10
Anapurna.300k_Assembly_Contig_70_1 53 1607 RFL 10 R0932E.300k_Assembly_Contig_34_2
54 1608 RFL 10 Primepii.300k_Assembly_Contig_191_1 55 1609 RFL 10
R197.300k_Assembly_Contig_189_1 56 1610 RFL 10 Triticum- 57 1611
timopheevii.300k_Assembly_Contig_13_1 RFL 10 R197.300k_Assembly_Contig_64_1 58 1612
RFL 10 Wheat-Rye-6R.300k_Assembly_Contig_62_2 59 1613 RFL 10 Wheat-Rye-
6R.300k_Assembly_Contig_190_1 60 1614 RFL 10 R0934F.300k_Assembly_Contig_29_1 61
1615 RFL 11 Anapurna.300k_Assembly_Contig_14_2 62 1616 RFL 11
R0934F.300k_Assembly_Contig_14_2 63 1617 RFL 11 R0934F.300k_Assembly_Contig_14_1 64
1618 RFL 11 Triticum- 65 1619 timopheevii.300k_Assembly_Contig_22_1 RFL 11
Primepii.300k_Assembly_Contig_38_2 66 1620 RFL 11 Triticum- 67 1621
timopheevii.300k_Assembly_Contig_22_2 RFL 11 R197.300k_Assembly_Contig_33_1 68 1622
RFL 11 R0932E.300k_Assembly_Contig_27_2 69 1623 RFL 11
R197.300k_Assembly_Contig_33_2 70 1624 RFL 11 R0932E.300k_Assembly_Contig_27_1 71
1625 RFL 11 Wheat-Rye-6R.300k_Assembly_Contig_12_2 72 1626 RFL 12
R197.300k_Assembly_Contig_119_1 73 1627 RFL 12 Wheat-Rye-
6R.300k_Assembly_Contig_115_1 74 1628 RFL 12 R0932E.300k_Assembly_Contig_120_2 75
1629 RFL 13 Anapurna.300k_Assembly_Contig_8_3 76 1630 RFL 13
Primepii.300k_Assembly_Contig_8_2 77 1631 RFL 13 R0932E.300k_Assembly_Contig_8_2 78
1632 RFL 13 R197.300k_Assembly_Contig_25_2 79 1633 RFL 13 Wheat-Rye-
6R.300k_Assembly_Contig_21_2 80 1634 RFL 13 R0934F.300k_Assembly_Contig_18_2 81 1635
RFL 14 R0932E.300k_Assembly_Contig_24_1 82 1636 RFL 14
R0934F.300k_Assembly_Contig_10_1 83 1637 RFL 14 Wheat-Rye-
6R.300k_Assembly_Contig_26_1 84 1638 RFL 14 R197.300k_Assembly_Contig_23_1 85 1639
RFL 14 Primepii.300k_Assembly_Contig_11_1 86 1640 RFL 14
Anapurna.300k_Assembly_Contig_13_1 87 1641 RFL 15 R197.300k_Assembly_Contig_29_1 88
1642 RFL 15 Anapurna.300k_Assembly_Contig_28_1 89 1643 RFL 15
R0934F.300k_Assembly_Contig_20_1 90 1644 RFL 15 Triticum- 91 1645
timopheevii.300k_Assembly_Contig_11_2 RFL 15 R0932E.300k_Assembly_Contig_38_1 92
1646 RFL 15 Wheat-Rye-6R.300k_Assembly_Contig_6_1 93 1647 RFL 15
Primepii.300k_Assembly_Contig_22_1 94 1648 RFL 16 Primepii.300k_Assembly_Contig_23_2
95 1649 RFL 16 R197.300k_Assembly_Contig_43_2 96 1650 RFL 16 Wheat-Rye-
6R.300k_Assembly_Contig_11_2 97 1651 RFL 16 Anapurna.300k_Assembly_Contig_10_2 98

1652 RFL 16 R0932E.300k_Assembly_Contig_18_2 99 1653 RFL 16
R0934F.300k_Assembly_Contig_15_2 100 1654 RFL 17 Triticum- 101 1655
timopheevii.300k_Assembly_Contig_7_1 RFL 17 R197.300k_Assembly_Contig_58_1 102 1656
RFL 17 Wheat-Rye-6R.300k_Assembly_Contig_59_1 103 1657 RFL 17
R0934F.300k_Assembly_Contig_51_1 104 1658 RFL 17 Triticum- 105 1659
timopheevii.300k_Assembly_Contig_29_2 RFL 18 R0934F.300k_Assembly_Contig_25_2 106
1660 RFL 18 R0932E.300k_Assembly_Contig_21_2 107 1661 RFL 18
R197.300k_Assembly_Contig_14_2 108 1662 RFL 18 Anapurna.300k_Assembly_Contig_6_2 109
1663 RFL 18 Triticum- 110 1664 timopheevii.300k_Assembly_Contig_16_3 RFL 18
Primepii.300k_Assembly_Contig_20_2 111 1665 RFL 18 Wheat-Rye-
6R.300k_Assembly_Contig_8_2 112 1666 RFL 19 Anapurna.300k_Assembly_Contig_68_1 113
1667 RFL 20 Triticum- 114 1668 timopheevii.300k_Assembly_Contig_47_1 RFL 21
R197.300k_Assembly_Contig_18_1 115 1669 RFL 21 Anapurna.300k_Assembly_Contig_16_1
116 1670 RFL 21 R0932E.300k_Assembly_Contig_58_1 117 1671 RFL 21 Wheat-Rye-
6R.300k_Assembly_Contig_23_3 118 1672 RFL 21 Wheat-Rye-6R.300k_Assembly_Contig_23_2
119 1673 RFL 21 Primepii.300k_Assembly_Contig_42_2 120 1674 RFL 21
R0934F.300k_Assembly_Contig_28_3 121 1675 RFL 21 Primepii.300k_Assembly_Contig_42_3
122 1676 RFL 21 R0934F.300k_Assembly_Contig_28_2 123 1677 RFL 22
Primepii.300k_Assembly_Contig_63_1 124 1678 RFL 22 R0934F.300k_Assembly_Contig_61_1
125 1679 RFL 23 R0934F.300k_Assembly_Contig_46_2 126 1680 RFL 23 Wheat-Rye-
6R.300k_Assembly_Contig_30_1 127 1681 RFL 23 R197.300k_Assembly_Contig_15_3 128 1682
RFL 23 R0932E.300k_Assembly_Contig_40_1 129 1683 RFL 23
R0934F.300k_Assembly_Contig_46_1 130 1684 RFL 23 R197.300k_Assembly_Contig_15_2 131
1685 RFL 23 Anapurna.300k_Assembly_Contig_42_1 132 1686 RFL 23
Primepii.300k_Assembly_Contig_14_2 133 1687 RFL 24 Anapurna.300k_Assembly_Contig_17_1
134 1688 RFL 24 R0934F.300k_Assembly_Contig_22_1 135 1689 RFL 24
R0932E.300k_Assembly_Contig_44_1 136 1690 RFL 24 Primepii.300k_Assembly_Contig_46_1
137 1691 RFL 24 R197.300k_Assembly_Contig_30_1 138 1692 RFL 24 Wheat-Rye-
6R.300k_Assembly_Contig_40_1 139 1693 RFL 25 R197.300k_Assembly_Contig_92_2 140 1694
RFL 25 Wheat-Rye-6R.300k_Assembly_Contig_109_2 141 1695 RFL 26
Anapurna.300k_Assembly_Contig_38_1 142 1696 RFL 26 Wheat-Rye-
6R.300k_Assembly_Contig_41_1 143 1697 RFL 27 Triticum- 144 1698
timopheevii.300k_Assembly_Contig_63_3 RFL 28 R0932E.300k_Assembly_Contig_23_2 145
1699 RFL 28 R0932E.300k_Assembly_Contig_23_3 146 1700 RFL 28
Primepii.300k_Assembly_Contig_13_1 147 1701 RFL 28 R197.300k_Assembly_Contig_10_3 148
1702 RFL 28 R197.300k_Assembly_Contig_10_2 149 1703 RFL 28
R0934F.300k_Assembly_Contig_17_1 150 1704 RFL 28 Anapurna.300k_Assembly_Contig_2_3
151 1705 RFL 28 Wheat-Rye-6R.300k_Assembly_Contig_7_3 152 1706 RFL 28
Anapurna.300k_Assembly_Contig_2_2 153 1707 RFL 28 Wheat-Rye-
6R.300k_Assembly_Contig_7_2 154 1708 RFL 29 Wheat-Rye-6R.300k_Assembly_Contig_77_2
155 1709 RFL 29 R0934F.300k_Assembly_Contig_78_1 156 1710 RFL 29 Wheat-Rye-
6R.300k_Assembly_Contig_77_1 157 1711 RFL 29 Primepii.300k_Assembly_Contig_67_1 158
1712 RFL 30 Primepii.300k_Assembly_Contig_91_1 159 1713 RFL 30
R0934F.300k_Assembly_Contig_64_1 160 1714 RFL 30 R0932E.300k_Assembly_Contig_110_1
161 1715 RFL 31 Triticum- 162 1716 timopheevii.300k_Assembly_Contig_14_1 RFL 32 Triticum-
163 1717 timopheevii.300k_Assembly_Contig_6_2 RFL 32 Primepii.300k_Assembly_Contig_6_2
164 1718 RFL 32 R0934F.300k_Assembly_Contig_31_2 165 1719 RFL 32
R197.300k_Assembly_Contig_7_2 166 1720 RFL 32 Anapurna.300k_Assembly_Contig_31_2 167
1721 RFL 32 Wheat-Rye-6R.300k_Assembly_Contig_5_2 168 1722 RFL 32
R0932E.300k_Assembly_Contig_29_2 169 1723 RFL 33 Triticum- 170 1724

timopheevii.300k_Assembly_Contig_8_1 RFL 33 Anapurna.300k_Assembly_Contig_5_2 171
1725 RFL 33 R197.300k_Assembly_Contig_6_3 172 1726 RFL 33 Wheat-Rye-
6R.300k_Assembly_Contig_25_1 173 1727 RFL 33 R0932E.300k_Assembly_Contig_19_1 174
1728 RFL 33 Primepii.300k_Assembly_Contig_21_2 175 1729 RFL 33
R0934F.300k_Assembly_Contig_104_1 176 1730 RFL 34 R0932E.300k_Assembly_Contig_5_1
177 1731 RFL 34 R0932E.300k_Assembly_Contig_2_4 178 1732 RFL 35 Triticum- 179 1733
timopheevii.300k_Assembly_Contig_21_2 RFL 36 Triticum- 180 1734
timopheevii.300k_Assembly_Contig_20_2 RFL 37 Anapurna.300k_Assembly_Contig_37_2 181
1735 RFL 37 Wheat-Rye-6R.300k_Assembly_Contig_19_2 182 1736 RFL 37
R0932E.300k_Assembly_Contig_12_2 183 1737 RFL 37 R197.300k_Assembly_Contig_20_2 184
1738 RFL 37 R0934F.300k_Assembly_Contig_37_2 185 1739 RFL 37
Primepii.300k_Assembly_Contig_10_2 186 1740 RFL 38 Triticum- 187 1741
timopheevii.300k_Assembly_Contig_43_2 RFL 39 R0932E.300k_Assembly_Contig_53_2 188
1742 RFL 39 Anapurna.300k_Assembly_Contig_29_2 189 1743 RFL 39
R0932E.300k_Assembly_Contig_22_1 190 1744 RFL 39 Wheat-Rye-
6R.300k_Assembly_Contig_28_2 191 1745 RFL 39 Triticum- 192 1746
timopheevii.300k_Assembly_Contig_25_2 RFL 39 Wheat-Rye-6R.300k_Assembly_Contig_15_1
193 1747 RFL 39 R0934F.300k_Assembly_Contig_27_1 194 1748 RFL 39
R0934F.300k_Assembly_Contig_26_2 195 1749 RFL 39 Primepii.300k_Assembly_Contig_40_2
196 1750 RFL 39 Primepii.300k_Assembly_Contig_30_1 197 1751 RFL 39
R197.300k_Assembly_Contig_36_2 198 1752 RFL 39 R197.300k_Assembly_Contig_27_1 199
1753 RFL 39 Anapurna.300k_Assembly_Contig_43_1 200 1754 RFL 40
R0934F.300k_Assembly_Contig_12_1 201 1755 RFL 40 Primepii.300k_Assembly_Contig_18_1
202 1756 RFL 40 Triticum- 203 1757 timopheevii.300k_Assembly_Contig_5_1 RFL 40
R0932E.300k_Assembly_Contig_7_1 204 1758 RFL 40 Anapurna.300k_Assembly_Contig_20_1
205 1759 RFL 40 R197.300k_Assembly_Contig_32_1 206 1760 RFL 41
R0934F.300k_Assembly_Contig_24_2 207 1761 RFL 41 Wheat-Rye-
6R.300k_Assembly_Contig_45_2 208 1762 RFL 41 Anapurna.300k_Assembly_Contig_39_2 209
1763 RFL 41 R0932E.300k_Assembly_Contig_56_2 210 1764 RFL 41
Primepii.300k_Assembly_Contig_47_2 211 1765 RFL 41 R197.300k_Assembly_Contig_40_2 212
1766 RFL 42 Triticum- 213 1767 timopheevii.300k_Assembly_Contig_10_3 RFL 43
R0934F.300k_Assembly_Contig_8_1 214 1768 RFL 43 R0932E.300k_Assembly_Contig_13_1
215 1769 RFL 43 Triticum- 216 1770 timopheevii.300k_Assembly_Contig_12_2 RFL 43
R197.300k_Assembly_Contig_22_1 217 1771 RFL 43 Primepii.300k_Assembly_Contig_31_1 218
1772 RFL 43 Anapurna.300k_Assembly_Contig_23_1 219 1773 RFL 43 Wheat-Rye-
6R.300k_Assembly_Contig_10_1 220 1774 RFL 44 R0932E.300k_Assembly_Contig_32_1 221
1775 RFL 44 Primepii.300k_Assembly_Contig_7_1 222 1776 RFL 44
Anapurna.300k_Assembly_Contig_33_1 223 1777 RFL 45 R0932E.300k_Assembly_Contig_73_2
224 1778 RFL 45 R197.300k_Assembly_Contig_70.sub.——2 225 1779 RFL 45 Wheat-Rye-
6R.300k_Assembly_Contig_72_2 226 1780 RFL 46 Wheat-Rye-6R.300k_Assembly_Contig_35_1
227 1781 RFL 47 R0934F.300k_Assembly_Contig_57_2 228 1782 RFL 47 Wheat-Rye-
6R.300k_Assembly_Contig_67_2 229 1783 RFL 47 Anapurna.300k_Assembly_Contig_85_2 230
1784 RFL 47 R0932E.300k_Assembly_Contig_71_2 231 1785 RFL 47
R197.300k_Assembly_Contig_62_2 232 1786 RFL 47 Primepii.300k_Assembly_Contig_86_3 233
1787 RFL 48 Primepii.300k_Assembly_Contig_51_1 234 1788 RFL 48
R197.300k_Assembly_Contig_57_1 235 1789 RFL 48 Anapurna.300k_Assembly_Contig_74_1
236 1790 RFL 48 R0934F.300k_Assembly_Contig_71_1 237 1791 RFL 48
R0932E.300k_Assembly_Contig_72_1 238 1792 RFL 48 Wheat-Rye-
6R.300k_Assembly_Contig_82_1 239 1793 RFL 49 R197.300k_Assembly_Contig_83_1 240 1794
RFL 49 R197.300k_Assembly_Contig_95_2 241 1795 RFL 49 Triticum- 242 1796

timopheevii.300k_Assembly_Contig_37_1 RFL 49 Triticum- 243 1797
timopheevii.300k_Assembly_Contig_38_1 RFL 50 Anapurna.300k_Assembly_Contig_24_2 244
1798 RFL 50 Primepii.300k_Assembly_Contig_24_2 245 1799 RFL 50
R0932E.300k_Assembly_Contig_25_2 246 1800 RFL 50 R197.300k_Assembly_Contig_21_2 247
1801 RFL 50 Wheat-Rye-6R.300k_Assembly_Contig_22_2 248 1802 RFL 50
R0934F.300k_Assembly_Contig_39_2 249 1803 RFL 51 Anapurna.300k_Assembly_Contig_35_2
250 1804 RFL 51 R0934F.300k_Assembly_Contig_5_4 251 1805 RFL 51
Primepii.300k_Assembly_Contig_17_3 252 1806 RFL 51
Anapurna.300k_Assembly_Contig_109_1 253 1807 RFL 51
Primepii.300k_Assembly_Contig_19_2 254 1808 RFL 51 R0932E.300k_Assembly_Contig_55_1
255 1809 RFL 51 Wheat-Rye-6R.300k_Assembly_Contig_2_1 256 1810 RFL 51
R0934F.300k_Assembly_Contig_55_2 257 1811 RFL 51 Triticum- 258 1812
timopheevii.300k_Assembly_Contig_4_2 RFL 51 R197.300k_Assembly_Contig_46_2 259 1813
RFL 51 Wheat-Rye-6R.300k_Assembly_Contig_37_2 260 1814 RFL 51
Anapurna.300k_Assembly_Contig_106_1 261 1815 RFL 51 R197.300k_Assembly_Contig_5_3
262 1816 RFL 51 R0932E.300k_Assembly_Contig_6_2 263 1817 RFL 52
R197.300k_Assembly_Contig_37_1 264 1818 RFL 52 R197.300k_Assembly_Contig_37_2 265
1819 RFL 52 R0932E.300k_Assembly_Contig_45_1 266 1820 RFL 52 Wheat-Rye-
6R.300k_Assembly_Contig_39_1 267 1821 RFL 52 Wheat-Rye-6R.300k_Assembly_Contig_39_2
268 1822 RFL 52 Primepii.300k_Assembly_Contig_49_1 269 1823 RFL 52
Anapurna.300k_Assembly_Contig_22_1 270 1824 RFL 52 R0934F.300k_Assembly_Contig_49_1
271 1825 RFL 52 R0934F.300k_Assembly_Contig_49_2 272 1826 RFL 52 Triticum- 273 1827
timopheevii.300k_Assembly_Contig_29_1 RFL 53 R0934F.300k_Assembly_Contig_75_2 274
1828 RFL 53 R0932E.300k_Assembly_Contig_64_2 275 1829 RFL 53
Anapurna.300k_Assembly_Contig_72_2 276 1830 RFL 53 Primepii.300k_Assembly_Contig_57_2
277 1831 RFL 53 Triticum- 278 1832 timopheevii.300k_Assembly_Contig_55_3 RFL 54 Wheat-
Rye-6R.300k_Assembly_Contig_80_1 279 1833 RFL 54 R0932E.300k_Assembly_Contig_209_1
280 1834 RFL 54 Anapurna.300k_Assembly_Contig_77_1 281 1835 RFL 55
R0932E.300k_Assembly_Contig_54_1 282 1836 RFL 55 R197.300k_Assembly_Contig_39_1 283
1837 RFL 55 R0934F.300k_Assembly_Contig_54_1 284 1838 RFL 55
Anapurna.300k_Assembly_Contig_41_1 285 1839 RFL 55 Primepii.300k_Assembly_Contig_44_1
286 1840 RFL 55 Wheat-Rye-6R.300k_Assembly_Contig_31_1 287 1841 RFL 56
R197.300k_Assembly_Contig_86_1 288 1842 RFL 56 R0932E.300k_Assembly_Contig_74_1 289
1843 RFL 56 Triticum- 290 1844 timopheevii.300k_Assembly_Contig_39_1 RFL 58 Wheat-Rye-
6R.300k_Assembly_Contig_66_1 291 1845 RFL 58 Anapurna.300k_Assembly_Contig_12_1 292
1846 RFL 59 Triticum- 293 1847 timopheevii.300k_Assembly_Contig_60_1 RFL 59
R197.300k_Assembly_Contig_115_1 294 1848 RFL 59 R0932E.300k_Assembly_Contig_161_1
295 1849 RFL 59 R0932E.300k_Assembly_Contig_48_1 296 1850 RFL 60
Primepii.300k_Assembly_Contig_94_1 297 1851 RFL 60 R197.300k_Assembly_Contig_95_1 298
1852 RFL 60 R0934F.300k_Assembly_Contig_73_2 299 1853 RFL 60 Wheat-Rye-
6R.300k_Assembly_Contig_48_2 300 1854 RFL 61 Triticum- 301 1855
timopheevii.300k_Assembly_Contig_48_1 RFL 62 R0934F.300k_Assembly_Contig_138_1 302
1856 RFL 63 R197.300k_Assembly_Contig_84_1 303 1857 RFL 63
R197.300k_Assembly_Contig_84_2 304 1858 RFL 63 Triticum- 305 1859
timopheevii.300k_Assembly_Contig_34_1 RFL 64 Primepii.300k_Assembly_Contig_50_1 306
1860 RFL 64 Wheat-Rye-6R.300k_Assembly_Contig_54_1 307 1861 RFL 64
R0934F.300k_Assembly_Contig_45_1 308 1862 RFL 64 Anapurna.300k_Assembly_Contig_61_1
309 1863 RFL 64 R197.300k_Assembly_Contig_42_1 310 1864 RFL 64
R0932E.300k_Assembly_Contig_59_1 311 1865 RFL 65 Triticum- 312 1866
timopheevii.300k_Assembly_Contig_42_2 RFL 66 Triticum- 313 1867

timopheevii.300k_Assembly_Contig_27_1 RFL 66 Triticum- 314 1868
timopheevii.300k_Assembly_Contig_80_1 RFL 67 Primepii.300k_Assembly_Contig_213_1 315
1869 RFL 67 Primepii.300k_Assembly_Contig_80_2 316 1870 RFL 67
Primepii.300k_Assembly_Contig_80_1 317 1871 RFL 67 R0934F.300k_Assembly_Contig_111_2
318 1872 RFL 67 Primepii.300k_Assembly_Contig_2_1 319 1873 RFL 67
R0934F.300k_Assembly_Contig_111_1 320 1874 RFL 67 R0934F.300k_Assembly_Contig_4_1
321 1875 RFL 68 Wheat-Rye-6R.300k_Assembly_Contig_53_1 322 1876 RFL 68
R0932E.300k_Assembly_Contig_68_1 323 1877 RFL 68 Wheat-Rye-
6R.300k_Assembly_Contig_53_3 324 1878 RFL 68 R197.300k_Assembly_Contig_19_1 325 1879
RFL 68 R0934F.300k_Assembly_Contig_66_1 326 1880 RFL 68 Triticum- 327 1881
timopheevii.300k_Assembly_Contig_33_1 RFL 68 Anapurna.300k_Assembly_Contig_71_1 328
1882 RFL 68 Anapurna.300k_Assembly_Contig_71_2 329 1883 RFL 68
Primepii.300k_Assembly_Contig_37_1 330 1884 RFL 68 R0932E.300k_Assembly_Contig_68_3
331 1885 RFL 68 R197.300k_Assembly_Contig_19_3 332 1886 RFL 69 Triticum- 333 1887
timopheevii.300k_Assembly_Contig_45_1 RFL 70 R0934F.300k_Assembly_Contig_68_1 334
1888 RFL 70 R197.300k_Assembly_Contig_65_1 335 1889 RFL 70
R0932E.300k_Assembly_Contig_94_1 336 1890 RFL 70 Primepii.300k_Assembly_Contig_55_1
337 1891 RFL 70 Anapurna.300k_Assembly_Contig_111_1 338 1892 RFL 70 Wheat-Rye-
6R.300k_Assembly_Contig_74_1 339 1893 RFL 70 Anapurna.300k_Assembly_Contig_98_1 340
1894 RFL 71 R0932E.300k_Assembly_Contig_114_1 341 1895 RFL 72
R0932E.300k_Assembly_Contig_98_1 342 1896 RFL 73 R0932E.300k_Assembly_Contig_89_2
343 1897 RFL 73 R197.300k_Assembly_Contig_78_2 344 1898 RFL 73 Triticum- 345 1899
timopheevii.300k_Assembly_Contig_31_2 RFL 74 R197.300k_Assembly_Contig_194_1 346
1900 RFL 74 R0934F.300k_Assembly_Contig_36_2 347 1901 RFL 74
R0934F.300k_Assembly_Contig_150_2 348 1902 RFL 74 R0932E.300k_Assembly_Contig_185_1
349 1903 RFL 74 R0932E.300k_Assembly_Contig_201_1 350 1904 RFL 74
R0932E.300k_Assembly_Contig_92_1 351 1905 RFL 74 R197.300k_Assembly_Contig_179_1
352 1906 RFL 74 R0932E.300k_Assembly_Contig_83_1 353 1907 RFL 74
R197.300k_Assembly_Contig_4_1 354 1908 RFL 74 R0934F.300k_Assembly_Contig_126_1 355
1909 RFL 75 R0932E.300k_Assembly_Contig_88_1 356 1910 RFL 77
R0932E.300k_Assembly_Contig_214_2 357 1911 RFL 78
R0932E.300k_Assembly_Contig_112_1 358 1912 RFL 79
R0932E.300k_Assembly_Contig_103_1 359 1913 RFL 79 R0934F.300k_Assembly_Contig_80_1
360 1914 RFL 79 R197.300k_Assembly_Contig_120_1 361 1915 RFL 79 Triticum- 362 1916
timopheevii.300k_Assembly_Contig_57_1 RFL 80 R197.300k_Assembly_Contig_59_1 363 1917
RFL 81 R0932E.300k_Assembly_Contig_4_1 364 1918 RFL 81
R0932E.300k_Assembly_Contig_199_1 365 1919 RFL 82 Triticum- 366 1920
timopheevii.300k_Assembly_Contig_64_2 RFL 82 R0932E.300k_Assembly_Contig_123_2 367
1921 RFL 83 Primepii.300k_Assembly_Contig_52_1 368 1922 RFL 83
R0934F.300k_Assembly_Contig_43_1 369 1923 RFL 83 R0932E.300k_Assembly_Contig_46_1
370 1924 RFL 83 Anapurna.300k_Assembly_Contig_62_1 371 1925 RFL 83
R197.300k_Assembly_Contig_54_1 372 1926 RFL 83 Wheat-Rye-
6R.300k_Assembly_Contig_60_1 373 1927 RFL 84 R0932E.300k_Assembly_Contig_116_1 374
1928 RFL 85 R197.300k_Assembly_Contig_80_3 375 1929 RFL 85 Triticum- 376 1930
timopheevii.300k_Assembly_Contig_36_3 RFL 85 R197.300k_Assembly_Contig_80_2 377 1931
RFL 87 Wheat-Rye-6R.300k_Assembly_Contig_73_1 378 1932 RFL 89
Primepii.300k_Assembly_Contig_83_1 379 1933 RFL 89 Primepii.300k_Assembly_Contig_183_1
380 1934 RFL 89 R0934F.300k_Assembly_Contig_99_1 381 1935 RFL 90
Primepii.300k_Assembly_Contig_82_2 382 1936 RFL 90 Wheat-Rye-
6R.300k_Assembly_Contig_65_2 383 1937 RFL 90 R0934F.300k_Assembly_Contig_63_1 384

1938 RFL 90 R197.300k_Assembly_Contig_66_2 385 1939 RFL 90
Anapurna.300k_Assembly_Contig_51_1 386 1940 RFL 90 R0932E.300k_Assembly_Contig_90_1
387 1941 RFL 92 Wheat-Rye-6R.300k_Assembly_Contig_55_2 388 1942 RFL 92
R197.300k_Assembly_Contig_47_2 389 1943 RFL 92 Triticum- 390 1944
timopheevii.300k_Assembly_Contig_30_2 RFL 92 R0932E.300k_Assembly_Contig_43_2 391
1945 RFL 92 Triticum- 392 1946 timopheevii.300k_Assembly_Contig_30_3 RFL 92
Anapurna.300k_Assembly_Contig_50_2 393 1947 RFL 92 R0934F.300k_Assembly_Contig_48_2
394 1948 RFL 92 Primepii.300k_Assembly_Contig_53_2 395 1949 RFL 93
R197.300k_Assembly_Contig_117_1 396 1950 RFL 93 R0932E.300k_Assembly_Contig_113_1
397 1951 RFL 94 Wheat-Rye-6R.300k_Assembly_Contig_41_2 398 1952 RFL 94
Anapurna.300k_Assembly_Contig_38_2 399 1953 RFL 94 R197.300k_Assembly_Contig_24_1
400 1954 RFL 94 R0934F.300k_Assembly_Contig_32_1 401 1955 RFL 94
R0932E.300k_Assembly_Contig_17_1 402 1956 RFL 94 Primepii.300k_Assembly_Contig_41_2
403 1957 RFL 95 Triticum- 404 1958 timopheevii.300k_Assembly_Contig_51_1 RFL 96
Primepii.300k_Assembly_Contig_72_1 405 1959 RFL 97 Anapurna.300k_Assembly_Contig_92_2
406 1960 RFL 97 Anapurna.300k_Assembly_Contig_19_1 407 1961 RFL 97 Wheat-Rye-
6R.300k_Assembly_Contig_103_2 408 1962 RFL 97 Wheat-Rye-
6R.300k_Assembly_Contig_63_1 409 1963 RFL 98 R0932E.300k_Assembly_Contig_79_1 410
1964 RFL 98 Primepii.300k_Assembly_Contig_70_1 411 1965 RFL 98
R0934F.300k_Assembly_Contig_70_1 412 1966 RFL 98 R197.300k_Assembly_Contig_63_1 413
1967 RFL 98 Wheat-Rye-6R.300k_Assembly_Contig_71_1 414 1968 RFL 99
R0932E.300k_Assembly_Contig_15_2 415 1969 RFL 99 R0932E.300k_Assembly_Contig_153_2
416 1970 RFL 100 R197.300k_Assembly_Contig_94_1 417 1971 RFL 100 Wheat-Rye-
6R.300k_Assembly_Contig_113_1 418 1972 RFL 100 R0932E.300k_Assembly_Contig_100_1
419 1973 RFL 100 Anapurna.300k_Assembly_Contig_113_1 420 1974 RFL 101
Anapurna.300k_Assembly_Contig_59_1 421 1975 RFL 101 Triticum- 422 1976
timopheevii.300k_Assembly_Contig_54_1 RFL 102 Triticum- 423 1977
timopheevii.300k_Assembly_Contig_71_1 RFL 103 Primepii.300k_Assembly_Contig_106_1 424
1978 RFL 103 Wheat-Rye-6R.300k_Assembly_Contig_98_1 425 1979 RFL 103
R197.300k_Assembly_Contig_79_1 426 1980 RFL 104 R0934F.300k_Assembly_Contig_69_1
427 1981 RFL 104 Triticum- 428 1982 timopheevii.300k_Assembly_Contig_35_1 RFL 104
R0932E.300k_Assembly_Contig_82_1 429 1983 RFL 104 R197.300k_Assembly_Contig_72_1
430 1984 RFL 105 Primepii.300k_Assembly_Contig_85_1 431 1985 RFL 106 Wheat-Rye-
6R.300k_Assembly_Contig_87_2 432 1986 RFL 106 Primepii.300k_Assembly_Contig_81_2 433
1987 RFL 106 Anapurna.300k_Assembly_Contig_80_1 434 1988 RFL 106
R0932E.300k_Assembly_Contig_76_1 435 1989 RFL 106 R197.300k_Assembly_Contig_67_1
436 1990 RFL 106 R0934F.300k_Assembly_Contig_103_2 437 1991 RFL 107
R0934F.300k_Assembly_Contig_117_1 438 1992 RFL 107 Triticum- 439 1993
timopheevii.300k_Assembly_Contig_69_1 RFL 108 Wheat-Rye-6R.300k_Assembly_Contig_86_1
440 1994 RFL 108 R0932E.300k_Assembly_Contig_87_1 441 1995 RFL 108
R197.300k_Assembly_Contig_77_1 442 1996 RFL 108 Anapurna.300k_Assembly_Contig_78_1
443 1997 RFL 108 R0934F.300k_Assembly_Contig_115_1 444 1998 RFL 109 Triticum- 445 1999
timopheevii.300k_Assembly_Contig_78_2 RFL 110 Triticum- 446 2000
timopheevii.300k_Assembly_Contig_15_2 RFL 111 R0932E.300k_Assembly_Contig_78_1 447
2001 RFL 111 R0934F.300k_Assembly_Contig_23_1 448 2002 RFL 111
Anapurna.300k_Assembly_Contig_58_1 449 2003 RFL 111 R197.300k_Assembly_Contig_69_1
450 2004 RFL 111 Primepii.300k_Assembly_Contig_35_1 451 2005 RFL 111 Wheat-Rye-
6R.300k_Assembly_Contig_83_1 452 2006 RFL 112 R0932E.300k_Assembly_Contig_20_1 453
2007 RFL 112 R197.300k_Assembly_Contig_17_1 454 2008 RFL 112 Wheat-Rye-
6R.300k_Assembly_Contig_17_1 455 2009 RFL 112 Anapurna.300k_Assembly_Contig_4_1 456

R0934F.300k_Assembly_Contig_6_1 457 2011 RFL 112 Primepii.300k_Assembly_Contig_12_1 457 2011 RFL 112
R0934F.300k_Assembly_Contig_6_1 458 2012 RFL 113 R0932E.300k_Assembly_Contig_216_1
459 2013 RFL 113 R0934F.300k_Assembly_Contig_202_1 460 2014 RFL 113
Primepii.300k_Assembly_Contig_92_1 461 2015 RFL 113 R197.300k_Assembly_Contig_98_1
462 2016 RFL 114 R0932E.300k_Assembly_Contig_86_1 463 2017 RFL 115
R0934F.300k_Assembly_Contig_128_1 464 2018 RFL 115 Triticum- 465 2019
timopheevii.300k_Assembly_Contig_74_1 RFL 116 Triticum- 466 2020
timopheevii.300k_Assembly_Contig_44_2 RFL 118 R197.300k_Assembly_Contig_75_1 467 2021
RFL 118 Wheat-Rye-6R.300k_Assembly_Contig_93_1 468 2022 RFL 118
Anapurna.300k_Assembly_Contig_87_1 469 2023 RFL 119 R197.300k_Assembly_Contig_1_1
470 2024 RFL 119 Wheat-Rye-6R.300k_Assembly_Contig_1_1 471 2025 RFL 119 Wheat-Rye-
6R.300k_Assembly_Contig_194_1 472 2026 RFL 119 R0934F.300k_Assembly_Contig_2_1 473
2027 RFL 119 Primepii.300k_Assembly_Contig_1_1 474 2028 RFL 119
Anapurna.300k_Assembly_Contig_1_1 475 2029 RFL 119 R0932E.300k_Assembly_Contig_1_1
476 2030 RFL 120 Triticum- 477 2031 timopheevii.300k_Assembly_Contig_67_1 RFL 121
R197.300k_Assembly_Contig_96_1 478 2032 RFL 121 R0932E.300k_Assembly_Contig_93_1
479 2033 RFL 121 Triticum- 480 2034 timopheevii.300k_Assembly_Contig_70_1 RFL 121
R0934F.300k_Assembly_Contig_102_1 481 2035 RFL 121 Wheat-Rye-
6R.300k_Assembly_Contig_95_1 482 2036 RFL 122 Primepii.300k_Assembly_Contig_90_1 483
2037 RFL 122 R197.300k_Assembly_Contig_68_1 484 2038 RFL 122
Anapurna.300k_Assembly_Contig_63_1 485 2039 RFL 122
R0932E.300k_Assembly_Contig_91_1 486 2040 RFL 122 R0934F.300k_Assembly_Contig_97_1
487 2041 RFL 122 Triticum- 488 2042 timopheevii.300k_Assembly_Contig_59_1 RFL 122
Wheat-Rye-6R.300k_Assembly_Contig_85_1 489 2043 RFL 123 Triticum- 490 2044
timopheevii.300k_Assembly_Contig_65_1 RFL 124 Primepii.300k_Assembly_Contig_96_1 491
2045 RFL 124 R197.300k_Assembly_Contig_116_1 492 2046 RFL 124
R0934F.300k_Assembly_Contig_110_1 493 2047 RFL 124 Wheat-Rye-
6R.300k_Assembly_Contig_92_1 494 2048 RFL 124 Anapurna.300k_Assembly_Contig_84_1 495
2049 RFL 124 R0932E.300k_Assembly_Contig_107_1 496 2050 RFL 125 Triticum- 497 2051
timopheevii.300k_Assembly_Contig_40_1 RFL 125 Triticum- 498 2052
timopheevii.300k_Assembly_Contig_41_3 RFL 125 R197.300k_Assembly_Contig_3_1 499 2053
RFL 126 R0932E.300k_Assembly_Contig_35_1 500 2054 RFL 126
R197.300k_Assembly_Contig_9_3 501 2055 RFL 126 R0932E.300k_Assembly_Contig_35_2 502
2056 RFL 126 Anapurna.300k_Assembly_Contig_18_3 503 2057 RFL 126
Primepii.300k_Assembly_Contig_5_3 504 2058 RFL 126 Wheat-Rye-
6R.300k_Assembly_Contig_29_1 505 2059 RFL 126 R0934F.300k_Assembly_Contig_19_3 506
2060 RFL 126 Wheat-Rye-6R.300k_Assembly_Contig_14_3 507 2061 RFL 126 Triticum- 508
2062 timopheevii.300k_Assembly_Contig_24_1 RFL 126 Primepii.300k_Assembly_Contig_43_1
509 2063 RFL 126 R0934F.300k_Assembly_Contig_30_2 510 2064 RFL 126
R0934F.300k_Assembly_Contig_30_1 511 2065 RFL 126 R197.300k_Assembly_Contig_16_1
512 2066 RFL 126 R0932E.300k_Assembly_Contig_16_3 513 2067 RFL 126
Anapurna.300k_Assembly_Contig_11_1 514 2068 RFL 127
Anapurna.300k_Assembly_Contig_73_1 515 2069 RFL 128 R197.300k_Assembly_Contig_113_2
516 2070 RFL 129 R0932E.300k_Assembly_Contig_104_1 517 2071 RFL 129
R0934F.300k_Assembly_Contig_91_1 518 2072 RFL 129 R197.300k_Assembly_Contig_101_1
519 2073 RFL 130 Triticum- 520 2074 timopheevii.300k_Assembly_Contig_53_2 RFL 131
R0934F.300k_Assembly_Contig_106_1 521 2075 RFL 131
Anapurna.300k_Assembly_Contig_94_1 522 2076 RFL 131
Primepii.300k_Assembly_Contig_112_1 523 2077 RFL 132 R197.300k_Assembly_Contig_82_1
524 2078 RFL 132 Wheat-Rye-6R.300k_Assembly_Contig_96_1 525 2079 RFL 132

Primepii.300k_Assembly_Contig_87_1 526 2080 RFL 133 R0932E.300k_Assembly_Contig_28_2
527 2081 RFL 133 Anapurna.300k_Assembly_Contig_7_2 528 2082 RFL 133
Primepii.300k_Assembly_Contig_26_2 529 2083 RFL 134 Wheat-Rye-
6R.300k_Assembly_Contig_106_1 530 2084 RFL 134 R197.300k_Assembly_Contig_100_1 531
2085 RFL 134 Primepii.300k_Assembly_Contig_102_1 532 2086 RFL 135 Triticum- 533 2087
timopheevii.300k_Assembly_Contig_93_1 RFL 136 Primepii.300k_Assembly_Contig_73_1 534
2088 RFL 136 R0934F.300k_Assembly_Contig_77_1 535 2089 RFL 136
R0932E.300k_Assembly_Contig_65_1 536 2090 RFL 136
Anapurna.300k_Assembly_Contig_52_1 537 2091 RFL 136 Wheat-Rye-
6R.300k_Assembly_Contig_84_1 538 2092 RFL 136 R197.300k_Assembly_Contig_81_1 539
2093 RFL 137 R0934F.300k_Assembly_Contig_21_3 540 2094 RFL 137
R0932E.300k_Assembly_Contig_10_3 541 2095 RFL 137 Primepii.300k_Assembly_Contig_48_3
542 2096 RFL 137 Anapurna.300k_Assembly_Contig_9_3 543 2097 RFL 137 Wheat-Rye-
6R.300k_Assembly_Contig_20_3 544 2098 RFL 137 R197.300k_Assembly_Contig_2_3 545 2099
RFL 138 R0934F.300k_Assembly_Contig_201_1 546 2100 RFL 138 Triticum- 547 2101
timopheevii.300k_Assembly_Contig_1_1 RFL 138 R0934F.300k_Assembly_Contig_1_1 548 2102
RFL 139 R0934F.300k_Assembly_Contig_74_2 549 2103 RFL 139
Primepii.300k_Assembly_Contig_79_2 550 2104 RFL 139 Wheat-Rye-
6R.300k_Assembly_Contig_70_2 551 2105 RFL 139 Anapurna.300k_Assembly_Contig_64_2 552
2106 RFL 140 Primepii.300k_Assembly_Contig_105_2 553 2107 RFL 140
R0934F.300k_Assembly_Contig_56_2 554 2108 RFL 141 R0932E.300k_Assembly_Contig_75_1
555 2109 RFL 141 R0934F.300k_Assembly_Contig_50_1 556 2110 RFL 142
R0934F.300k_Assembly_Contig_76_1 557 2111 RFL 142 Primepii.300k_Assembly_Contig_75_1
558 2112 RFL 143 Wheat-Rye-6R.300k_Assembly_Contig_3_1 559 2113 RFL 143
R0934F.300k_Assembly_Contig_9_1 560 2114 RFL 143 Primepii.300k_Assembly_Contig_3_1
561 2115 RFL 143 R197.300k_Assembly_Contig_8_1 562 2116 RFL 143
R0932E.300k_Assembly_Contig_3_1 563 2117 RFL 143 Anapurna.300k_Assembly_Contig_3_1
564 2118 RFL 144 Anapurna.300k_Assembly_Contig_49_1 565 2119 RFL 144 Wheat-Rye-
6R.300k_Assembly_Contig_58_1 566 2120 RFL 144 R197.300k_Assembly_Contig_56_1 567
2121 RFL 144 R0934F.300k_Assembly_Contig_40_1 568 2122 RFL 144
Primepii.300k_Assembly_Contig_58_1 569 2123 RFL 144 R0932E.300k_Assembly_Contig_50_1
570 2124 RFL 145 R0932E.300k_Assembly_Contig_62_1 571 2125 RFL 145
R0934F.300k_Assembly_Contig_52_1 572 2126 RFL 145
Anapurna.300k_Assembly_Contig_53_1 573 2127 RFL 145
Primepii.300k_Assembly_Contig_34_2 574 2128 RFL 145 R197.300k_Assembly_Contig_44_2
575 2129 RFL 145 Wheat-Rye-6R.300k_Assembly_Contig_36_2 576 2130 RFL 146
Anapurna.300k_Assembly_Contig_46_3 577 2131 RFL 146 R197.300k_Assembly_Contig_147_3
578 2132 RFL 146 Wheat-Rye-6R.300k_Assembly_Contig_18_3 579 2133 RFL 146
R0932E.300k_Assembly_Contig_42_3 580 2134 RFL 147
Anapurna.300k_Assembly_Contig_36_4 581 2135 RFL 147
R0934F.300k_Assembly_Contig_35_3 582 2136 RFL 147
Anapurna.300k_Assembly_Contig_36_3 583 2137 RFL 147 Wheat-Rye-
6R.300k_Assembly_Contig_43_3 584 2138 RFL 147 R0932E.300k_Assembly_Contig_49_3 585
2139 RFL 147 R197.300k_Assembly_Contig_2_6 586 2140 RFL 147
Primepii.300k_Assembly_Contig_32_3 587 2141 RFL 148 R197.300k_Assembly_Contig_28_2
588 2142 RFL 148 R0934F.300k_Assembly_Contig_13_2 589 2143 RFL 148
Primepii.300k_Assembly_Contig_4_2 590 2144 RFL 148 R0932E.300k_Assembly_Contig_218_1
591 2145 RFL 149 Triticum- 592 2146 timopheevii.300k_Assembly_Contig_18_2 RFL 150
Triticum- 593 2147 timopheevii.300k_Assembly_Contig_66_3 RFL 151
R0934F.300k_Assembly_Contig_215_1 594 2148 RFL 151

Primepii.300k_Assembly_Contig_119_1 595 2149 RFL 151 R197.300k_Assembly_Contig_114_1
596 2150 RFL 152 Triticum- 597 2151 timopheevii.300k_Assembly_Contig_114_1 RFL 152
R0934F.300k_Assembly_Contig_177_1 598 2152 RFL 153
R0932E.300k_Assembly_Contig_121_1 599 2153 RFL 153
Anapurna.300k_Assembly_Contig_124_1 600 2154 RFL 153
Primepii.300k_Assembly_Contig_126_1 601 2155 RFL 153
R0934F.300k_Assembly_Contig_206_1 602 2156 RFL 153
R0934F.300k_Assembly_Contig_147_1 603 2157 RFL 154
R0934F.300k_Assembly_Contig_174_1 604 2158 RFL 154
Primepii.300k_Assembly_Contig_143_1 605 2159 RFL 154
Primepii.300k_Assembly_Contig_120_1 606 2160 RFL 154
R0934F.300k_Assembly_Contig_134_1 607 2161 RFL 154
Primepii.300k_Assembly_Contig_125_1 608 2162 RFL 154 Wheat-Rye-
6R.300k_Assembly_Contig_42_1 609 2163 RFL 154 R0934F.300k_Assembly_Contig_139_1 610
2164 RFL 154 R197.300k_Assembly_Contig_123_1 611 2165 RFL 154
R0932E.300k_Assembly_Contig_168_1 612 2166 RFL 154
R0932E.300k_Assembly_Contig_37_1 613 2167 RFL 154
Anapurna.300k_Assembly_Contig_135_1 614 2168 RFL 154
R197.300k_Assembly_Contig_140_1 615 2169 RFL 154 R197.300k_Assembly_Contig_161_1
616 2170 RFL 154 Anapurna.300k_Assembly_Contig_112_1 617 2171 RFL 154 Wheat-Rye-
6R.300k_Assembly_Contig_160_1 618 2172 RFL 154 R0934F.300k_Assembly_Contig_182_1
619 2173 RFL 154 R0934F.300k_Assembly_Contig_200_1 620 2174 RFL 154 Wheat-Rye-
6R.300k_Assembly_Contig_105_1 621 2175 RFL 154 Primepii.300k_Assembly_Contig_176_1
622 2176 RFL 154 R0934F.300k_Assembly_Contig_165_1 623 2177 RFL 154 Wheat-Rye-
6R.300k_Assembly_Contig_200_1 624 2178 RFL 154 R197.300k_Assembly_Contig_13_1 625
2179 RFL 154 R197.300k_Assembly_Contig_158_1 626 2180 RFL 154
Primepii.300k_Assembly_Contig_160_1 627 2181 RFL 154
R0932E.300k_Assembly_Contig_156_1 628 2182 RFL 154
R0932E.300k_Assembly_Contig_179_1 629 2183 RFL 154
Anapurna.300k_Assembly_Contig_128_1 630 2184 RFL 154
Anapurna.300k_Assembly_Contig_104_1 631 2185 RFL 154
R0932E.300k_Assembly_Contig_177_1 632 2186 RFL 155 R197.300k_Assembly_Contig_127_1
633 2187 RFL 155 R0934F.300k_Assembly_Contig_109_1 634 2188 RFL 155 Wheat-Rye-
6R.300k_Assembly_Contig_121_1 635 2189 RFL 155 Anapurna.300k_Assembly_Contig_108_1
636 2190 RFL 155 R0932E.300k_Assembly_Contig_122_1 637 2191 RFL 155
Primepii.300k_Assembly_Contig_110_1 638 2192 RFL 156 Wheat-Rye-
6R.300k_Assembly_Contig_130_1 639 2193 RFL 156 Anapurna.300k_Assembly_Contig_117_2
640 2194 RFL 156 R0934F.300k_Assembly_Contig_176_1 641 2195 RFL 156
Anapurna.300k_Assembly_Contig_171_1 642 2196 RFL 156
R0932E.300k_Assembly_Contig_154_2 643 2197 RFL 156
Primepii.300k_Assembly_Contig_134_2 644 2198 RFL 157
Primepii.300k_Assembly_Contig_108_1 645 2199 RFL 157 R197.300k_Assembly_Contig_97_1
646 2200 RFL 157 Wheat-Rye-6R.300k_Assembly_Contig_117_1 647 2201 RFL 157
Anapurna.300k_Assembly_Contig_105_1 648 2202 RFL 157
R0932E.300k_Assembly_Contig_126_1 649 2203 RFL 157
R0934F.300k_Assembly_Contig_116_1 650 2204 RFL 158 R197.300k_Assembly_Contig_105_3
651 2205 RFL 158 Anapurna.300k_Assembly_Contig_89_3 652 2206 RFL 158 Wheat-Rye-
6R.300k_Assembly_Contig_88_3 653 2207 RFL 158 R0934F.300k_Assembly_Contig_83_3 654
2208 RFL 158 Primepii.300k_Assembly_Contig_77_3 655 2209 RFL 158
R0932E.300k_Assembly_Contig_80_3 656 2210 RFL 159 Wheat-Rye-

6R.300k_Assembly_Contig_177_1 657 2211 RFL 159 Wheat-Rye-
6R.300k_Assembly_Contig_185_1 658 2212 RFL 159 Anapurna.300k_Assembly_Contig_176_1
659 2213 RFL 159 R197.300k_Assembly_Contig_191_1 660 2214 RFL 160 Triticum- 661 2215
timopheevii.300k_Assembly_Contig_111_1 RFL 161 R197.300k_Assembly_Contig_28_3 662
2216 RFL 161 R0934F.300k_Assembly_Contig_13_3 663 2217 RFL 161
Anapurna.300k_Assembly_Contig_27_3 664 2218 RFL 161 R0932E.300k_Assembly_Contig_9_3
665 2219 RFL 161 Primepii.300k_Assembly_Contig_4_3 666 2220 RFL 161
R0932E.300k_Assembly_Contig_218_2 667 2221 RFL 161 Wheat-Rye-
6R.300k_Assembly_Contig_4_4 668 2222 RFL 162 Triticum- 669 2223
timopheevii.300k_Assembly_Contig_94_1 RFL 163 Primepii.300k_Assembly_Contig_128_1 670
2224 RFL 163 Wheat-Rye-6R.300k_Assembly_Contig_133_1 671 2225 RFL 163
R197.300k_Assembly_Contig_133_2 672 2226 RFL 163 R0932E.300k_Assembly_Contig_133_2
673 2227 RFL 163 R0934F.300k_Assembly_Contig_125_1 674 2228 RFL 163
Anapurna.300k_Assembly_Contig_122_2 675 2229 RFL 164
Primepii.300k_Assembly_Contig_127_1 676 2230 RFL 164
R0934F.300k_Assembly_Contig_124_1 677 2231 RFL 165
Anapurna.300k_Assembly_Contig_145_1 678 2232 RFL 165
R197.300k_Assembly_Contig_149_1 679 2233 RFL 165 Primepii.300k_Assembly_Contig_139_1
680 2234 RFL 165 R0932E.300k_Assembly_Contig_140_1 681 2235 RFL 165 Wheat-Rye-
6R.300k_Assembly_Contig_139_1 682 2236 RFL 165 R0934F.300k_Assembly_Contig_132_1
683 2237 RFL 166 Primepii.300k_Assembly_Contig_130_1 684 2238 RFL 166
R0934F.300k_Assembly_Contig_122_2 685 2239 RFL 167 R197.300k_Assembly_Contig_118_1
686 2240 RFL 167 Anapurna.300k_Assembly_Contig_93_1 687 2241 RFL 167
R0932E.300k_Assembly_Contig_96_1 688 2242 RFL 167 R0934F.300k_Assembly_Contig_100_1
689 2243 RFL 167 Wheat-Rye-6R.300k_Assembly_Contig_107_1 690 2244 RFL 167
Primepii.300k_Assembly_Contig_100_1 691 2245 RFL 168 Triticum- 692 2246
timopheevii.300k_Assembly_Contig_88_1 RFL 168 Triticum- 693 2247
timopheevii.300k_Assembly_Contig_28_1 RFL 169 Triticum- 694 2248
timopheevii.300k_Assembly_Contig_86_2 RFL 170 R197.300k_Assembly_Contig_94_2 695
2249 RFL 170 R0934F.300k_Assembly_Contig_67_2 696 2250 RFL 170
Primepii.300k_Assembly_Contig_60_2 697 2251 RFL 170 Wheat-Rye-
6R.300k_Assembly_Contig_113_2 698 2252 RFL 170 Anapurna.300k_Assembly_Contig_174_3
699 2253 RFL 171 Triticum- 700 2254 timopheevii.300k_Assembly_Contig_84_1 RFL 171
Triticum- 701 2255 timopheevii.300k_Assembly_Contig_46_1 RFL 172
R0932E.300k_Assembly_Contig_67_1 702 2256 RFL 172 Wheat-Rye-
6R.300k_Assembly_Contig_57_1 703 2257 RFL 172 R0934F.300k_Assembly_Contig_82_1 704
2258 RFL 172 Primepii.300k_Assembly_Contig_68_1 705 2259 RFL 172
Anapurna.300k_Assembly_Contig_32_1 706 2260 RFL 172 R197.300k_Assembly_Contig_61_1
707 2261 RFL 173 Triticum- 708 2262 timopheevii.300k_Assembly_Contig_83_1 RFL 174
Triticum- 709 2263 timopheevii.300k_Assembly_Contig_41_1 RFL 174
R197.300k_Assembly_Contig_102_1 710 2264 RFL 174 Triticum- 711 2265
timopheevii.300k_Assembly_Contig_41_2 RFL 175 Primepii.300k_Assembly_Contig_200_1 712
2266 RFL 175 Primepii.300k_Assembly_Contig_162_1 713 2267 RFL 175
R0932E.300k_Assembly_Contig_144_1 714 2268 RFL 175
R0934F.300k_Assembly_Contig_156_1 715 2269 RFL 175
Anapurna.300k_Assembly_Contig_123_1 716 2270 RFL 176 R197.300k_Assembly_Contig_93_1
717 2271 RFL 176 R0934F.300k_Assembly_Contig_90_1 718 2272 RFL 176 Wheat-Rye-
6R.300k_Assembly_Contig_100_1 719 2273 RFL 176 Anapurna.300k_Assembly_Contig_86_1
720 2274 RFL 176 R0932E.300k_Assembly_Contig_108_1 721 2275 RFL 177
R197.300k_Assembly_Contig_128_1 722 2276 RFL 177 Wheat-Rye-

6R.300k_Assembly_Contig_126_1 723 2277 RFL 177 Anapurna.300k_Assembly_Contig_127_1
724 2278 RFL 177 R0932E.300k_Assembly_Contig_135_1 725 2279 RFL 178
Primepii.300k_Assembly_Contig_97_1 726 2280 RFL 179 Triticum- 727 2281
timopheevii.300k_Assembly_Contig_89_1 RFL 180 R0932E.300k_Assembly_Contig_160_1 728
2282 RFL 180 Triticum- 729 2283 timopheevii.300k_Assembly_Contig_97_1 RFL 180
Primepii.300k_Assembly_Contig_141_1 730 2284 RFL 180 R197.300k_Assembly_Contig_141_1
731 2285 RFL 180 Wheat-Rye-6R.300k_Assembly_Contig_149_1 732 2286 RFL 180
Anapurna.300k_Assembly_Contig_142_1 733 2287 RFL 180
R0934F.300k_Assembly_Contig_146_1 734 2288 RFL 181 Wheat-Rye-
6R.300k_Assembly_Contig_104_1 735 2289 RFL 181 Primepii.300k_Assembly_Contig_104_1
736 2290 RFL 182 Wheat-Rye-6R.300k_Assembly_Contig_53_2 737 2291 RFL 182
R197.300k_Assembly_Contig_19_2 738 2292 RFL 182 R0932E.300k_Assembly_Contig_68_2
739 2293 RFL 183 R0932E.300k_Assembly_Contig_31_2 740 2294 RFL 183
R197.300k_Assembly_Contig_49_2 741 2295 RFL 183 R0934F.300k_Assembly_Contig_41_2
742 2296 RFL 183 Wheat-Rye-6R.300k_Assembly_Contig_27_2 743 2297 RFL 183 Triticum- 744
2298 timopheevii.300k_Assembly_Contig_26_1 RFL 183 Anapurna.300k_Assembly_Contig_40_2
745 2299 RFL 183 Primepii.300k_Assembly_Contig_65_2 746 2300 RFL 184
R0932E.300k_Assembly_Contig_63_1 747 2301 RFL 184 R197.300k_Assembly_Contig_35_1
748 2302 RFL 184 Anapurna.300k_Assembly_Contig_170_1 749 2303 RFL 184 Wheat-Rye-
6R.300k_Assembly_Contig_61_1 750 2304 RFL 185 R0934F.300k_Assembly_Contig_92_3 751
2305 RFL 185 Triticum- 752 2306 timopheevii.300k_Assembly_Contig_52_3 RFL 185
R0932E.300k_Assembly_Contig_109_3 753 2307 RFL 185 R197.300k_Assembly_Contig_90_3
754 2308 RFL 186 Wheat-Rye-6R.300k_Assembly_Contig_131_1 755 2309 RFL 186
R0932E.300k_Assembly_Contig_141_1 756 2310 RFL 186 R197.300k_Assembly_Contig_143_1
757 2311 RFL 186 R0934F.300k_Assembly_Contig_133_1 758 2312 RFL 186
Anapurna.300k_Assembly_Contig_139_1 759 2313 RFL 186
Primepii.300k_Assembly_Contig_144_1 760 2314 RFL 187
R0934F.300k_Assembly_Contig_53_3 761 2315 RFL 187 Primepii.300k_Assembly_Contig_28_3
762 2316 RFL 187 R197.300k_Assembly_Contig_53_3 763 2317 RFL 187
R0932E.300k_Assembly_Contig_14_3 764 2318 RFL 187 Wheat-Rye-
6R.300k_Assembly_Contig_49_3 765 2319 RFL 187 Anapurna.300k_Assembly_Contig_55_3 766
2320 RFL 189 Triticum- 767 2321 timopheevii.300k_Assembly_Contig_23_1 RFL 191
R197.300k_Assembly_Contig_136_1 768 2322 RFL 192 R0934F.300k_Assembly_Contig_119_1
769 2323 RFL 192 R197.300k_Assembly_Contig_134_1 770 2324 RFL 192 Wheat-Rye-
6R.300k_Assembly_Contig_125_1 771 2325 RFL 192 Primepii.300k_Assembly_Contig_133_1
772 2326 RFL 192 Anapurna.300k_Assembly_Contig_125_1 773 2327 RFL 192
R0932E.300k_Assembly_Contig_136_1 774 2328 RFL 193
R0932E.300k_Assembly_Contig_130_1 775 2329 RFL 193 R0934F.300k_Assembly_Contig_95_1
776 2330 RFL 193 Anapurna.300k_Assembly_Contig_107_1 777 2331 RFL 193
R197.300k_Assembly_Contig_110_1 778 2332 RFL 194 Primepii.300k_Assembly_Contig_107_1
779 2333 RFL 194 R197.300k_Assembly_Contig_99_1 780 2334 RFL 194 Wheat-Rye-
6R.300k_Assembly_Contig_156_1 781 2335 RFL 194 R197.300k_Assembly_Contig_132_1 782
2336 RFL 194 Primepii.300k_Assembly_Contig_121_1 783 2337 RFL 194 Wheat-Rye-
6R.300k_Assembly_Contig_51_1 784 2338 RFL 195 Wheat-Rye-
6R.300k_Assembly_Contig_165_1 785 2339 RFL 195 R0934F.300k_Assembly_Contig_163_1
786 2340 RFL 195 R0932E.300k_Assembly_Contig_95_1 787 2341 RFL 195
Primepii.300k_Assembly_Contig_171_1 788 2342 RFL 195
Anapurna.300k_Assembly_Contig_2_4 789 2343 RFL 195 R197.300k_Assembly_Contig_176_1
790 2344 RFL 196 Anapurna.300k_Assembly_Contig_81_2 791 2345 RFL 196
R197.300k_Assembly_Contig_88_2 792 2346 RFL 196 Wheat-Rye-

6R.300k_Assembly_Contig_114_2 793 2347 RFL 196 R0934F.300k_Assembly_Contig_112_2
794 2348 RFL 196 Primepii.300k_Assembly_Contig_103_2 795 2349 RFL 196
R0932E.300k_Assembly_Contig_106_2 796 2350 RFL 197 Wheat-Rye-
6R.300k_Assembly_Contig_89_1 797 2351 RFL 197 R0932E.300k_Assembly_Contig_84_1 798
2352 RFL 197 R197.300k_Assembly_Contig_89_1 799 2353 RFL 197
Primepii.300k_Assembly_Contig_78_1 800 2354 RFL 197
Anapurna.300k_Assembly_Contig_65_1 801 2355 RFL 197
R0934F.300k_Assembly_Contig_105_1 802 2356 RFL 198 Wheat-Rye-
6R.300k_Assembly_Contig_155_1 803 2357 RFL 198 R0934F.300k_Assembly_Contig_151_1
804 2358 RFL 198 R197.300k_Assembly_Contig_157_1 805 2359 RFL 198
Anapurna.300k_Assembly_Contig_150_1 806 2360 RFL 198
Primepii.300k_Assembly_Contig_157_1 807 2361 RFL 198
R0932E.300k_Assembly_Contig_169_1 808 2362 RFL 199
R0934F.300k_Assembly_Contig_170_1 809 2363 RFL 199
Anapurna.300k_Assembly_Contig_166_1 810 2364 RFL 199
R0932E.300k_Assembly_Contig_181_1 811 2365 RFL 199 R197.300k_Assembly_Contig_178_1
812 2366 RFL 199 Primepii.300k_Assembly_Contig_175_2 813 2367 RFL 199
Primepii.300k_Assembly_Contig_175_1 814 2368 RFL 199 Triticum- 815 2369
timopheevii.300k_Assembly_Contig_110_1 RFL 199 Wheat-Rye-
6R.300k_Assembly_Contig_171_1 816 2370 RFL 200 Wheat-Rye-
6R.300k_Assembly_Contig_154_1 817 2371 RFL 200 Primepii.300k_Assembly_Contig_149_1
818 2372 RFL 200 R197.300k_Assembly_Contig_148_1 819 2373 RFL 200
Anapurna.300k_Assembly_Contig_151_1 820 2374 RFL 200
R0932E.300k_Assembly_Contig_158_1 821 2375 RFL 200
R0934F.300k_Assembly_Contig_135_1 822 2376 RFL 201
Anapurna.300k_Assembly_Contig_156_1 823 2377 RFL 201 Wheat-Rye-
6R.300k_Assembly_Contig_152_1 824 2378 RFL 201 R0934F.300k_Assembly_Contig_162_1
825 2379 RFL 201 Triticum- 826 2380 timopheevii.300k_Assembly_Contig_95_1 RFL 201
Primepii.300k_Assembly_Contig_153_1 827 2381 RFL 201 R197.300k_Assembly_Contig_162_1
828 2382 RFL 201 R0932E.300k_Assembly_Contig_159_1 829 2383 RFL 202
Anapurna.300k_Assembly_Contig_157_1 830 2384 RFL 202 Wheat-Rye-
6R.300k_Assembly_Contig_164_1 831 2385 RFL 202 R197.300k_Assembly_Contig_166_1 832
2386 RFL 202 R0932E.300k_Assembly_Contig_165_1 833 2387 RFL 202
R0934F.300k_Assembly_Contig_154_1 834 2388 RFL 202
Primepii.300k_Assembly_Contig_167_1 835 2389 RFL 203 Triticum- 836 2390
timopheevii.300k_Assembly_Contig_105_1 RFL 203 Triticum- 837 2391
timopheevii.300k_Assembly_Contig_119_1 RFL 203 Wheat-Rye-
6R.300k_Assembly_Contig_147_1 838 2392 RFL 203 R0932E.300k_Assembly_Contig_173_2
839 2393 RFL 203 R0932E.300k_Assembly_Contig_173_1 840 2394 RFL 203 Triticum- 841 2395
timopheevii.300k_Assembly_Contig_3_1 RFL 203 Triticum- 842 2396
timopheevii.300k_Assembly_Contig_120_1 RFL 203 R197.300k_Assembly_Contig_150_1 843
2397 RFL 203 Anapurna.300k_Assembly_Contig_153_1 844 2398 RFL 203
R0934F.300k_Assembly_Contig_157_2 845 2399 RFL 203
Primepii.300k_Assembly_Contig_156_1 846 2400 RFL 203
R0934F.300k_Assembly_Contig_157_1 847 2401 RFL 204 Triticum- 848 2402
timopheevii.300k_Assembly_Contig_103_1 RFL 204 Triticum- 849 2403
timopheevii.300k_Assembly_Contig_100_1 RFL 205 Triticum- 850 2404
timopheevii.300k_Assembly_Contig_96_1 RFL 205 R0932E.300k_Assembly_Contig_155_1 851
2405 RFL 205 Anapurna.300k_Assembly_Contig_140_1 852 2406 RFL 205
Primepii.300k_Assembly_Contig_148_1 853 2407 RFL 205 R197.300k_Assembly_Contig_151_1

854 2408 RFL 205 R0934F.300k_Assembly_Contig_159_1 855 2409 RFL 205 Wheat-Rye-
6R.300k_Assembly_Contig_151_1 856 2410 RFL 206 R0932E.300k_Assembly_Contig_117_1
857 2411 RFL 207 Triticum- 858 2412 timopheevii.300k_Assembly_Contig_62_1 RFL 208
Wheat-Rye-6R.300k_Assembly_Contig_136_1 859 2413 RFL 209 Wheat-Rye-
6R.300k_Assembly_Contig_143_1 860 2414 RFL 209 R0934F.300k_Assembly_Contig_140_1
861 2415 RFL 209 R0932E.300k_Assembly_Contig_174_1 862 2416 RFL 209
R197.300k_Assembly_Contig_154_1 863 2417 RFL 209 Primepii.300k_Assembly_Contig_152_1
864 2418 RFL 209 Anapurna.300k_Assembly_Contig_141_1 865 2419 RFL 210 Wheat-Rye-
6R.300k_Assembly_Contig_163_1 866 2420 RFL 210 Anapurna.300k_Assembly_Contig_149_1
867 2421 RFL 210 R0934F.300k_Assembly_Contig_152_1 868 2422 RFL 210
R197.300k_Assembly_Contig_168_1 869 2423 RFL 210 R0932E.300k_Assembly_Contig_172_1
870 2424 RFL 210 Primepii.300k_Assembly_Contig_166_1 871 2425 RFL 211 Triticum- 872
2426 timopheevii.300k_Assembly_Contig_99_1 RFL 212 R0932E.300k_Assembly_Contig_99_1
873 2427 RFL 212 R197.300k_Assembly_Contig_85_2 874 2428 RFL 212
R0932E.300k_Assembly_Contig_97_2 875 2429 RFL 212 Triticum- 876 2430
timopheevii.300k_Assembly_Contig_73_1 RFL 212 R0934F.300k_Assembly_Contig_94_2 877
2431 RFL 212 Wheat-Rye-6R.300k_Assembly_Contig_91_1 878 2432 RFL 212
Primepii.300k_Assembly_Contig_76_2 879 2433 RFL 212 R197.300k_Assembly_Contig_109_1
880 2434 RFL 212 Primepii.300k_Assembly_Contig_99_1 881 2435 RFL 212
R0934F.300k_Assembly_Contig_107_1 882 2436 RFL 212 Wheat-Rye-
6R.300k_Assembly_Contig_68_2 883 2437 RFL 212 Anapurna.300k_Assembly_Contig_96_1 884
2438 RFL 212 Triticum- 885 2439 timopheevii.300k_Assembly_Contig_132_1 RFL 212
Anapurna.300k_Assembly_Contig_75_2 886 2440 RFL 213
Primepii.300k_Assembly_Contig_61_2 887 2441 RFL 213 Wheat-Rye-
6R.300k_Assembly_Contig_64_2 888 2442 RFL 213 Anapurna.300k_Assembly_Contig_69_2 889
2443 RFL 213 R0934F.300k_Assembly_Contig_65_2 890 2444 RFL 213
R197.300k_Assembly_Contig_48_2 891 2445 RFL 213 R0932E.300k_Assembly_Contig_70_2
892 2446 RFL 215 Triticum- 893 2447 timopheevii.300k_Assembly_Contig_98_1 RFL 216
R197.300k_Assembly_Contig_96_2 894 2448 RFL 216 R0932E.300k_Assembly_Contig_93_2
895 2449 RFL 216 Anapurna.300k_Assembly_Contig_126_1 896 2450 RFL 216
R0934F.300k_Assembly_Contig_102_2 897 2451 RFL 217 Triticum- 898 2452
timopheevii.300k_Assembly_Contig_16_2 RFL 218 Primepii.300k_Assembly_Contig_95_2 899
2453 RFL 218 Wheat-Rye-6R.300k_Assembly_Contig_166_1 900 2454 RFL 218
R197.300k_Assembly_Contig_112_2 901 2455 RFL 219 R0932E.300k_Assembly_Contig_105_1
902 2456 RFL 219 Wheat-Rye-6R.300k_Assembly_Contig_110_1 903 2457 RFL 219
R197.300k_Assembly_Contig_87_1 904 2458 RFL 219 Anapurna.300k_Assembly_Contig_91_1
905 2459 RFL 219 R0934F.300k_Assembly_Contig_101_1 906 2460 RFL 219
Primepii.300k_Assembly_Contig_98_1 907 2461 RFL 220
Primepii.300k_Assembly_Contig_115_2 908 2462 RFL 220 Wheat-Rye-
6R.300k_Assembly_Contig_108_2 909 2463 RFL 221 R0934F.300k_Assembly_Contig_47_2 910
2464 RFL 221 R0932E.300k_Assembly_Contig_33_2 911 2465 RFL 221 Wheat-Rye-
6R.300k_Assembly_Contig_16_2 912 2466 RFL 221 Anapurna.300k_Assembly_Contig_30_2 913
2467 RFL 221 R197.300k_Assembly_Contig_52_2 914 2468 RFL 221
Primepii.300k_Assembly_Contig_15_2 915 2469 RFL 222 Wheat-Rye-
6R.300k_Assembly_Contig_97_1 916 2470 RFL 222 R197.300k_Assembly_Contig_108_1 917
2471 RFL 222 Anapurna.300k_Assembly_Contig_101_1 918 2472 RFL 223
R0934F.300k_Assembly_Contig_58_2 919 2473 RFL 223
Anapurna.300k_Assembly_Contig_25_3 920 2474 RFL 223 Wheat-Rye-
6R.300k_Assembly_Contig_46_3 921 2475 RFL 223 R197.300k_Assembly_Contig_45_3 922
2476 RFL 223 R0932E.300k_Assembly_Contig_219_3 923 2477 RFL 223

Primepii.300k_Assembly_Contig_36_2 924 2478 RFL 223 R0932E.300k_Assembly_Contig_57_3
925 2479 RFL 224 R0932E.300k_Assembly_Contig_212_1 926 2480 RFL 225
R0932E.300k_Assembly_Contig_117_2 927 2481 RFL 226
R0932E.300k_Assembly_Contig_67_2 928 2482 RFL 226
Primepii.300k_Assembly_Contig_118_1 929 2483 RFL 226 Wheat-Rye-
6R.300k_Assembly_Contig_101_1 930 2484 RFL 226 Wheat-Rye-
6R.300k_Assembly_Contig_57_2 931 2485 RFL 226 R0934F.300k_Assembly_Contig_82_2 932
2486 RFL 226 Anapurna.300k_Assembly_Contig_95_1 933 2487 RFL 226
R0934F.300k_Assembly_Contig_98_1 934 2488 RFL 226 Primepii.300k_Assembly_Contig_68_2
935 2489 RFL 226 Anapurna.300k_Assembly_Contig_32_2 936 2490 RFL 226
R197.300k_Assembly_Contig_61_2 937 2491 RFL 227 R0934F.300k_Assembly_Contig_62_1
938 2492 RFL 227 Primepii.300k_Assembly_Contig_196_1 939 2493 RFL 228
R197.300k_Assembly_Contig_85_1 940 2494 RFL 228 Wheat-Rye-
6R.300k_Assembly_Contig_68_1 941 2495 RFL 228 R0934F.300k_Assembly_Contig_94_1 942
2496 RFL 228 R0932E.300k_Assembly_Contig_97_1 943 2497 RFL 228
Anapurna.300k_Assembly_Contig_75_1 944 2498 RFL 228
Primepii.300k_Assembly_Contig_76_1 945 2499 RFL 229 R0932E.300k_Assembly_Contig_30_3
946 2500 RFL 229 R197.300k_Assembly_Contig_26_5 947 2501 RFL 229
Anapurna.300k_Assembly_Contig_26_4 948 2502 RFL 229 Wheat-Rye-
6R.300k_Assembly_Contig_24_5 949 2503 RFL 229 Primepii.300k_Assembly_Contig_33_3 950
2504 RFL 229 R0934F.300k_Assembly_Contig_16_5 951 2505 RFL 230
R0932E.300k_Assembly_Contig_209_2 952 2506 RFL 231
R0932E.300k_Assembly_Contig_36_1 953 2507 RFL 231 R0934F.300k_Assembly_Contig_42_1
954 2508 RFL 231 R197.300k_Assembly_Contig_55_1 955 2509 RFL 231
Anapurna.300k_Assembly_Contig_44_1 956 2510 RFL 231 Wheat-Rye-
6R.300k_Assembly_Contig_33_1 957 2511 RFL 231 Primepii.300k_Assembly_Contig_54_1 958
2512 RFL 232 R0932E.300k_Assembly_Contig_171_1 959 2513 RFL 232
Primepii.300k_Assembly_Contig_173_1 960 2514 RFL 232 Wheat-Rye-
6R.300k_Assembly_Contig_168_1 961 2515 RFL 232 R197.300k_Assembly_Contig_177_1 962
2516 RFL 232 Anapurna.300k_Assembly_Contig_165_1 963 2517 RFL 232
R0934F.300k_Assembly_Contig_158_1 964 2518 RFL 234 Triticum- 965 2519
timopheevii.300k_Assembly_Contig_101_1 RFL 235 Triticum- 966 2520
timopheevii.300k_Assembly_Contig_75_1 RFL 236 R0934F.300k_Assembly_Contig_33_2 967
2521 RFL 236 R0932E.300k_Assembly_Contig_52_1 968 2522 RFL 236
Primepii.300k_Assembly_Contig_29_3 969 2523 RFL 236 Wheat-Rye-
6R.300k_Assembly_Contig_38_1 970 2524 RFL 236 R197.300k_Assembly_Contig_50_2 971
2525 RFL 236 Anapurna.300k_Assembly_Contig_34_2 972 2526 RFL 237
R0934F.300k_Assembly_Contig_118_1 973 2527 RFL 237 Triticum- 974 2528
timopheevii.300k_Assembly_Contig_82_1 RFL 238 Primepii.300k_Assembly_Contig_29_4 975
2529 RFL 238 R0934F.300k_Assembly_Contig_33_3 976 2530 RFL 238
Anapurna.300k_Assembly_Contig_34_3 977 2531 RFL 238
R0932E.300k_Assembly_Contig_52_2 978 2532 RFL 238 Wheat-Rye-
6R.300k_Assembly_Contig_38_2 979 2533 RFL 238 R197.300k_Assembly_Contig_50_3 980
2534 RFL 239 Anapurna.300k_Assembly_Contig_92_1 981 2535 RFL 239 Wheat-Rye-
6R.300k_Assembly_Contig_103_1 982 2536 RFL 240 Primepii.300k_Assembly_Contig_140_1
983 2537 RFL 240 R197.300k_Assembly_Contig_156_1 984 2538 RFL 240
R0932E.300k_Assembly_Contig_148_1 985 2539 RFL 240
R0934F.300k_Assembly_Contig_136_1 986 2540 RFL 241
Primepii.300k_Assembly_Contig_138_1 987 2541 RFL 241 Wheat-Rye-
6R.300k_Assembly_Contig_134_2 988 2542 RFL 241 R197.300k_Assembly_Contig_164_2 989

2543 RFL 241 R0932E.300k_Assembly_Contig_143_1 990 2544 RFL 241
Anapurna.300k_Assembly_Contig_136_1 991 2545 RFL 241
R0934F.300k_Assembly_Contig_137_1 992 2546 RFL 242
Anapurna.300k_Assembly_Contig_72_1 993 2547 RFL 243
Anapurna.300k_Assembly_Contig_8_2 994 2548 RFL 244
Anapurna.300k_Assembly_Contig_164_1 995 2549 RFL 245
R0934F.300k_Assembly_Contig_58_1 996 2550 RFL 245
Anapurna.300k_Assembly_Contig_25_2 997 2551 RFL 245
R0932E.300k_Assembly_Contig_57_2 998 2552 RFL 245 Wheat-Rye-
6R.300k_Assembly_Contig_46_2 999 2553 RFL 245 R197.300k_Assembly_Contig_45_2 1000
2554 RFL 245 Primepii.300k_Assembly_Contig_36_1 1001 2555 RFL 245
R0932E.300k_Assembly_Contig_219_2 1002 2556 RFL 246
Primepii.300k_Assembly_Contig_116_2 1003 2557 RFL 246
Anapurna.300k_Assembly_Contig_90_3 1004 2558 RFL 246
R197.300k_Assembly_Contig_125_2 1005 2559 RFL 246 Wheat-Rye-
6R.300k_Assembly_Contig_102_2 1006 2560 RFL 246 R0934F.300k_Assembly_Contig_96_2
1007 2561 RFL 246 R0932E.300k_Assembly_Contig_125_2 1008 2562 RFL 247
R0932E.300k_Assembly_Contig_53_1 1009 2563 RFL 247 R0934F.300k_Assembly_Contig_26_1
1010 2564 RFL 247 Primepii.300k_Assembly_Contig_40_1 1011 2565 RFL 247
Anapurna.300k_Assembly_Contig_29_1 1012 2566 RFL 247 R197.300k_Assembly_Contig_36_1
1013 2567 RFL 247 Wheat-Rye-6R.300k_Assembly_Contig_28_1 1014 2568 RFL 247 Triticum-
1015 2569 timopheevii.300k_Assembly_Contig_25_1 RFL 248
R0934F.300k_Assembly_Contig_183_1 1016 2570 RFL 248 Wheat-Rye-
6R.300k_Assembly_Contig_176_1 1017 2571 RFL 248 R0932E.300k_Assembly_Contig_180_1
1018 2572 RFL 249 Anapurna.300k_Assembly_Contig_130_1 1019 2573 RFL 249 Wheat-Rye-
6R.300k_Assembly_Contig_128_1 1020 2574 RFL 249 R197.300k_Assembly_Contig_144_1
1021 2575 RFL 249 R0934F.300k_Assembly_Contig_131_1 1022 2576 RFL 249
R0932E.300k_Assembly_Contig_137_1 1023 2577 RFL 249
Primepii.300k_Assembly_Contig_135_1 1024 2578 RFL 250 Wheat-Rye-
6R.300k_Assembly_Contig_47_2 1025 2579 RFL 250 Primepii.300k_Assembly_Contig_69_2
1026 2580 RFL 250 R0932E.300k_Assembly_Contig_69_2 1027 2581 RFL 250
Anapurna.300k_Assembly_Contig_56_2 1028 2582 RFL 250
R0934F.300k_Assembly_Contig_59_2 1029 2583 RFL 250 R197.300k_Assembly_Contig_91_2
1030 2584 RFL 251 R197.300k_Assembly_Contig_26_6 1031 2585 RFL 251 Triticum- 1032 2586
timopheevii.300k_Assembly_Contig_108_1 RFL 251 Anapurna.300k_Assembly_Contig_26_5
1033 2587 RFL 251 R0932E.300k_Assembly_Contig_30_4 1034 2588 RFL 251 Wheat-Rye-
6R.300k_Assembly_Contig_24_6 1035 2589 RFL 251 Primepii.300k_Assembly_Contig_33_4
1036 2590 RFL 251 R0934F.300k_Assembly_Contig_16_6 1037 2591 RFL 252
Primepii.300k_Assembly_Contig_189_1 1038 2592 RFL 252
R0934F.300k_Assembly_Contig_169_1 1039 2593 RFL 253 Wheat-Rye-
6R.300k_Assembly_Contig_112_2 1040 2594 RFL 253 R0932E.300k_Assembly_Contig_111_2
1041 2595 RFL 253 R0934F.300k_Assembly_Contig_130_2 1042 2596 RFL 253
Anapurna.300k_Assembly_Contig_88_2 1043 2597 RFL 254
R0932E.300k_Assembly_Contig_119_2 1044 2598 RFL 254
R0934F.300k_Assembly_Contig_108_2 1045 2599 RFL 254 Triticum- 1046 2600
timopheevii.300k_Assembly_Contig_77_2 RFL 254 R197.300k_Assembly_Contig_122_2 1047
2601 RFL 254 Wheat-Rye-6R.300k_Assembly_Contig_116_2 1048 2602 RFL 254
Anapurna.300k_Assembly_Contig_103_2 1049 2603 RFL 254
Primepii.300k_Assembly_Contig_101_2 1050 2604 RFL 255
R0934F.300k_Assembly_Contig_212_1 1051 2605 RFL 256 Wheat-Rye-

6R.300k_Assembly_Contig_76_1 1052 2606 RFL 256 R0934F.300k_Assembly_Contig_79_1
1053 2607 RFL 256 Triticum- 1054 2608 timopheevii.300k_Assembly_Contig_32_1 RFL 256
R0932E.300k_Assembly_Contig_81_1 1055 2609 RFL 256
Anapurna.300k_Assembly_Contig_67_1 1056 2610 RFL 256
Primepii.300k_Assembly_Contig_89_1 1057 2611 RFL 256 R197.300k_Assembly_Contig_74_1
1058 2612 RFL 257 Primepii.300k_Assembly_Contig_71_1 1059 2613 RFL 257 Wheat-Rye-
6R.300k_Assembly_Contig_78_1 1060 2614 RFL 257 R0932E.300k_Assembly_Contig_101_1
1061 2615 RFL 257 R0934F.300k_Assembly_Contig_85_1 1062 2616 RFL 257
R197.300k_Assembly_Contig_76_1 1063 2617 RFL 257 Anapurna.300k_Assembly_Contig_79_1
1064 2618 RFL 258 Primepii.300k_Assembly_Contig_41_1 1065 2619 RFL 259 Triticum- 1066
2620 timopheevii.300k_Assembly_Contig_76_2 RFL 260
Anapurna.300k_Assembly_Contig_155_1 1067 2621 RFL 260
R0932E.300k_Assembly_Contig_163_2 1068 2622 RFL 260
R0934F.300k_Assembly_Contig_160_1 1069 2623 RFL 260 Wheat-Rye-
6R.300k_Assembly_Contig_162_1 1070 2624 RFL 260 R197.300k_Assembly_Contig_171_1
1071 2625 RFL 260 Primepii.300k_Assembly_Contig_146_1 1072 2626 RFL 261 Triticum- 1073
2627 timopheevii.300k_Assembly_Contig_90_2 RFL 262
Primepii.300k_Assembly_Contig_172_1 1074 2628 RFL 262 Wheat-Rye-
6R.300k_Assembly_Contig_159_1 1075 2629 RFL 262 R197.300k_Assembly_Contig_167_1
1076 2630 RFL 263 R197.300k_Assembly_Contig_126_1 1077 2631 RFL 263 Wheat-Rye-
6R.300k_Assembly_Contig_157_1 1078 2632 RFL 263 Wheat-Rye-
6R.300k_Assembly_Contig_189_1 1079 2633 RFL 263 R0932E.300k_Assembly_Contig_132_1
1080 2634 RFL 263 Anapurna.300k_Assembly_Contig_114_1 1081 2635 RFL 264 Triticum- 1082
2636 timopheevii.300k_Assembly_Contig_64_1 RFL 265 R197.300k_Assembly_Contig_129_1
1083 2637 RFL 265 R0934F.300k_Assembly_Contig_120_1 1084 2638 RFL 265 Wheat-Rye-
6R.300k_Assembly_Contig_118_1 1085 2639 RFL 265 Anapurna.300k_Assembly_Contig_102_1
1086 2640 RFL 265 Triticum- 1087 2641 timopheevii.300k_Assembly_Contig_79_1 RFL 265
Primepii.300k_Assembly_Contig_123_1 1088 2642 RFL 265
R0932E.300k_Assembly_Contig_127_1 1089 2643 RFL 266 Triticum- 1090 2644
timopheevii.300k_Assembly_Contig_17_2 RFL 267 Triticum- 1091 2645
timopheevii.300k_Assembly_Contig_90_3 RFL 268 R197.300k_Assembly_Contig_90_4 1092
2646 RFL 268 Triticum- 1093 2647 timopheevii.300k_Assembly_Contig_52_4 RFL 268
R0934F.300k_Assembly_Contig_92_4 1094 2648 RFL 268
R0932E.300k_Assembly_Contig_109_4 1095 2649 RFL 269 Triticum- 1096 2650
timopheevii.300k_Assembly_Contig_84_2 RFL 270 Triticum- 1097 2651
timopheevii.300k_Assembly_Contig_56_1 RFL 271 R0932E.300k_Assembly_Contig_167_1 1098
2652 RFL 272 Triticum- 1099 2653 timopheevii.300k_Assembly_Contig_17_1 RFL 273
Primepii.300k_Assembly_Contig_84_2 1100 2654 RFL 273
R0932E.300k_Assembly_Contig_212_2 1101 2655 RFL 273
R0934F.300k_Assembly_Contig_62_2 1102 2656 RFL 274 Triticum- 1103 2657
timopheevii.300k_Assembly_Contig_87_1 RFL 275 Anapurna.300k_Assembly_Contig_208_1
1104 2658 RFL 276 R0934F.300k_Assembly_Contig_42_2 1105 2659 RFL 276
R197.300k_Assembly_Contig_55_2 1106 2660 RFL 276 R0932E.300k_Assembly_Contig_36_2
1107 2661 RFL 276 Primepii.300k_Assembly_Contig_54_2 1108 2662 RFL 276 Wheat-Rye-
6R.300k_Assembly_Contig_33_2 1109 2663 RFL 276 Anapurna.300k_Assembly_Contig_44_2
1110 2664 RFL 277 Triticum- 1111 2665 timopheevii.300k_Assembly_Contig_2_1 RFL 277
R0934F.300k_Assembly_Contig_3_1 1112 2666 RFL 278 Triticum- 1113 2667
timopheevii.300k_Assembly_Contig_85_2 RFL 279 Wheat-Rye-6R.300k_Assembly_Contig_56_2
1114 2668 RFL 279 R0934F.300k_Assembly_Contig_60_2 1115 2669 RFL 279
Anapurna.300k_Assembly_Contig_60_3 1116 2670 RFL 279

Primepii.300k_Assembly_Contig_56_3 1117 2671 RFL 279 R197.300k_Assembly_Contig_60_2
1118 2672 RFL 279 R0932E.300k_Assembly_Contig_2_2 1119 2673 RFL 280
R0934F.300k_Assembly_Contig_129_1 1120 2674 RFL 280 R197.300k_Assembly_Contig_139_1
1121 2675 RFL 280 Anapurna.300k_Assembly_Contig_129_1 1122 2676 RFL 280 Wheat-Rye-
6R.300k_Assembly_Contig_127_1 1123 2677 RFL 280 Primepii.300k_Assembly_Contig_136_2
1124 2678 RFL 280 R0932E.300k_Assembly_Contig_138_1 1125 2679 RFL 281 Triticum- 1126
2680 timopheevii.300k_Assembly_Contig_23_2 RFL 282 R197.300k_Assembly_Contig_181_1
1127 2681 RFL 282 Wheat-Rye-6R.300k_Assembly_Contig_186_1 1128 2682 RFL 282
R0932E.300k_Assembly_Contig_95_2 1129 2683 RFL 282
Primepii.300k_Assembly_Contig_177_1 1130 2684 RFL 282
R0934F.300k_Assembly_Contig_155_1 1131 2685 RFL 282 Wheat-Rye-
6R.300k_Assembly_Contig_119_1 1132 2686 RFL 282 R197.300k_Assembly_Contig_106_1
1133 2687 RFL 282 Primepii.300k_Assembly_Contig_161_1 1134 2688 RFL 282
Anapurna.300k_Assembly_Contig_160_1 1135 2689 RFL 282
R197.300k_Assembly_Contig_219_1 1136 2690 RFL 282
Anapurna.300k_Assembly_Contig_198_1 1137 2691 RFL 282
Anapurna.300k_Assembly_Contig_100_1 1138 2692 RFL 282 Wheat-Rye-
6R.300k_Assembly_Contig_170_1 1139 2693 RFL 283 R0934F.300k_Assembly_Contig_79_2
1140 2694 RFL 283 Wheat-Rye-6R.300k_Assembly_Contig_76_2 1141 2695 RFL 283 Triticum-
1142 2696 timopheevii.300k_Assembly_Contig_32_2 RFL 283
Primepii.300k_Assembly_Contig_89_2 1143 2697 RFL 283 R197.300k_Assembly_Contig_74_2
1144 2698 RFL 283 R0932E.300k_Assembly_Contig_81_2 1145 2699 RFL 283
Anapurna.300k_Assembly_Contig_67_2 1146 2700 RFL 284
Primepii.300k_Assembly_Contig_84_1 1147 2701 RFL 285
Primepii.300k_Assembly_Contig_111_2 1148 2702 RFL 286 Triticum- 1149 2703
timopheevii.300k_Assembly_Contig_55_2 RFL 287 Anapurna.300k_Assembly_Contig_154_2
1150 2704 RFL 287 Wheat-Rye-6R.300k_Assembly_Contig_142_2 1151 2705 RFL 287
R197.300k_Assembly_Contig_153_2 1152 2706 RFL 287
Primepii.300k_Assembly_Contig_158_2 1153 2707 RFL 288
Primepii.300k_Assembly_Contig_60_1 1154 2708 RFL 288
R0934F.300k_Assembly_Contig_67_1 1155 2709 RFL 289
R0932E.300k_Assembly_Contig_206_2 1156 2710 RFL 290
R0932E.300k_Assembly_Contig_166_2 1157 2711 RFL 290
R0934F.300k_Assembly_Contig_166_2 1158 2712 RFL 290
Primepii.300k_Assembly_Contig_174_2 1159 2713 RFL 290
R197.300k_Assembly_Contig_174_2 1160 2714 RFL 290
Anapurna.300k_Assembly_Contig_161_2 1161 2715 RFL 290 Wheat-Rye-
6R.300k_Assembly_Contig_161_2 1162 2716 RFL 291 Anapurna.300k_Assembly_Contig_27_2
1163 2717 RFL 291 R0932E.300k_Assembly_Contig_9_2 1164 2718 RFL 291 Wheat-Rye-
6R.300k_Assembly_Contig_4_3 1165 2719 RFL 292 Triticum- 1166 2720
timopheevii.300k_Assembly_Contig_68_2 RFL 293 R197.300k_Assembly_Contig_73_1 1167
2721 RFL 294 Primepii.300k_Assembly_Contig_116_1 1168 2722 RFL 294 Wheat-Rye-
6R.300k_Assembly_Contig_102_1 1169 2723 RFL 294 Anapurna.300k_Assembly_Contig_90_2
1170 2724 RFL 294 R197.300k_Assembly_Contig_125_1 1171 2725 RFL 294
R0934F.300k_Assembly_Contig_96_1 1172 2726 RFL 294
R0932E.300k_Assembly_Contig_125_1 1173 2727 RFL 295 Triticum- 1174 2728
timopheevii.300k_Assembly_Contig_112_1 RFL 296 R197.300k_Assembly_Contig_159_1 1175
2729 RFL 296 Wheat-Rye-6R.300k_Assembly_Contig_146_1 1176 2730 RFL 296
Primepii.300k_Assembly_Contig_154_1 1177 2731 RFL 296
R0934F.300k_Assembly_Contig_144_1 1178 2732 RFL 296

Anapurna.300k_Assembly_Contig_152_1 1179 2733 RFL 296
R0932E.300k_Assembly_Contig_149_1 1180 2734 RFL 297
R0932E.300k_Assembly_Contig_129_1 1181 2735 RFL 297
Anapurna.300k_Assembly_Contig_120_1 1182 2736 RFL 297
Primepii.300k_Assembly_Contig_129_1 1183 2737 RFL 297
R197.300k_Assembly_Contig_145_1 1184 2738 RFL 297 R0934F.300k_Assembly_Contig_121_1
1185 2739 RFL 297 Wheat-Rye-6R.300k_Assembly_Contig_129_1 1186 2740 RFL 298 Wheat-
Rye-6R.300k_Assembly_Contig_120_1 1187 2741 RFL 298
Anapurna.300k_Assembly_Contig_110_1 1188 2742 RFL 298
R197.300k_Assembly_Contig_124_1 1189 2743 RFL 298 R0932E.300k_Assembly_Contig_124_1
1190 2744 RFL 298 Primepii.300k_Assembly_Contig_124_1 1191 2745 RFL 298
R0934F.300k_Assembly_Contig_127_1 1192 2746 RFL 299 R197.300k_Assembly_Contig_195_2
1193 2747 RFL 299 R0934F.300k_Assembly_Contig_192_1 1194 2748 RFL 299 Wheat-Rye-
6R.300k_Assembly_Contig_192_1 1195 2749 RFL 300 Wheat-Rye-
6R.300k_Assembly_Contig_111_2 1196 2750 RFL 300 R197.300k_Assembly_Contig_121_2 1197
2751 RFL 300 Primepii.300k_Assembly_Contig_113_2 1198 2752 RFL 301
Anapurna.300k_Assembly_Contig_69_3 1199 2753 RFL 301
Primepii.300k_Assembly_Contig_61_3 1200 2754 RFL 301 R197.300k_Assembly_Contig_48_3
1201 2755 RFL 301 R0934F.300k_Assembly_Contig_65_3 1202 2756 RFL 301 Wheat-Rye-
6R.300k_Assembly_Contig_64_3 1203 2757 RFL 301 R0932E.300k_Assembly_Contig_70_3
1204 2758 RFL 302 Triticum- 1205 2759 timopheevii.300k_Assembly_Contig_24_2 RFL 303
Wheat-Rye-6R.300k_Assembly_Contig_199_1 1206 2760 RFL 303
Anapurna.300k_Assembly_Contig_195_1 1207 2761 RFL 304
Anapurna.300k_Assembly_Contig_120_2 1208 2762 RFL 304
R0932E.300k_Assembly_Contig_129_2 1209 2763 RFL 304
Primepii.300k_Assembly_Contig_129_2 1210 2764 RFL 304
R197.300k_Assembly_Contig_145_2 1211 2765 RFL 304 Wheat-Rye-
6R.300k_Assembly_Contig_129_2 1212 2766 RFL 304 R0934F.300k_Assembly_Contig_121_2
1213 2767 RFL 305 Wheat-Rye-6R.300k_Assembly_Contig_135_1 1214 2768 RFL 306
R197.300k_Assembly_Contig_107_2 1215 2769 RFL 306 R0934F.300k_Assembly_Contig_88_2
1216 2770 RFL 306 Anapurna.300k_Assembly_Contig_76_2 1217 2771 RFL 306
Primepii.300k_Assembly_Contig_74_2 1218 2772 RFL 306
R0932E.300k_Assembly_Contig_115_2 1219 2773 RFL 306 Wheat-Rye-
6R.300k_Assembly_Contig_81_2 1220 2774 RFL 307 Triticum- 1221 2775
timopheevii.300k_Assembly_Contig_104_1 RFL 308 R197.300k_Assembly_Contig_198_1 1222
2776 RFL 308 R0932E.300k_Assembly_Contig_178_2 1223 2777 RFL 308 Wheat-Rye-
6R.300k_Assembly_Contig_193_1 1224 2778 RFL 308 Anapurna.300k_Assembly_Contig_183_1
1225 2779 RFL 308 R0934F.300k_Assembly_Contig_175_1 1226 2780 RFL 308
Primepii.300k_Assembly_Contig_198_1 1227 2781 RFL 309 Wheat-Rye-
6R.300k_Assembly_Contig_52_2 1228 2782 RFL 309 Primepii.300k_Assembly_Contig_194_1
1229 2783 RFL 309 R0932E.300k_Assembly_Contig_51_1 1230 2784 RFL 309
R0934F.300k_Assembly_Contig_191_2 1231 2785 RFL 309 R197.300k_Assembly_Contig_190_2
1232 2786 RFL 310 Triticum- 1233 2787 timopheevii.300k_Assembly_Contig_19_1 RFL 311
Wheat-Rye-6R.300k_Assembly_Contig_123_1 1234 2788 RFL 311
Anapurna.300k_Assembly_Contig_118_1 1235 2789 RFL 311
R197.300k_Assembly_Contig_131_1 1236 2790 RFL 311 R0932E.300k_Assembly_Contig_128_1
1237 2791 RFL 312 Anapurna.300k_Assembly_Contig_146_1 1238 2792 RFL 312
R0932E.300k_Assembly_Contig_175_1 1239 2793 RFL 312 Wheat-Rye-
6R.300k_Assembly_Contig_158_1 1240 2794 RFL 312 R197.300k_Assembly_Contig_155_1
1241 2795 RFL 312 R0934F.300k_Assembly_Contig_153_1 1242 2796 RFL 312

Primepii.300k_Assembly_Contig_145_1 1243 2797 RFL 313
R0934F.300k_Assembly_Contig_59_1 1244 2798 RFL 313 R0932E.300k_Assembly_Contig_69_1
1245 2799 RFL 313 Wheat-Rye-6R.300k_Assembly_Contig_47_1 1246 2800 RFL 313
Primepii.300k_Assembly_Contig_69_1 1247 2801 RFL 313
Anapurna.300k_Assembly_Contig_56_1 1248 2802 RFL 313 R197.300k_Assembly_Contig_91_1
1249 2803 RFL 314 Anapurna.300k_Assembly_Contig_143_2 1250 2804 RFL 314
R197.300k_Assembly_Contig_169_2 1251 2805 RFL 314
R0932E.300k_Assembly_Contig_150_2 1252 2806 RFL 314 Wheat-Rye-
6R.300k_Assembly_Contig_148_2 1253 2807 RFL 314 Primepii.300k_Assembly_Contig_147_2
1254 2808 RFL 314 R0934F.300k_Assembly_Contig_148_2 1255 2809 RFL 315
R197.300k_Assembly_Contig_129_2 1256 2810 RFL 315 R0934F.300k_Assembly_Contig_120_2
1257 2811 RFL 315 Wheat-Rye-6R.300k_Assembly_Contig_118_2 1258 2812 RFL 315
Primepii.300k_Assembly_Contig_123_2 1259 2813 RFL 315
Anapurna.300k_Assembly_Contig_102_2 1260 2814 RFL 315 Triticum- 1261 2815
timopheevii.300k_Assembly_Contig_79_2 RFL 315 R0932E.300k_Assembly_Contig_127_2 1262
2816 RFL 316 R197.300k_Assembly_Contig_208_2 1263 2817 RFL 317
R197.300k_Assembly_Contig_9_2 1264 2818 RFL 317 Anapurna.300k_Assembly_Contig_18_2
1265 2819 RFL 317 Primepii.300k_Assembly_Contig_5_2 1266 2820 RFL 317
R0934F.300k_Assembly_Contig_19_2 1267 2821 RFL 317 Wheat-Rye-
6R.300k_Assembly_Contig_14_2 1268 2822 RFL 317 R0932E.300k_Assembly_Contig_16_2
1269 2823 RFL 318 Anapurna.300k_Assembly_Contig_57_1 1270 2824 RFL 319
Primepii.300k_Assembly_Contig_117_1 1271 2825 RFL 319
R0934F.300k_Assembly_Contig_123_1 1272 2826 RFL 320
R0932E.300k_Assembly_Contig_105_2 1273 2827 RFL 320 Wheat-Rye-
6R.300k_Assembly_Contig_110_2 1274 2828 RFL 320 Anapurna.300k_Assembly_Contig_91_2
1275 2829 RFL 320 R197.300k_Assembly_Contig_87_2 1276 2830 RFL 320
R0934F.300k_Assembly_Contig_101_2 1277 2831 RFL 320
Primepii.300k_Assembly_Contig_98_2 1278 2832 RFL 321
R0934F.300k_Assembly_Contig_105_2 1279 2833 RFL 321
R0932E.300k_Assembly_Contig_84_2 1280 2834 RFL 321 Wheat-Rye-
6R.300k_Assembly_Contig_89_2 1281 2835 RFL 321 R197.300k_Assembly_Contig_89_2 1282
2836 RFL 321 Primepii.300k_Assembly_Contig_78_2 1283 2837 RFL 321
Anapurna.300k_Assembly_Contig_65_2 1284 2838 RFL 322
R0934F.300k_Assembly_Contig_47_1 1285 2839 RFL 322
Primepii.300k_Assembly_Contig_15_1 1286 2840 RFL 323
R0932E.300k_Assembly_Contig_21_1 1287 2841 RFL 323 R0934F.300k_Assembly_Contig_25_1
1288 2842 RFL 323 R197.300k_Assembly_Contig_14_1 1289 2843 RFL 323
Anapurna.300k_Assembly_Contig_6_1 1290 2844 RFL 323
Primepii.300k_Assembly_Contig_20_1 1291 2845 RFL 323 Wheat-Rye-
6R.300k_Assembly_Contig_8_1 1292 2846 RFL 324 Anapurna.300k_Assembly_Contig_178_1
1293 2847 RFL 324 Triticum- 1294 2848 timopheevii.300k_Assembly_Contig_116_1 RFL 324
R197.300k_Assembly_Contig_193_1 1295 2849 RFL 324
R0932E.300k_Assembly_Contig_191_1 1296 2850 RFL 324 Wheat-Rye-
6R.300k_Assembly_Contig_184_1 1297 2851 RFL 324 Primepii.300k_Assembly_Contig_186_1
1298 2852 RFL 324 R0934F.300k_Assembly_Contig_188_1 1299 2853 RFL 325
Primepii.300k_Assembly_Contig_74_1 1300 2854 RFL 325 R197.300k_Assembly_Contig_107_1
1301 2855 RFL 325 Wheat-Rye-6R.300k_Assembly_Contig_81_1 1302 2856 RFL 325
R0934F.300k_Assembly_Contig_88_1 1303 2857 RFL 325
R0932E.300k_Assembly_Contig_115_1 1304 2858 RFL 325
Anapurna.300k_Assembly_Contig_76_1 1305 2859 RFL 326

R0932E.300k_Assembly_Contig_77_1 1306 2860 RFL 326 Wheat-Rye-
6R.300k_Assembly_Contig_75_1 1307 2861 RFL 326 R197.300k_Assembly_Contig_103_1 1308
2862 RFL 326 R0934F.300k_Assembly_Contig_72_1 1309 2863 RFL 327
Primepii.300k_Assembly_Contig_192_1 1310 2864 RFL 327 R197.300k_Assembly_Contig_1921
1311 2865 RFL 327 Wheat-Rye-6R.300k_Assembly_Contig_187_1 1312 2866 RFL 327
R0932E.300k_Assembly_Contig_190_1 1313 2867 RFL 327
R0934F.300k_Assembly_Contig_185_1 1314 2868 RFL 327
Anapurna.300k_Assembly_Contig_182_1 1315 2869 RFL 328
Primepii.300k_Assembly_Contig_95_1 1316 2870 RFL 328 R197.300k_Assembly_Contig_112_1
1317 2871 RFL 328 Wheat-Rye-6R.300k_Assembly_Contig_173_1 1318 2872 RFL 329 Triticum-
1319 2873 timopheevii.300k_Assembly_Contig_88_2 RFL 329 Wheat-Rye-
6R.300k_Assembly_Contig_124_1 1320 2874 RFL 329 Anapurna.300k_Assembly_Contig_115_1
1321 2875 RFL 329 Primepii.300k_Assembly_Contig_122_1 1322 2876 RFL 329
R197.300k_Assembly_Contig_130_1 1323 2877 RFL 329 R0934F.300k_Assembly_Contig_114_1
1324 2878 RFL 330 Wheat-Rye-6R.300k_Assembly_Contig_141_2 1325 2879 RFL 330
R197.300k_Assembly_Contig_146_2 1326 2880 RFL 330
Primepii.300k_Assembly_Contig_155_1 1327 2881 RFL 330
Anapurna.300k_Assembly_Contig_138_2 1328 2882 RFL 330 Triticum- 1329 2883
timopheevii.300k_Assembly_Contig_92_1 RFL 330 R0934F.300k_Assembly_Contig_142_2 1330
2884 RFL 330 R0932E.300k_Assembly_Contig_146_2 1331 2885 RFL 331
R0932E.300k_Assembly_Contig_26_1 1332 2886 RFL 331 R197.300k_Assembly_Contig_41_1
1333 2887 RFL 331 R0934F.300k_Assembly_Contig_38_1 1334 2888 RFL 331 Wheat-Rye-
6R.300k_Assembly_Contig_50_1 1335 2889 RFL 331 Primepii.300k_Assembly_Contig_45_1
1336 2890 RFL 331 Anapurna.300k_Assembly_Contig_45_1 1337 2891 RFL 332 Triticum- 1338
2892 timopheevii.300k_Assembly_Contig_72_1 RFL 333 R0934F.300k_Assembly_Contig_53_2
1339 2893 RFL 333 Primepii.300k_Assembly_Contig_28_2 1340 2894 RFL 333 Wheat-Rye-
6R.300k_Assembly_Contig_49_2 1341 2895 RFL 333 R0932E.300k_Assembly_Contig_14_2
1342 2896 RFL 333 R197.300k_Assembly_Contig_53_2 1343 2897 RFL 333
Anapurna.300k_Assembly_Contig_55_2 1344 2898 RFL 334
R197.300k_Assembly_Contig_208_1 1345 2899 RFL 335 Wheat-Rye-
6R.300k_Assembly_Contig_112_1 1346 2900 RFL 335 R0934F.300k_Assembly_Contig_130_1
1347 2901 RFL 335 R0932E.300k_Assembly_Contig_111_1 1348 2902 RFL 335
Anapurna.300k_Assembly_Contig_88_1 1349 2903 RFL 335
R0932E.300k_Assembly_Contig_206_1 1350 2904 RFL 336 Triticum- 1351 2905
timopheevii.300k_Assembly_Contig_18_1 RFL 337 R197.300k_Assembly_Contig_184_1 1352
2906 RFL 337 Triticum- 1353 2907 timopheevii.300k_Assembly_Contig_115_1 RFL 337
R0932E.300k_Assembly_Contig_188_1 1354 2908 RFL 337
Primepii.300k_Assembly_Contig_185_1 1355 2909 RFL 337
R0934F.300k_Assembly_Contig_186_1 1356 2910 RFL 337
Anapurna.300k_Assembly_Contig_173_1 1357 2911 RFL 337 Wheat-Rye-
6R.300k_Assembly_Contig_182_1 1358 2912 RFL 338 Primepii.300k_Assembly_Contig_111_1
1359 2913 RFL 339 Anapurna.300k_Assembly_Contig_36_2 1360 2914 RFL 339 Wheat-Rye-
6R.300k_Assembly_Contig_43_2 1361 2915 RFL 339 R0934F.300k_Assembly_Contig_35_2
1362 2916 RFL 339 R197.300k_Assembly_Contig_2_5 1363 2917 RFL 339
R0932E.300k_Assembly_Contig_49_2 1364 2918 RFL 339
Primepii.300k_Assembly_Contig_32_2 1365 2919 RFL 340 Triticum- 1366 2920
timopheevii.300k_Assembly_Contig_91_1 RFL 341 R0932E.300k_Assembly_Contig_118_1 1367
2921 RFL 341 R197.300k_Assembly_Contig_104_1 1368 2922 RFL 341
Anapurna.300k_Assembly_Contig_97_1 1369 2923 RFL 341 Wheat-Rye-
6R.300k_Assembly_Contig_94_1 1370 2924 RFL 342 Triticum- 1371 2925

timopheevii.300k_Assembly_Contig_11_1 RFL 343 Primepii.300k_Assembly_Contig_179_1 1372
2926 RFL 344 Triticum- 1373 2927 timopheevii.300k_Assembly_Contig_128_1 RFL 345
R0932E.300k_Assembly_Contig_120_1 1374 2928 RFL 346 Wheat-Rye-
6R.300k_Assembly_Contig_142_1 1375 2929 RFL 346 R197.300k_Assembly_Contig_153_1
1376 2930 RFL 346 Primepii.300k_Assembly_Contig_158_1 1377 2931 RFL 346
Anapurna.300k_Assembly_Contig_154_1 1378 2932 RFL 347
Anapurna.300k_Assembly_Contig_9_2 1379 2933 RFL 347
R0932E.300k_Assembly_Contig_10_2 1380 2934 RFL 347 R0934F.300k_Assembly_Contig_21_2
1381 2935 RFL 347 Primepii.300k_Assembly_Contig_48_2 1382 2936 RFL 347
R197.300k_Assembly_Contig_2_2 1383 2937 RFL 347 Wheat-Rye-
6R.300k_Assembly_Contig_20_2 1384 2938 RFL 348 Triticum- 1385 2939
timopheevii.300k_Assembly_Contig_58_2 RFL 349 Primepii.300k_Assembly_Contig_45_2 1386
2940 RFL 349 R197.300k_Assembly_Contig_41_2 1387 2941 RFL 349
R0932E.300k_Assembly_Contig_26_2 1388 2942 RFL 349 R0934F.300k_Assembly_Contig_38_2
1389 2943 RFL 349 Anapurna.300k_Assembly_Contig_45_2 1390 2944 RFL 349 Wheat-Rye-
6R.300k_Assembly_Contig_50_2 1391 2945 RFL 350 Anapurna.300k_Assembly_Contig_177_1
1392 2946 RFL 350 R0934F.300k_Assembly_Contig_184_1 1393 2947 RFL 350
R0932E.300k_Assembly_Contig_186_1 1394 2948 RFL 350
R197.300k_Assembly_Contig_187_1 1395 2949 RFL 350 Wheat-Rye-
6R.300k_Assembly_Contig_181_1 1396 2950 RFL 351 R0934F.300k_Assembly_Contig_93_1
1397 2951 RFL 351 Primepii.300k_Assembly_Contig_88_1 1398 2952 RFL 352 Triticum- 1399
2953 timopheevii.300k_Assembly_Contig_68_1 RFL 353 R197.300k_Assembly_Contig_138_2
1400 2954 RFL 353 Wheat-Rye-6R.300k_Assembly_Contig_122_2 1401 2955 RFL 353
Anapurna.300k_Assembly_Contig_119_2 1402 2956 RFL 353
Primepii.300k_Assembly_Contig_131_2 1403 2957 RFL 353
R0932E.300k_Assembly_Contig_131_2 1404 2958 RFL 353
R0934F.300k_Assembly_Contig_113_2 1405 2959 RFL 354
R0932E.300k_Assembly_Contig_106_1 1406 2960 RFL 354
Anapurna.300k_Assembly_Contig_81_1 1407 2961 RFL 354 R197.300k_Assembly_Contig_88_1
1408 2962 RFL 354 Wheat-Rye-6R.300k_Assembly_Contig_114_1 1409 2963 RFL 354
R0934F.300k_Assembly_Contig_112_1 1410 2964 RFL 354
Primepii.300k_Assembly_Contig_103_1 1411 2965 RFL 355
Anapurna.300k_Assembly_Contig_133_1 1412 2966 RFL 356 Triticum- 1413 2967
timopheevii.300k_Assembly_Contig_91_2 RFL 357 Wheat-Rye-6R.300k_Assembly_Contig_56_1
1414 2968 RFL 357 Anapurna.300k_Assembly_Contig_60_2 1415 2969 RFL 357
Primepii.300k_Assembly_Contig_56_2 1416 2970 RFL 357
R0934F.300k_Assembly_Contig_60_1 1417 2971 RFL 357 R0932E.300k_Assembly_Contig_2_1
1418 2972 RFL 357 R197.300k_Assembly_Contig_60_1 1419 2973 RFL 358
R0934F.300k_Assembly_Contig_193_1 1420 2974 RFL 358 Wheat-Rye-
6R.300k_Assembly_Contig_188_1 1421 2975 RFL 358 R197.300k_Assembly_Contig_197_1
1422 2976 RFL 358 Primepii.300k_Assembly_Contig_190_1 1423 2977 RFL 358
R0932E.300k_Assembly_Contig_194_1 1424 2978 RFL 358
Anapurna.300k_Assembly_Contig_184_1 1425 2979 RFL 359
Primepii.300k_Assembly_Contig_210_1 1426 2980 RFL 359
R0934F.300k_Assembly_Contig_208_1 1427 2981 RFL 360 Wheat-Rye-
6R.300k_Assembly_Contig_153_1 1428 2982 RFL 360 R197.300k_Assembly_Contig_160_1
1429 2983 RFL 361 Wheat-Rye-6R.300k_Assembly_Contig_97_2 1430 2984 RFL 361
Primepii.300k_Assembly_Contig_91_2 1431 2985 RFL 361
R0932E.300k_Assembly_Contig_110_2 1432 2986 RFL 361
R0934F.300k_Assembly_Contig_86_2 1433 2987 RFL 361

Anapurna.300k_Assembly_Contig_101_2 1434 2988 RFL 361
R197.300k_Assembly_Contig_108_2 1435 2989 RFL 362 Triticum- 1436 2990
timopheevii.300k_Assembly_Contig_61_1 RFL 363 R0932E.300k_Assembly_Contig_31_1 1437
2991 RFL 363 R197.300k_Assembly_Contig_49_1 1438 2992 RFL 363
R0934F.300k_Assembly_Contig_41_1 1439 2993 RFL 363
Anapurna.300k_Assembly_Contig_40_1 1440 2994 RFL 363 Wheat-Rye-
6R.300k_Assembly_Contig_27_1 1441 2995 RFL 363 Primepii.300k_Assembly_Contig_65_1
1442 2996 RFL 364 Wheat-Rye-6R.300k_Assembly_Contig_141_3 1443 2997 RFL 364
R197.300k_Assembly_Contig_146_3 1444 2998 RFL 364 Triticum- 1445 2999
timopheevii.300k_Assembly_Contig_92_2 RFL 364 Anapurna.300k_Assembly_Contig_138_3
1446 3000 RFL 364 R0934F.300k_Assembly_Contig_142_3 1447 3001 RFL 364
R0932E.300k_Assembly_Contig_146_3 1448 3002 RFL 364
Primepii.300k_Assembly_Contig_155_2 1449 3003 RFL 365
R0934F.300k_Assembly_Contig_5_3 1450 3004 RFL 365 Primepii.300k_Assembly_Contig_17_2
1451 3005 RFL 365 R197.300k_Assembly_Contig_5_2 1452 3006 RFL 366 Wheat-Rye-
6R.300k_Assembly_Contig_144_1 1453 3007 RFL 366 Anapurna.300k_Assembly_Contig_148_1
1454 3008 RFL 366 Primepii.300k_Assembly_Contig_142_1 1455 3009 RFL 366
R0932E.300k_Assembly_Contig_147_1 1456 3010 RFL 366
R197.300k_Assembly_Contig_170_1 1457 3011 RFL 366 R0934F.300k_Assembly_Contig_168_1
1458 3012 RFL 367 R197.300k_Assembly_Contig_138_1 1459 3013 RFL 367 Wheat-Rye-
6R.300k_Assembly_Contig_122_1 1460 3014 RFL 367 Anapurna.300k_Assembly_Contig_119_1
1461 3015 RFL 367 R0932E.300k_Assembly_Contig_131_1 1462 3016 RFL 367
Primepii.300k_Assembly_Contig_131_1 1463 3017 RFL 367
R0934F.300k_Assembly_Contig_113_1 1464 3018 RFL 368
R0932E.300k_Assembly_Contig_63_2 1465 3019 RFL 368 R197.300k_Assembly_Contig_35_2
1466 3020 RFL 368 Anapurna.300k_Assembly_Contig_170_2 1467 3021 RFL 368 Wheat-Rye-
6R.300k_Assembly_Contig_61_2 1468 3022 RFL 369 Anapurna.300k_Assembly_Contig_159_1
1469 3023 RFL 369 R0932E.300k_Assembly_Contig_162_1 1470 3024 RFL 369
R197.300k_Assembly_Contig_165_1 1471 3025 RFL 369 Wheat-Rye-
6R.300k_Assembly_Contig_138_1 1472 3026 RFL 369 R0934F.300k_Assembly_Contig_164_1
1473 3027 RFL 369 Primepii.300k_Assembly_Contig_168_1 1474 3028 RFL 370
R0932E.300k_Assembly_Contig_28_1 1475 3029 RFL 370
Primepii.300k_Assembly_Contig_26_1 1476 3030 RFL 370
Anapurna.300k_Assembly_Contig_7_1 1477 3031 RFL 371
Anapurna.300k_Assembly_Contig_117_1 1478 3032 RFL 371
R0932E.300k_Assembly_Contig_154_1 1479 3033 RFL 371
Primepii.300k_Assembly_Contig_134_1 1480 3034 RFL 372
R0932E.300k_Assembly_Contig_131_3 1481 3035 RFL 372
R0934F.300k_Assembly_Contig_113_3 1482 3036 RFL 373
Primepii.300k_Assembly_Contig_211_1 1483 3037 RFL 373
R197.300k_Assembly_Contig_206_1 1484 3038 RFL 373 R0934F.300k_Assembly_Contig_205_1
1485 3039 RFL 374 Triticum- 1486 3040 timopheevii.300k_Assembly_Contig_103_2 RFL 375
R0932E.300k_Assembly_Contig_42_2 1487 3041 RFL 375
Anapurna.300k_Assembly_Contig_46_2 1488 3042 RFL 375
R197.300k_Assembly_Contig_147_2 1489 3043 RFL 375 Wheat-Rye-
6R.300k_Assembly_Contig_18_2 1490 3044 RFL 376 R0932E.300k_Assembly_Contig_119_3
1491 3045 RFL 376 R197.300k_Assembly_Contig_122_3 1492 3046 RFL 376
R0934F.300k_Assembly_Contig_108_3 1493 3047 RFL 376 Wheat-Rye-
6R.300k_Assembly_Contig_116_3 1494 3048 RFL 376 Anapurna.300k_Assembly_Contig_103_3
1495 3049 RFL 376 Primepii.300k_Assembly_Contig_101_3 1496 3050 RFL 377

R0932E.300k_Assembly_Contig_74_2 1497 3051 RFL 378 R197.300k_Assembly_Contig_202_1
1498 3052 RFL 378 R0932E.300k_Assembly_Contig_197_1 1499 3053 RFL 378 Wheat-Rye-
6R.300k_Assembly_Contig_196_1 1500 3054 RFL 378 Anapurna.300k_Assembly_Contig_187_1
1501 3055 RFL 378 R0934F.300k_Assembly_Contig_199_1 1502 3056 RFL 378
Primepii.300k_Assembly_Contig_205_1 1503 3057 RFL 379
Primepii.300k_Assembly_Contig_223_1 1504 3058 RFL 379 Wheat-Rye-
6R.300k_Assembly_Contig_179_1 1505 3059 RFL 379 Anapurna.300k_Assembly_Contig_193_1
1506 3060 RFL 379 R0932E.300k_Assembly_Contig_178_1 1507 3061 RFL 380 Triticum- 1508
3062 timopheevii.300k_Assembly_Contig_85_1 RFL 381 R197.300k_Assembly_Contig_169_1
1509 3063 RFL 381 Wheat-Rye-6R.300k_Assembly_Contig_148_1 1510 3064 RFL 381
R0932E.300k_Assembly_Contig_150_1 1511 3065 RFL 381
Primepii.300k_Assembly_Contig_147_1 1512 3066 RFL 381
R0934F.300k_Assembly_Contig_148_1 1513 3067 RFL 381
Anapurna.300k_Assembly_Contig_143_1 1514 3068 RFL 382 Triticum- 1515 3069
timopheevii.300k_Assembly_Contig_109_1 RFL 383 Wheat-Rye-
6R.300k_Assembly_Contig_111_1 1516 3070 RFL 383 R197.300k_Assembly_Contig_121_1
1517 3071 RFL 383 Primepii.300k_Assembly_Contig_113_1 1518 3072 RFL 384 Triticum- 1519
3073 timopheevii.300k_Assembly_Contig_117_1 RFL 385
Anapurna.300k_Assembly_Contig_167_1 1520 3074 RFL 385
R0932E.300k_Assembly_Contig_182_1 1521 3075 RFL 385 Wheat-Rye-
6R.300k_Assembly_Contig_175_1 1522 3076 RFL 385 R0934F.300k_Assembly_Contig_172_1
1523 3077 RFL 385 Primepii.300k_Assembly_Contig_170_1 1524 3078 RFL 385
R197.300k_Assembly_Contig_175_1 1525 3079 RFL 386
Primepii.300k_Assembly_Contig_105_1 1526 3080 RFL 386
R0934F.300k_Assembly_Contig_56_1 1527 3081 RFL 387
Primepii.300k_Assembly_Contig_159_2 1528 3082 RFL 388 Wheat-Rye-
6R.300k_Assembly_Contig_175_2 1529 3083 RFL 388 Anapurna.300k_Assembly_Contig_167_2
1530 3084 RFL 388 R0934F.300k_Assembly_Contig_172_2 1531 3085 RFL 388
Primepii.300k_Assembly_Contig_170_2 1532 3086 RFL 388
R0932E.300k_Assembly_Contig_182_2 1533 3087 RFL 388
R197.300k_Assembly_Contig_175_2 1534 3088 RFL 389 R0934F.300k_Assembly_Contig_74_1
1535 3089 RFL 389 Anapurna.300k_Assembly_Contig_64_1 1536 3090 RFL 389 Wheat-Rye-
6R.300k_Assembly_Contig_70_1 1537 3091 RFL 389 Primepii.300k_Assembly_Contig_79_1
1538 3092 RFL 390 Wheat-Rye-6R.300k_Assembly_Contig_217_1 1539 3093 RFL 390
Anapurna.300k_Assembly_Contig_203_1 1540 3094 RFL 391
R197.300k_Assembly_Contig_159_2 1541 3095 RFL 391 Wheat-Rye-
6R.300k_Assembly_Contig_146_2 1542 3096 RFL 391 Primepii.300k_Assembly_Contig_154_2
1543 3097 RFL 391 R0934F.300k_Assembly_Contig_144_2 1544 3098 RFL 391
Anapurna.300k_Assembly_Contig_152_2 1545 3099 RFL 391
R0932E.300k_Assembly_Contig_149_2 1546 3100 RFL 392
R0934F.300k_Assembly_Contig_118_2 1547 3101 RFL 392 Triticum- 1548 3102
timopheevii.300k_Assembly_Contig_82_2 RFL 393 Primepii.300k_Assembly_Contig_115_3 1549
3103 RFL 393 Wheat-Rye-6R.300k_Assembly_Contig_108_3 1550 3104 RFL 394
R0934F.300k_Assembly_Contig_222_1 1551 3105 RFL 395
Primepii.300k_Assembly_Contig_159_1 1552 3106 RFL 396 Triticum- 1553 3107
timopheevii.300k_Assembly_Contig_58_1 RFL 397 Triticum- 1554 3133
timopheevii.300k_Assembly_Contig_46_2

Example 3: Mapping of the Genes Encoding Candidate Rf Proteins in the Chromosomal Intervals
Associated with Fertility Restoration

(41) A. Fine-Mapping of the Genomic Region Containing Rf1 Genetic Determinants

(42) Three F2 mapping populations segregating for Rf1 (R197xKalahari, R204xAlixan and R0932ExAltigo) encompassing 210, 218 and 212 individuals respectively were phenotyped and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform.

(43) Fertility tests were conducted indoors under controlled growth conditions, either in growth chambers or in greenhouses, enabling normal fertility of the tested wheat plants. The fertility scores indicated have been calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets. The t-tests conducted were done by comparing the fertility scores of F1s made with a restorer and the fertility scores of a panel of elite inbred lines grown under the same conditions.

(44) Rf1 was first mapped on the short arm of the chromosome 1A between 4 cM and 10.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1087371 and cfn0530841. These two SNP markers delimit the largest possible interval defined by the three mapping populations.

(45) Subsequently, joint analysis of the three mapping populations and phenotyping of the individual F2 recombinant plants on derived F3 families validated the QTL position and delimited the Rf1 interval between 7 CM and 8.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1082074 and cfn0523990. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the wheat genome reference physical map. The left border (cfn1082074) was anchored on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the right border (cfn0523990) was anchored on the IWGSCWGAV02_1AS_scaffold47238 scaffold.

(46) Next, the locus was fine-mapped by screening 2976 and 3072 F3 lines from R197xKalahari and R204xAlixan derived from F2 plants heterozygous at the locus. Phenotyping and analysis of recombinant plant progenies within the interval redefined a smaller mapping interval between 7.5 and 8.8 CM delimited by cfn0522096 and cfn0527067 SNP markers on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the IWGSCWGAV02_1AS_scaffold47238 scaffold, respectively.

(47) B. Fine-Mapping of the Genomic Region Containing Rf3 Genetic Determinants

(48) Three F2 mapping populations (TJB155xAnapurna, 2852xAltamira, and AH46xR0946E) encompassing 217, 135, and 246 individuals respectively and a doubled-haploid (DH) population (H46xR934F) consisting of 140 individual plants segregating for Rf3 were phenotyped as described in example 1, and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform. Rf3 was first mapped on the short arm of the chromosome 1B between 18.9 CM and 24.2 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn0554333 and cfn0560679. These two SNP markers delimit the largest possible interval defined by the four mapping populations.

(49) Subsequently, joint analysis of the four mapping populations and validation of the phenotype of the individual F2/DH recombinant plants on derived F3 families validated the QTL, genetically delimited the locus between 22.2 cM and 22.7 cM on Limagrain's internal consensus map, and physically delimited the Rf3 interval between SNP markers cfn0436720 and cfn0238384. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the physical map. The left border (cfn0436720) was anchored on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the right border (cfn0238384) was anchored on the IWGSCWGAV02_1BS_scaffold5117 scaffold.

(50) Next, the locus was fine-mapped by screening 2496 and 672 plants from TJB155xAnapurna and AH46xR0946E F2 plants heterozygous at the locus. Analysis of recombinant F3 plant progenies within the interval redefined a smaller mapping interval between 22.5 and 22.7 cM delimited by cfn1249269 and BS00090770 SNP markers on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the IWGSCWGAV02_1BS_scaffold5117 scaffold, respectively.

(51) C. Mapping of the Genomic Region Containing Rf7 Genetic Determinants

(52) We crossed R197 and Primepii and then derived a population of 176 plants from individuals that were rf1 and rf3, i.e. not carrying the restorer alleles at the loci Rf1 and Rf3. The plants were genotyped with 18100 SNP markers using Limagrain's internal genotyping platform and phenotyped as described in example 1. We mapped the Rf7 locus on chromosome 7BL. Moreover, internal genotyping data showing a strong genetic divergence suggests the presence of an exotic chromosomal fragment which is stably transmitted through generations. We identified a large QTL ranging from 45 cM to 88 cM on chromosome 7B on Limagrain's internal consensus map with a peak on 46.7 cM (cfn0919993 with LOD score of 3.37E-40). Initial analysis of the recombinant plants suggests the Rf7 gene could be located between cfn3407185 and W90K_RAC875_c33564_120 markers delimiting a mapping interval of 0.3 cM between 46.7 cM and 47 cM on Limagrain's internal consensus map.

Example 4: Identification of Candidate Orthologous RFL Groups

(53) For each of the 282 RFL groups identified in example 2, the captured RFL ORFs (in the following referred to as protein sequences) were identified and the total number of RFL protein sequences is reported for each accession (Table 7, Table 8).

(54) Only the following RFL clusters will be taken into consideration: 1. RFL cluster contains protein representatives for all seven accessions and the sequences show polymorphism and/or length differences. 2. RFL cluster contains representatives for only these accessions for which genetic characterization indicated that they may contain the same Rf gene or genes.

(55) Tables 8A, 9A, 10 and 11 show the lists of the orthologous RFL groups correlating respectively with Rf1, Rf3, Rf7 and Rf-Rye-6R genes after the first screen. *T. timopheevii* is known to be fertile and, as a consequence, to restore T-CMS. This line is added here as it could contain any of the target Rf1, Rf3, Rf7 or Rf-rye genes.

(56) Finally, only the orthologous RFL groups mapping in the chromosomal interval in wheat *Triticum aestivum* Chinese Spring reference genome are considered as candidates for further analyses. These orthologous RFL groups will be selected as "candidate Rf groups".

(57) Mapping was achieved using the tool tblastn from the BLAST+ Suite

(https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE_TYPE=BlastDocs&DOC_TYPE=Download).

Specific parameters (-evalue 1e-25-best_hit_score_edge 0.05-best_hit_overhang 0.25) were used in order to keep all best hits.

(58) A. Results for Rf1 Accessions:

(59) Table 8A shows the orthologous RFL groups comprising at least one sequence captured from an accession characterized as bearing the Rf1 gene (R197, R0932E and *T. timopheevii*). The mapping described in example 3 allows us to discard orthologous RFL groups mapped outside of the chromosomal interval genetically associated with Rf1 fertility on the short arm of chromosome 1A. In this way, four RFLs clusters (79, 104, 185 and 268) were identified as potentially corresponding to the Rf protein encoded by the Rf1 gene.

(60) All protein sequences from group RFL185 contain ~500 amino acids and only 8.5 PPR motifs. Typically, full-length functional RFL proteins are expected to contain 15-20 PPR motifs. In addition, the last PPR motif of RFL185 is composed of only 15 amino acids. This indicates that RFL185 is truncated. RFL268 is also truncated (382 amino acids). Detailed sequence analysis has shown that RFL185 and RFL268 are remnants of the same gene that was split by a frameshift. Thus, both proteins are unlikely to be functional.

(61) Hence RFL79 and RFL104 are considered as being the best candidate Rf groups for Rf1.

(62) TABLE-US-00012 TABLE 8A Selection of RFL based on accession CMS information for Rf1. Mapping Restorer from Rye Restorer Positionned CMS genotype MAINTAINER Rf3 Rf1 + Rf7 Rf1 Rf3 introgression *Triticum* in Rf1 mapping RFLGene ANAPURNA PRIMEPII R197 R0932E R0934F Wheat-Rye-6R *timopheevii* interval RFL1 0 0 1 1 0 0 0 NO RFL56 0 0 1 1 0 0 1 NO RFL59 0 0 1 2 0 0 1 NO RFL73 0 0 1 1 0 0 1 NO RFL74 0 0 3 4 3 0 0 NO RFL79 0 0 1 1 1 0 1

YES RFL93 0 0 1 1 0 0 0 NO RFL104 0 0 1 1 1 0 1 YES RFL129 0 0 1 1 1 0 0 NO RFL185 0 0 1 1 1 0 1 YES RFL268 0 0 1 1 1 0 1 YES

(63) It can be noted from Table 8A that the only accession lacking Rf1 containing sequences in these candidate Rf groups is the accession R0934F.

(64) TABLE-US-00013 TABLE 8B presents the proteins in the candidate Rf groups 79 and 104.

RFL Cluster 79 79 length ORF name 0 0 808aa R197.300k_Assembly_Contig_120_1 1 1 808aa

R0932E.300k_Assembly_Contig_103_1 2 2 808aa R0934F.300k_Assembly_Contig_80_1 3 3

808aa Triticum- timopheevii.300k_Assembly_Contig_57_1 RFL Cluster 104 104 length ORF

name 0 0 757aa R197.300k_Assembly_Contig_72_1 1 1 757aa

R0932E.300k_Assembly_Contig_82_1 2 2 757aa R0934F.300k_Assembly_Contig_69_1 3 3 757aa

Triticum timopheevii.300k_Assembly_Contig_35_1

(65) The DNA sequences derived from the contigs identified in example 2 and encoding RFL proteins from candidate Rf groups 79 and 104 were aligned with BWA-MEM software (Li H. and Durbin R., 2010). It was observed that these sequences differ in the 5' UTR region in R0934F compared to R0932E and R197. One hypothesis is that the DNA sequences in R0934F were generated by a recombination event between the DNA sequences from candidate Rf groups 79 and 104. This recombined sequence may not be functional in R0934F as this line is only known to carry Rf3.

(66) B. Results for Rf3 Accessions:

(67) The same rationale as for the Rf1 accessions was applied to the accessions (Primepii and R0934F) characterized as carrying the Rf3 gene. Table 9A shows that orthologous RFL groups 67, 89, 140, 166 and 252 are candidate Rf groups for the Rf protein encoded by Rf3 as they contain proteins identified in Primepii and R0934, the two accessions characterized as carrying Rf3 restorer gene and are located in the mapped genetic interval on the short arm on chromosome 1B.

(68) TABLE-US-00014 TABLE 9A selection of RFL clusters based on germplasm CMS information for Rf3 Mapping Restorer from Rye Restorer Positionned CMS genotype

MAINTAINER Rf3 Rf1 + Rf7 Rf1 Rf3 introgression *Triticum* in Rf3 mapping RFLGene

ANAPURNA PRIMEPII R197 R0932E R0934F Wheat-Rye-6R *timopheevii* interval RFL22 0 1 0

0 1 0 0 NO RFL67 0 4 0 0 3 0 0 YES* RFL89 0 2 0 0 1 0 0 YES RFL140 0 1 0 0 1 0 0 YES

RFL142 0 1 0 0 1 0 0 NO RFL164 0 1 0 0 1 0 0 NO RFL166 0 1 0 0 1 0 0 YES RFL227 0 1 0 0 1 0

0 NO RFL252 0 1 0 0 1 0 0 YES

(69) In order to achieve an exhaustive analysis for selection of Rf3 candidates, all 282 RFL clusters were carefully analyzed with regard to number of RFLs, their length and their origin in relation to Rf3 genotype information. RFL clusters composed of RFL sequences from multiple accessions were screened for full-length protein sequences originating only from Primepii and R0934F genotypes and partial/shorter sequences from the non-Rf3-carrying genotypes. This analysis allowed the identification of four additional Rf3-candidate RFL clusters: RFL28, RFL29, RFL60 and RFL170.

(70) TABLE-US-00015 TABLE 9B details of the proteins present in RFL cluster 28, 29, 60 and 170 including protein length (aa = amino acids) and name. Cluster RFL 28 28 Rf3 0 0 323aa

Anapurna.300k_Assembly_Contig_2_3 1 1 479aa Anapurna.300k_Assembly_Contig_2_2 2 2

857aa Primepii.300k_Assembly_Contig_13_1 3 3 323aa R197.300k_Assembly_Contig_10_3 4 4

479aa R197.300k_Assembly_Contig_10_2 5 5 479aa R0932E.300k_Assembly_Contig_23_2 6 6

323aa R0932E.300k_Assembly_Contig_23_3 7 7 857aa R0934F.300k_Assembly_Contig_17_1 8 8

323aa Wheat-Rye- 6R.300k_Assembly_Contig_7_3 9 9 479aa Wheat-Rye-

6R.300k_Assembly_Contig_7_2 Cluster RFL 29 29 Rf3 0 0 828aa

Primepii.300k_Assembly_Contig_67_1 1 1 828aa R0934F.300k_Assembly_Contig_78_1 3 2

536aa Wheat-Rye- 6R.300k_Assembly_Contig_77_2 4 3 295aa Wheat-Rye-

6R.300k_Assembly_Contig_77_1 Cluster RFL 60 60 Rf3 0 0 828aa

Primepii.300k_Assembly_Contig_94_1 1 1 287aa R197.300k_Assembly_Contig_95_1 2 2 828aa

R0934F.300k_Assembly_Contig_73_2 4 3 809aa Wheat-Rye- 6R.300k_Assembly_Contig_48_2
Cluster RFL 170 170 Rf3 0 0 219aa Anapurna.300k_Assembly_Contig_174_3 1 1 560aa
Primepii.300k_Assembly_Contig_60_2 2 2 369aa R197.300k_Assembly_Contig_94_2 3 3 560aa
R0934F.300k_Assembly_Contig_67_2 4 4 369aa Wheat-Rye- 6R.300k_Assembly_Contig_113_2

(71) Table 9B shows that:

(72) In candidate group RFL 28, the sequences from Rf3 genotypes (Primepii and R0934F) are 857 amino acids long whereas sequences from all other germplasms (non-Rf3) are truncated (sizes ranging from 323-479 amino acids) and thus are most probably nonfunctional.

(73) In candidate group RFL 29, the sequences from Rf3 accessions (Primepii and R0934F) are 828 amino acids long whereas sequences from all other (non-Rf3) germplasms are truncated and thus most probably nonfunctional.

(74) In candidate group RFL 60, the sequences in non-Rf3 genotypes (R197 and R0932E) are either absent (R0932E) or deemed nonfunctional due to their amino acid length (287 amino acids). The sequences from Rf3 genotypes Primepii and R0934F based on their sequence length appear to be full length and functional. In addition, our mapping analysis positioned RFL60 cluster within the Rf3 interval.

(75) In cluster RFL 170, the sequences from Rf3 genotypes (Primepii and R0934F) are significantly larger (560 amino acids) than sequences from non Rf3 genotypes that appear truncated (below 370 amino acids) and are considered as being nonfunctional. Our detailed sequence analysis has shown that RFL170 is actually a second ORF, in addition to RFL 288, encoded by the same contig. Both ORFs originate from the same RFL gene in which contiguity was disrupted by a frameshift.

(76) C. Rf7 and Rf-Rye Accessions:

(77) The same rationale as for the analysis of Rf1 and Rf3 carrying accessions was applied to the accessions carrying the Rf7 restorer gene (R197 and *T. timopheevii*). The Rf7 gene was mapped on chromosome 7BL.

(78) Table 10 shows that RFL 80, 128 and 191 are candidate Rf groups for the Rf protein encoded by the Rf7 gene as the proteins assigned to those clusters were found only in either R197 or *T. timopheevii*.

(79) In regard to a restorer gene that originates from the introgression of rye chromosome 6R into wheat genome, clusters composed of single proteins originating from the Wheat-Rye-6R restorer line are considered as good candidates for a Rye-6R-specific restorer gene. Those criteria are true for the RFL 46, 87 and 208 orthologous groups listed in Table 6. Due to their high sequence divergence compared to *Triticum* sequences these genes are great candidates for restorer genes originating from rye.

(80) TABLE-US-00016 TABLE 10 selection of RFL based on germplasm CMS information for Rf7 Mapping Restorer from Rye Restorer Positionned CMS genotype MAINTAINER Rf3 Rf1 + Rf7 Rf1 Rf3 introgression *Triticum* in Rf7 mapping RFLGene ANAPURNA PRIMEPII R197 R0932E R0934F Wheat-Rye-6R *timopheevii* interval RFL49 0 0 2 0 0 0 2 NO RFL63 0 0 2 0 0 0 1 NO RFL80 0 0 1 0 0 0 0 Not mapped RFL85 0 0 2 0 0 0 1 NO RFL125 0 0 1 0 0 0 2 NO RFL128 0 0 1 0 0 0 0 Not mapped RFL174 0 0 1 0 0 0 2 NO RFL191 0 0 1 0 0 0 0 Not mapped

(81) TABLE-US-00017 TABLE 11 selection of RFL based on germplasm CMS information for Rf-rye Restorer from Rye Restorer CMS genotype MAINTAINER Rf3 Rf1 + Rf7 Rf1 Rf3 introgression *Triticum* RFLGene ANAPURNA PRIMEPII R197 R0932E R0934F Wheat-Rye-6R *timopheevii* RFL46 0 0 0 0 0 1 0 RFL87 0 0 0 0 0 1 0 RFL208 0 0 0 0 0 1 0

Example 5: Cloning of Candidate Genes for Fertility Restoration of *T. timopheevii* CMS

(82) The nucleic acid sequence encoding any of the RFL proteins from a candidate Rf group could be used for cloning and transformation. However, in the present experiment for cloning and transformation purposes, a DNA sequence encoding the longest RFL protein which was characterized as having a start codon, mitochondrial targeting sequence and number of PPR motifs

between 15 and 20 was preferentially used. If at least two longest RFL proteins happen to have the same length, the nucleic acid encoding such RFL protein, and presenting the longest 5'-UTR sequence will be preferentially chosen to perform the cloning and transformation steps.

(83) Wheat-Rye-6R RFL46 sequence derived from Wheat-Rye-6R.300k_Assembly_Contig_35_1 was optimized to provide SEQ ID N°3115. This sequence was cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and a 3' termination sequence of the gene encoding a sorghum heat shock protein (accession number: Sb03g006880); The termination sequence, named terSbHSP, is depicted in SEQ ID N° 3110. The sequence of the recombinant construct is depicted in SEQ ID N°3125.

(84) Similarly as above, TaRFL104 sequence (derived from R0932E.300k_Assembly_Contig_82_1), TaRFL67 sequence (derived from Primepii.300k_Assembly_Contig_2_1), TaRFL79 sequence (derived from R197.300k_Assembly_Contig_120_1) and TaRFL89 sequence (derived from R0934F.300k_Assembly_Contig_99_1) were adapted for cloning purpose as depicted respectively in SEQ ID N°3117 to 3120. These sequences were cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746. The sequences of the recombinant constructs are respectively depicted in SEQ ID N°3131, 3128, 3129 and 3130.

(85) TaRFL104 sequence (depicted in SEQ ID N°3117) was cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL104, SEQ ID N° 3113) and the 3' termination sequence of *Triticum aestivum* RFL104 encoding gene (terTaRFL104, depicted in SEQ ID N° 3112), into the destination binary plasmid pBIOS10747. The sequence of the recombinant construct is depicted in SEQ ID N°3126.

(86) TaRFL79 sequence (depicted in SEQ ID N°3119) was cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL79, SEQ ID N°3123) and the 3' termination sequence of *T. aestivum* RFL79 encoding gene (terTaRFL79, depicted in SEQ ID N° 3124), into the destination binary plasmid pBIOS10747. The sequence of the recombinant construct is depicted in SEQ ID N°3122.

(87) Wheat-Rye RFL46 sequence was also cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL46, SEQ ID N° 3114) and the 3' termination sequence of *Triticum aestivum* RFL46 encoding gene (terTaRFL46, depicted in SEQ ID N° 3111), into the destination binary plasmid pBIOS10747. For this construct, Wheat-Rye RFL46 sequence is derived from Wheat-Rye-6R.300k_Assembly_Contig_35_1 coding sequence and modified for cloning purpose without any optimization steps. The coding sequence is depicted in SEQ ID N°3116. The sequence of the recombinant construct is depicted in SEQ ID N°3127.

(88) The binary destination vectors pBIOS10746 and pBIOS10747 are a derivative of the binary vector pMRT (WO2001018192A3).

(89) All the binary plasmids described above were transformed into *Agrobacterium* EHA105.

Example 6: Transformation & Fertility Restoration Phenotyping Assays

(90) In order to screen for candidate genes involved in fertility restoration, the BGA_Fielder_CMS wheat cultivar harboring both cytoplasmic male sterility and strong transformability and regeneration potential was developed. BGA_Fielder_CMS wheat cultivars were transformed with *Agrobacterium* strains obtained in example 5 essentially as described by WO 2000/063398. Wheat transgenic events were generated for each construct described above.

(91) For construct comprising RFL46, transformation was also performed with the cultivar Fielder.

(92) All wheat transgenic plants generated in example 6 and control fertile plants were grown in a glasshouse under standard wheat growth conditions (16 h of light period at 20° C. and 8 h of dark period at 15° C. with constant 60% humidity) until control grains of the wild type Fielder cultivar reached maturity stage.

- (93) Fertility of the transgenic plants was evaluated by counting the number of seeds and empty glumes per spikes on each plant and comparing with the wild type Fielder and BGA_Fielder_CMS control plants. Plants are also evaluated by observing anther extrusion.
- (94) 16 transformed CMS-Fielder plants overexpressing the RFL79 sequence recited in SEQ ID N°361 (as listed in table 7) under the ZmUbi promoter derived from 11 independent transformation events were analyzed.
- (95) All the plants present restoration of male fertility while 100% of untransformed CMS-Fielder plants grown in parallel are fully sterile with no anther extrusion and no seed produced, and 100% of WT-Fielder plants are fertile.
- (96) These results confirm that RFL79 can restore fertility of a CMS-T plant and that genetic transformation of CMS-Fielder is an efficient system to test the function of restorer-of-fertility genes.

Example 7: Cloning of Rf Gene Promoter, Transformation and GUS Assays

- (97) The *E. coli* beta-glucuronidase (EcGUS) sequence was optimized (as depicted in SEQ ID N°3121) and cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL46, SEQ ID N° 3114) and the 3' termination sequence of *T. aestivum* gene encoding RFL46 (terTaRFL46, depicted in SEQ ID N° 3111), into the destination binary plasmid pBIOS10743 forming pBIOS11468.
- (98) Fielder wheat cultivars were transformed with these *Agrobacterium* strains essentially as described by WO 2000/063398. Wheat transgenic events were generated for each construct described above.
- (99) After booting stage until anthesis, heads and floral organs were dissected and incubated in X-Gluc solution (Jefferson, 1987) at 37° C. for 16 hours, to assess GUS expression.

Example 8: Identification of Full Length RFL PPR Genes Potentially Involved in Rf4 Fertility Restoration

- (100) A second capture was achieved using a set of different accessions compared to the capture performed in example 2. The following accessions were used: Two Maintainer lines (Anapurna, Fielder) The *T. timopheevii* accession as described in example 2 Four accessions identified as restorer lines of *T. timopheevii*-type CMS and characterized by the presence of the Rf4 restorer locus: L13, R113, 17F3R-0377 and GSTR435. Rf1, Rf3 and Rf7 restorer accessions which are characterized by the absence of the Rf4 restorer locus: R197, R0934F.
- (101) GSTR435 is derived from an introgression of *Aegilops speltoides* into *Triticum aestivum* and at is available USDA (<https://npgsweb.ars-grin.gov/gringlobal/search.aspx>). The three other accessions, R113 (available via Australian Grains Genebank: 90819), L13 (available via Australian Grains Genebank: 90821) and 17F3R-0377 (derived from R113) are all derived from *Triticum timopheevi* introgressions into *Triticum aestivum*.
- (102) The bait design and hybridization with DNA fragments from the accessions were performed as in example 2. Then, a subset of 100K read pairs from each accession were mapped to the RFL groups identified in table 7 using Novoalign (version 3.04.06, <http://www.novocraft.com/products/novoalign/>) with settings allowing multiple hits with approximately 97% of identity (options: -r all -t 240). The average coverage per RFL was calculated using Bedtools utilities (version 2.26.0, then <http://github.com/arq5x/bedtools2>) coverageBed (option: -d) and groupBy (options: -o mean).
- (103) The relative coverage of all RFLs with the reads from each accession was assessed. A first ranking of the RFLs according to their coverage with reads from accessions derived from *T. timopheevii* was assessed. Only RFL groups showing no coverage (value from 0 to 10) with reads from “non-Rf” (maintainer) or non-Rf4 accessions but showing significant coverage (value>30) with reads from Rf4 accessions were considered. FIG. 3 shows the list of these RFL groups potentially corresponding to the Rf4 gene.
- (104) The coverage with accession GSTR435 was also assessed. FIG. 3 shows that only RFL120

shows significant coverage in accession GSTR435, although lower than for the other accessions. This could be explained by a greater phylogenetic distance between *T. aestivum* and *Aegilops speltoides* than between *T. aestivum* and *T. timopheevii*.

(105) In order to investigate further the sequences related to the RFL120 group, the reads mapping to RFL120 for each Rf4 accession were then assembled in two steps. The first step consisted of merging overlapping read pairs with the utility `bbmerge.sh` from the BBMAP package (version 36.59 <https://sourceforge.net/projects/bbmap/>) and assembling them with the utility `tadwrapper.sh` from the same package (options: `k=150,180,210,240,270,300,330,360,390,420,450 bisect=t`). The contigs from the first step were deduplicated with the tool `dedupe.sh` also from the same package and given to another assembler, SPADes (version 3.10.1 <http://bioinf.spbau.ru/spades>), as “trusted contigs” along with all read pairs from the same accession (options: `--cov-cutoff 5-careful`) to generate the final assembly of the accession. Then for each accession the protein sequence RFL120 (SEQ ID N° 477 as listed in table 7) best hit was searched using the `tblastn` utility from the BLAST+ package (version 2.2.30 https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE_TYPE=BlastDocs&DOC_TYPE=Download) using default settings.

(106) The following sequences were finally identified to be included in the RFL120 group:

(107) RFL120-R113 which is depicted in SEQ ID N°3138 and is encoded by SEQ ID N°3142, RFL120-L13 which is depicted in SEQ ID N°3137 and is encoded by SEQ ID N°3141, RFL120-17F3R-0377 is depicted in SEQ ID N°3135 and is encoded by SEQ ID N°3139 and finally RFL120-GSTR 435 (called here RFL120-spelt) is depicted in SEQ ID N°3136 and is encoded by SEQ ID N°3140.

(108) Alignment between the above amino acid or nucleotide sequences with the corresponding sequences from RFL120, called here “RFL120_timo”, and recited in SEQ ID N°477 and SEQ ID N°2031 (see table 7) shows that RFL120-17F3R-0377 is truncated. Regarding RFL120-R113 and RFL120-L13, the nucleotide sequences are both identical to RFL120-timo except that they are respectively 357 and 349 nucleotides longer on the 5'UTR region.

(109) FIGS. 4a and 4b shows respectively the alignment between nucleotide and amino acid sequences of RFL120-spelt with RFL120-timo. This shows that RFL120-timo nucleotide sequence is 95% identical to the “RFL120_spelt” one which explains the low coverage previously observed with GSTR435 in FIG. 3.

(110) In conclusion, the results confirm that the RFL120 group is the strongest candidate for Rf4.

Example 9: Cloning, Transformation and Fertility Restoration Assays

(111) Following the same methods as described in Examples 5 and 6, the nucleotide sequences of RFL120-timo, RFL120-spelt, RFL120-R113 and RFLK120-L13 are, when appropriate, optimized for ensuring proper expression in wheat and adapted for cloning purposes.

(112) These sequences are cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (`proZmUbi` depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (`intZmUbi`, depicted in SEQ ID N° 3109, Christensen et al 1992) and a 3' termination sequence of the gene encoding a sorghum heat shock protein (accession number: Sb03g006880); The termination sequence, named `terSbHSP`, is depicted in SEQ ID N° 3110.

(113) Transformation and fertility assays are performed as in Example 6.

Example 10: Evaluation of the CMS *T. timopheevii* Rf3 Restorer Lines Fertility

(114) Eleven wheat elite lines classified as carrying Rf3 restorer gene known to be involved in restoration of the T-type cytoplasmic male sterility in *Triticum timopheevii* (T-CMS) were assessed for their capacity to restore the T-CMS cytoplasm and characterized for their fertility genotype. Hybrids resulting from crosses between a sterile CMS wheat line, used as a female parent, and a given wheat restorer line, used as a pollen donor, were studied regarding their ability to produce grain. The elite lines included both commercially available lines such as Altigo, Aristote, Cellule, Altamira, Rubisko, Primepii or Premio as well as Limagrain proprietary lines (Table 12). In

addition, Chinese Springlines were included in the study (Table 12).

(115) Fertility tests have been conducted indoors, either in growth chamber or in greenhouse, under controlled growth conditions enabling a normal expression of the fertility phenotype of the tested wheat plants. The plants were grown under 16 hours light period and temperature between 2° and 25° C. and 8 hours dark period at a temperature between 15° C. and 20° C., with humidity between 50 and 70%. The observed restoration of pollen fertility may be partial or complete.

(116) The fertility score of F1 wheat plants carrying T-CMS cytoplasm may be calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets and may be compared with the fertility scores of a panel of control fertile plants which in this study consists of elite inbred lines bearing a normal wheat cytoplasm, grown in the same area and under the same agro-environmental conditions. It is preferred that such panel of lines comprises a set of at least 5 elite inbred lines. Besides, it is preferred that at least 10 spikes from different F1 individual plant will be assessed for a given experiment.

(117) Fertility score i higher than zero (>0) indicates that the plant has acquired partial or full fertility restoration. For each fertility score, a statistical test is achieved to obtain a p-value. Examples of statistical tests are the Anova or mean comparison tests. A p-value below a 5% threshold will indicate that the two distributions are statistically different. Therefore, a significantly lower fertility score of the tested wheat plant as compared to the fertility score of the fertile control plant is indicative that the F1 plant has not acquired full restoration fertility (i.e. partial restoration). A significantly similar or higher fertility score is indicative that the F1 plant has acquired full restoration of fertility. The fertility scores indicated were calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets.

(118) The t-tests were conducted by comparing the fertility scores of F1 plants carrying Rf3 restorer gene and the fertility scores of a panel of elite inbred lines grown under the same conditions (633 spikes from 37 winter and spring elite lines; $\mu=2.36$, $\alpha=0.59$).

(119) The results presented in Table 12 show that there are two types of partial Rf3 restorer of fertility in the CMS hybrids. One that can be referred to as “Rf3” and which fertility scores are comprised between 1 and 1.8, and a second type referred to as “Rf3 weak” which fertility scores are less than 1. For example, the CMS-hybrids made with Primepii produced an average fertility score of 1.7 grains/spikelet over 10 individual F1 spikes (“Rf3” phenotype) while hybrids made with Altigo produced an average fertility score of 0.7 grains/spikelet over 47 individual F1 spikes (“Rf3 weak” phenotype) (Table 12). Different Chinese Spring lines were shown by genetic mapping and marker assisted selection to harbor an Rf3 restorer locus (data not shown). CMS-hybrids made with Chinese Spring lines were evaluated for fertility score. They show a mean estimated fertility score of 0.6 grains/spikelet and a “Rf3 weak” phenotype.

(120) TABLE-US-00018 TABLE 12 Wheat elite lines used in the study and the fertility scores of CMS-hybrids generated with their pollen. Number of analysed Fertility Elite variety Genotype Phenotype spikes score STD CHINESE RFL29b Rf3 weak 20 0.6 0.7 SPRING ALTIGO RFL29b Rf3 weak 47 0.7 0.5 ARISTOTE RFL29b Rf3 weak 43 0.8 0.8 CELLULE RFL29a Rf3 10 1.2 1.0 ALTAMIRA RFL29a Rf3 20 1.5 1.1 PREMIO RFL29a Rf3 12 1.2 1.0 PRIMEPII RFL29a Rf3 10 1.7 0.8 R0946E RFL29a Rf3 35 1.8 0.6 RUBISKO RFL29a Rf3 7 1.2 0.6 TJB155 RFL29a Rf3 27 1.7 0.7 ATOMO RFL29c Maintainer 22 0.0 0.0 CONTROL 633 2.4 0.6 ELITES *NA: Not available, STD: Standard Deviation

Example 11: Comparison of Genotypes Between the Analyzed Rf3 Restorer Lines

(121) The RFL gene capture was achieved with accessions listed in Table 12 as described in Example 2. For each RFL identified in Example 4B, the corresponding protein sequences from each accession were aligned for comparison.

(122) The results show that for RFL29, RFL164 and RFL166, strong association between the phenotype and the genotype exist. For RFL29, three different alleles referred to as “a”, “b” and “c” were identified while two different alleles “a” and “b” are identified either for RFL164 or RFL166.

All accessions with an “Rf3” phenotype carry RFL29a, RFL164a and RFL166a alleles while all the “Rf3 weak” accessions carry RFL29b, RFL166b and RFL164b alleles in their genotype (123) For RFL29, the maintainer line Atomo is characterized by the presence of two truncated ORFs, probably due to a frameshift mutation, RFL29c_1 and RFL29c_2, encoding proteins consisting of 258 and 535 amino acids, respectively. This genotype form is only present in maintainer lines (data not shown).

(124) FIG. 5A shows the protein sequence alignment of RFL29a, RFL29b, RFL29c_1 and RFL29c_2. FIGS. 5B and 5C, respectively, show the protein sequence alignments of RFL164a and RFL164b (depicted in SEQ ID N°3144), and RFL166a and RFL166b (depicted in SEQ ID N°3145).

Example 12: Cloning, Transformation and Fertility Restoration Assays

(125) Following the same methods as described in Examples 5 and 6, the nucleotide sequences of RFL29a (depicted in SEQ ID N° 3146 or SEQ ID N°1712 and encoding a sequence identical to SEQ ID N°158), RFL29b (depicted in SEQ ID N° 3149 and encoding a sequence identical to SEQ ID N°3143), RFL164a (depicted in SEQ ID N° 3147 or SEQ ID NO° 2230 and encoding a sequence identical to SEQ ID N°676), and RFL166a (depicted in SEQ ID N° 3148 or SEQ ID NO:2238 and encoding a sequence identical to SEQ ID N°684), were cloned via a Golden Gate reaction into destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and a 3' termination sequence of the gene encoding a sorghum heat shock protein (accession number: Sb03g006880 terSbHSP depicted in SEQ ID N° 3110). The sequences of each of the recombinant constructs are respectively depicted in SEQ ID N°3150, SEQ ID N°3151, 3152 and 3153.

(126) Similarly, the RFL29a and RFL29b sequences were cloned downstream of their endogenous promoter pRFL29a (depicted in SEQ ID N°3154) and pRFL29b (depicted in SEQ ID N°3155) and the terminator sequence terSbHSP. The sequences of each of the recombinant construct are respectively depicted in SEQ ID N°3156 and SEQ ID N°3157.

(127) Finally, two other cassettes were made identically as described previously with the only exception that the corresponding endogenous terminator sequences terRFL29a (depicted in SEQ ID N°3160) and terRFL29b (depicted in SEQ ID N°3161) are used. The corresponding expression cassettes are respectively depicted in SEQ ID N°3158 and SEQ ID N°3159. Transformation and fertility assays are performed as in Example 6.

Example 13: Comparison of the 5'UTR Sequence of the RFL29a and RFL29b Encoding Gene

(128) In order to analyze whether the variation of the level of fertility could be explained by a variation in the gene expression level, the 5'UTR sequence of RFL29a gene was isolated from BACs generated from TJB155 line (Table 12) classified as “Rf3” and the 5'UTR sequence of RFL29b was identified from the Chinese Spring classified as “Rf3weak” line (IWGSC RefSeq v1.0 assembly).

(129) The alignment of the 5'UTR regions identified in the RFL29a and RFL29b genes is shown in FIG. 6.

(130) Sequence comparison shows that the 5'UTR sequence of the RFL29a gene comprises a deletion of the 163 bp-long region identified in the 5'UTR of RFL29b corresponding sequence (SEQ ID NO: 3174). Part of this region has been identified in the patent application WO2018015403 as being putatively involved in miRNA-mediated repression of the expression of the PPR gene identified downstream.

(131) Sequence comparison between the different accessions listed in Table 12 shows that all “Rf3weak” accessions harbor the 163 bp and all the “Rf3” accessions harbor the 163 bp deletion.

(132) Because of the 163 bp deletion in the 5'UTR sequence of RFL29a gene, it is expected that the 163 bp region impairs the expression of RFL29b gene such that the fertility level is weak in lines harboring the RFL29b allele compared to lines harboring the RFL29a allele.

Example 14: Cloning, Transformation and Fertility Assays with a Deleted TaRFL29b Promoter (133) Following the same methods as described in Example 5, the nucleotide sequence of RFL29b was expressed under the modified promoter pRFL29b~~del~~ (depicted in SEQ ID N°3162) which is bearing a deletion of the 163 bp region, from the nucleotide 1876 to nucleotide 2038 of the RFL29b promoter sequence depicted in SEQ ID N°3155. The termination sequence, terSbHSP, depicted in SEQ ID N°3110 is used. The recombinant construct is depicted in SEQ ID N°3163. (134) Transformation of CMS*Fielder wheat line (which is either not “Rf3” or “Rf3 weak”) is performed as in Example 6 except that the following controls are added: all of the “Rf3” phenotype lines as listed in Table 12, all of the “Rf3 weak” lines as listed in Table 12 and finally the transformed lines with the cassette harboring the pTaRFL29b promoter upstream to RFL29b as described in example 12.

Example 15: Modification of the Endogenous Promoter of RFL29b by CRISPR Technology to Revert “Rf3Weak” Lines to “Rf3” Lines

(135) In order to increase the expression of RL29b, different deletions of the 163 bp are performed in the promoter of RFL29b using endonuclease for site-directed mutagenesis.

(136) Table 13 provides the different endonucleases with the associated PAM motif and the corresponding target sequences.

(137) FIG. 7 shows the position of the different target sequences around and within the 163 bp region identified for different endonucleases. The designed guide sequences that can be used in combination to perform a deletion in the 163 bp region are listed in Table 13.

(138) TABLE-US-00019 TABLE 13 Endonuc leases Target_ID Target name guide sequence

Endonuc	leases	Target_ID	Target name	guide sequence
LbCPF1 23	LbCpf1-100-	TAATTTCTACTAAGTGTAGATCGAGCGGAGGG	Target-23	AGTACTAGATAA(SEQ ID NO: 3175)
LbCPF1 42	LbCpf1-100-	TAATTTCTACTAAGTGTAGATGGAACGGAGGG	Target-42	AGTATTATCTAG(SEQ ID NO: 3176)
LbCPF1 67	LbCpf1-100-	TAATTTCTACTAAGTGTAGATAGATAGCTAGAA	Target-67	AGACAATTATT(SEQ ID NO: 3177)
LbCPF1 71	LbCpf1-100-	TAATTTCTACTAAGTGTAGATTTTGAGATAGCT	Target-71	AGAAAGACAAT(SEQ ID NO: 3178)
SpCAS9 14	SpCas9-100-	TGACAAGTATTTCCGAGCGGGTTTTAGAGCTA	Target-14	GAAATAGCAAGTTAAAATAAGGCTAGTCCGTTA
SpCAS9 54	SpCas9-100-	TCAACTTGAAAAAGTGGCACCGAGTCGGTGCT TTT(SEQ ID NO: 3179)	Target-54	GACAATTATTTAGGAACGGAGTTTTAGAGCTAG
SpCAS9 55	SpCas9-100-	AAATAGCAAGTTAAAATAAGGCTAGTCCGTTAT	Target-55	CAACTTGAAAAAGTGGCACCGAGTCGGTGCTT TT(SEQ ID NO: 3180)
SpCAS9 58	SpCas9-100-	AAATAGCAAGTTAAAATAAGGCTAGTCCGTTAT	Target-58	CAACTTGAAAAAGTGGCACCGAGTCGGTGCTT TT(SEQ ID NO: 3181)
SpCAS9 63	SpCas9-100-	AAATAGCAAGTTAAAATAAGGCTAGTCCGTTAT	Target-63	CAACTTGAAAAAGTGGCACCGAGTCGGTGCTT TT(SEQ ID NO: 3182)
SpCAS9 155	SpCas9-100-	AAATAGCAAGTTAAAATAAGGCTAGTCCGTTAT	Target-155	CAACTTGAAAAAGTGGCACCGAGTCGGTGCTT TT(SEQ ID NO: 3183)
SpCAS9 179	SpCas9-100-	AAATAGCAAGTTAAAATAAGGCTAGTCCGTTAT	Target-179	CAACTTGAAAAAGTGGCACCGAGTCGGTGCTT TT(SEQ ID NO: 3184)
SpCAS9 184	SpCas9-100-	AAATAGCAAGTTAAAATAAGGCTAGTCCGTTAT	Target-184	CAACTTGAAAAAGTGGCACCGAGTCGGTGCTT TT(SEQ ID NO: 3185)

(139) The nucleotide sequence encoding for LbCpf1 endonuclease is optimized (as depicted in SEQ ID N°3164) and cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and in the 3' region, the mouse nuclear import NLS sequence (as depicted in SEQ ID N° 3172) and the 3' termination sequence terZmHSP depicted in SEQ ID N° 3170.

(140) The nucleotide sequence encoding for SpCas9 endonuclease is optimized (as depicted in SEQ ID N° 3165) and cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, downstream to the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and the SV40NLS sequence (as depicted in SEQ ID N° 3173) and upstream of the mouse nuclear import NLS sequence and the 3' termination sequence terAtNos depicted in SEQ ID N° 3171.

(141) Each guide sequence is cloned between the pTaU6 promoter (depicted in SEQ ID N° 3168) and the termination sequence TerRNApolIII (depicted in SEQ ID N°3169).

(142) Each cassette expressing the endonuclease is cloned consecutively to the cassette expressing the corresponding guide sequence. A recombinant cassette expressing LbCpf1 and guide sequences directed to target-23 and target-71 is depicted in SEQ ID N°3167. A recombinant cassette expressing SpCas9 and guide sequences directed to target-58 and target-54 is depicted in SEQ ID N°3166.

(143) Transformation is performed as in Example 6 except that all “Rf3 weak” lines as listed in Table 12 are transformed. Fertility assays to select lines with an “Rf3” phenotype are performed as in example 14.

Example 16: Evaluation of the CMS *T. timopheevii* Restorer Lines Fertility

(144) Some wheat elite lines possess an ability to partially restore the fertility of the cytoplasmic male sterility *Triticum timopheevii* (T-CMS). The hybrid formed between a sterile CMS wheat line taken as female and the wheat restorer line taken as male can produce grain. Commercial lines as Allezy, Altamira, Altigo, Aristote, Osado, Cellule or Premio and Limagrain proprietary lines have been tested for their capacity to restore T-CMS. All lines have also been characterized for their fertility genotype.

(145) Fertility tests have been conducted indoor, either in growth chamber or in greenhouse, under controlled growth conditions enabling a normal expression of fertility of the tested wheat plants. The fertility scores indicated have been calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets. The t-tests conducted were done by comparing the fertility scores of F1s made with a restorer and the fertility scores of a panel of elite inbred lines grown under the same conditions (633 spikes from 37 winter and spring elite lines; $\mu=2.36$, $\alpha=0.59$).

(146) The results in Table 14 show that these lines act as partial restorer of fertility in a CMS hybrid. For example, the hybrids made with Allezy produce an average fertility of 0.99 grains/spikelets over 33 individual F1 spikes. This native ability to partially restore the fertility of the CMS *T. timopheevii* was previously identified in wheat accessions as Primepii or Maris Hunstmann (Bahl and Maan, 1973), and it is confirmed by our own internal data (See Table 14). Beside this supposedly wheat intrinsic source of restoration, the hybridizations between *Triticum timopheevii* and *Triticum aestivum* conducted by Wilson (Wilson and Ross 1962) contributed to the introduction into wheat of further restorer genes which chromosome arm localization was identified through monosomic analysis (Bahl and Maan, 1973; Maan and al, 1984).

(147) Following these initial works many breeding programs implemented worldwide aimed at creating restorer lines for the T-CMS such as TJB155 (BBSRC Small Grain Cereals Collection: 2072 to 2075), which ability to restore is also confirmed by our own internal data (Table 14).

(148) L13 (available via Australian Grains Genebank: 90821), is a wheat restorer line selected from R113 (available via Australian Grains Genebank: 90819) which carries the restorer allele Rf4. It produces a low level of fertility restoration when crossed with CMS lines (0.85 seeds/spikelet on average). When compared with the fertility scores of a panel of elite lines grown under the same conditions it appears that none of the tested putative restorer make possible a full restoration of fertility of the hybrid (p-values<0.05. table 14).

(149) Table 14 also shows that restorer lines bearing two different Rf loci, are not able to fully restore the sterility induced by T-CMS. Therefore, all restorer lines tested, with single or two Rf loci are only partial T-CMS restorer lines.

(150) TABLE-US-00020 TABLE 14 mean fertility scores and standard deviation from the indicated number of spikes. The first column indicates the name of the male line taken as pollinator for the F1 cross. All the data (except for Elite inbred lines) presented are from the created F1 crosses between the listed lines and a CMS tester. "Elite inbred lines" data refers to a panel consisting of 37 winter and spring elite lines. The t-test was implemented comparing the mean of fertility for each of the F1 crosses with the mean of the fertility scores of a panel of 633 spikes of elite inbred lines bearing fertile cytoplasm. Haplotypes SPIKES FERTILITY STD DEV P-VALUE ALLEZY Rf3 33 0.99 0.89 3.00E-33 ALTAMIRA Rf3 20 1.48 1.1 4.00E-10 ALTIGO Rf3 47 0.71 0.55 1.00E-63 ARISTOTE Rf3 37 1 0.73 3.00E-37 CELLULE Rf3 10 1.25 0.99 5.00E-09 MARISHUNSTMANN Rf3 16 1.26 0.88 7.00E-13 OSADO Rf3 9 1.52 0.68 2.00E-05 PREMIO Rf3 12 1.21 1.03 8.00E-11 PRIMEPII Rf3 51 1.55 0.75 2.00E-19 TJB155 Rf3 27 1.7 0.66 2.00E-08 R204 Rf1 + Rf7 59 2.09 0.51 6.00E-04 L13 Rf4 12 0.85 0.38 4.00E-18 R0929D Rf3 + Rf7 18 1.83 0.62 1.63E-04 R0936T Rf1 + Rf3 4 1.76 0.11 4.06E-02 ELITE inbred lines 633 2.36 0.59

Example 17: Fine Mapping of the Genomic Region Containing Rf1, Rf3 and Genetic Mapping of Rf4 and Rf7 Genetic Determinants

(151) A. Fine-Mapping of the Genomic Region Containing Rf1 Genetic Determinants

(152) Three F2 mapping populations segregating for Rf1 (R197xKalahari, R204xAlixan and R0932ExAltigo) encompassing 210, 218 and 212 individuals respectively were phenotyped as described in example 1 and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform. Fertility of R204 and R197 lines is genetically associated to Rf1 and Rf7 locus. Fertility in R0932E line is associated to Rf1 locus.

(153) Rf1 was first mapped on the short arm of the chromosome 1A between 4 cM and 10.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1087371 and cfn0530841. These two SNP markers delimit the largest possible interval defined by the three mapping populations (see FIG. 8).

(154) Following, joint analysis of the three mapping populations and phenotyping of the individual F2 recombinant plants on derived F3 families validated the QTL position and delimited Rf1 interval between 7 cM and 8.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1082074 and cfn0523990. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the wheat genome reference physical map. The left border (cfn1082074) was anchored on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the right border (cfn0523990) was anchored on the IWGSCWGAV02_1AS_scaffold47238 scaffold.

(155) Next, we decided to enlarge the population sizes to fine-map the locus and screened 2976 and 3072 F3 lines from R197xKalahari and R204xAlixan derived from F2 plants heterozygote at the locus respectively. Phenotyping and analysis of recombinant plant progenies within the interval redefined a smaller mapping interval between 7.5 and 8.8 cM delimited by cfn0522096 and cfn0527067 SNP markers on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the IWGSCWGAV02_1AS_scaffold47238 scaffold respectively.

(156) B. Fine-Mapping of the Genomic Region Containing Rf3 Genetic Determinants

(157) Three F2 mapping populations (TJB155xAnapurna, 2852xAltamira, and AH46xR0946E) encompassing 217, 135, and 246 individuals respectively and a doubled-haploid (DH) population (H46xR934F) consisting of 140 individual plants segregating for Rf3 were phenotyped as described in example 1 and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform. Sources of Rf3 locus are TJB155, Altamira and R0946E.

(158) Rf3 was first mapped on the short arm of the chromosome 1B between 18.9 cM and 24.2 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn0554333 and cfn0560679. These two SNP markers delimit the largest possible interval defined by the four mapping populations (see FIG. 9).

(159) Following, joint analysis of the four mapping populations and validation of the phenotype of the individual F2/DH recombinant plants on derived F3 families validated the QTL, genetically delimited the locus between 22.2 cM and 22.7 cM on Limagrain's internal consensus map and physically delimited the Rf3 interval between SNP markers cfn0436720 and cfn0238384. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the physical map. The left border (cfn0436720) was anchored on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the right border (cfn0238384) was anchored on the IWGSCWGAV02_1BS_scaffold5117 scaffold.

(160) Next, we decided to enlarge the population sizes to fine-map the locus and screened 2496 and 672 plants from TJB155xAnapurna and AH46xR0946E derived F2 plants heterozygote at the locus. Analysis of recombinant F3 plant progenies within the interval redefined a smaller mapping interval between 22.5 and 22.7 cM delimited by cfn1249269 and BS00090770 SNP markers on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the IWGSCWGAV02_1BS_scaffold5117 scaffold respectively.

(161) C. Mapping of the Genomic Region Containing Rf7 Genetic Determinants

(162) We crossed R197 (harboring Rf1 and Rf7 locus) and Primepii (harboring Rf3 locus) and then derived a population of 176 plants from individuals that were rf1 and rf3, which means not carrying the restorer alleles at the loci Rf1 and Rf3. The plants were genotyped with 18100 SNP markers using Limagrain's internal genotyping platform and phenotyped as described in example 1. We mapped the Rf7 locus on chromosome 7BL. Moreover, internal genotyping data showing a strong genetic divergence suggests the presence of an exotic chromosomal fragment which is stably transmitted through generation. We identified a large QTL ranging from 45 cM to 88 cM on chromosome 7B on Limagrain's internal consensus map with a peak on 46.7 cM (cfn0919993 with LOD score of 3.37E-40). First expertise of the recombinant plants suggests the Rf7 gene could be located between cfn3407185 and W90K_RAC875_c33564_120 markers delimiting a mapping interval of 0.3 cM between 46.7 cM and 47 cM on Limagrain's internal consensus map.

(163) D. Mapping of the Genomic Region Containing Rf4 Genetic Determinant

(164) A mapping population from the cross between AH46 and L13 (harboring Rf4 locus) consisting of 124 individual plants segregating for Rf4 was genotyped with 18100 SNP markers using Limagrain's internal genotyping platform and phenotyped as described in example 1. A QTL that we named Rf4 was identified on chromosome 6B between 0 cM to 65 cM on Limagrain's internal consensus map with a peak on 43.3 cM (cfn0393953 with LOD score of 1.08E-13). Moreover, internal genotyping data showing a strong genetic divergence suggests the presence of an exotic chromosomal fragment which is stably transmitted through generation. It is expected that the rate of recombination be very low within this chromosomal region and, consequently, any marker in linkage with cfn0393953 is considered to be associated with Rf4 locus.

Example 18: Identification of SNP Associated with the Rf Genes and its Use in MAS

(165) We constructed a BAC library with a DH line comprising Rf1, Rf3 and Rf7 alleles. This library was used for the identification of BAC clones within the Rf1 and Rf3 QTL intervals. More specifically, we identified and sequenced and genetically validated 3 BAC clones within the Rf1 region and 3 BAC clones within the Rf3 region. The BAC sequences were compared to Chinese

Spring reference genome for SNP discovery. We then saturated Rf1 and Rf3 mapping intervals by mining available SNPs from public genomic resources and the newly discovered SNPs from sequenced BAC clones.

(166) Screening for polymorphic and informative SNP markers on a diversity panel consisting of 83 wheat elite plants and known restorers for Rf1, Rf3, Rf4 and Rf7 identified a set of tightly linked markers localized within or in the immediate flanking regions of the Rf1, Rf3, Rf4 and Rf7 mapping intervals. These SNP markers can be used alone and/or in haplotype to select for Rf or rf plants in MAS breeding schemes (FIG. 10A and FIG. 11A). Allele information, marker sequences and primer information are provided in Table 15. Following, smaller sets of 2-3 markers of high quality were chosen to follow the traits in MAS breeding schemes.

(167) As an example, the identification of the presence of the Rf1 locus in the genome of a plant is achieved by using either the marker 276113_96B22_97797 or 104A4_105588 (FIGS. 10A and B).

(168) Similarly, the identification of the presence of the Rf3 locus in the genome of a plant is achieved by using either the marker 136H5_3M5_7601 or 136H5_3M5_89176 (FIG. 11A, 11B, 11C, 11D and Table 15). However, any marker listed on the FIG. 11A can be used to distinguish the maintainer lines from the restorer lines if they are polymorphic in the germplasm. As an example, in the Table 16, the presence vs absence of the Rf3 locus is achieved by using either the marker cfn1246088 or IWB72107

(169) The identification of the presence of the Rf7 locus in the genome of a plant is achieved by using the markers cfn0917304, cfn0919993 and cfn0920459 (Table 17). In this case the 3 markers might be used in haplotype to distinguish restorer plants from maintainer plants. Thus the haplotype TGC would identify the restorer plants.

(170) Finally, the identification of the presence of the Rf4 locus in the genome of a plant is achieved by using the markers cfn0393953 and cfn0856945 (Table 18).

(171) TABLE-US-00021 TABLE 15 Haplotypes of a series of restorer lines and maintainer lines at the locus Rf3 for 2 SNP markers. Those 2 SNP markers makes possible to fully distinguish the maintainer lines from the restorer lines (including the elite lines Altamira, Cellule, Premio and the accessions TJB155 and Primepi). R: restorer/ CODE M: maintainer Rf alleles 136H5_3M5_7601 136H5_3M5_89176 LGWR16-0016 R homozygous Rf3 T A LGWR16-0026 R homozygous Rf3 T A ALTAMIRA R homozygous Rf3 T A CELLULE R homozygous Rf3 T A TJB155 R homozygous Rf3 T A ALLEZY R homozygous Rf3 T A PREMIO R homozygous Rf3 T A PRIMEPI R homozygous Rf3 T A AIGLE M homozygous rf3 C G AIRBUS M homozygous rf3 C G ALHAMBRA M homozygous rf3 C G ALIXAN M homozygous rf3 C G AMADEUS M homozygous rf3 C G ANAPURNA M homozygous rf3 C G APACHE M homozygous rf3 C G ARKEOS M homozygous rf3 C G ARLEQUIN M homozygous rf3 C G ARTDECO M homozygous rf3 C G ARTURNICK M homozygous rf3 C G ATOMO M homozygous rf3 C G AVENUE M homozygous rf3 C G CEZANNE M homozygous rf3 C G CROISADE M homozygous rf3 C G FRUCTIDOR M homozygous rf3 C G GAZUL M homozygous rf3 C G HERMANN M homozygous rf3 C G HORATIO M homozygous rf3 C G KALAHARI M homozygous rf3 C G

(172) TABLE-US-00022 TABLE 16 Haplotypes of a series of restorer lines and maintainer lines at the locus Rf3 for 2 SNP markers. Those 2 SNP markers makes possible to fully distinguish the maintainer lines from the restorer lines. R: restorer/ CODE M: maintainer Rf alleles cfn1246088 IWB72107 ALTIGO R homozygous Rf3 A A ARISTOTE R homozygous Rf3 A A OSADO R homozygous Rf3 A A AIGLE M homozygous rf3 C G AIRBUS M homozygous rf3 C G ALHAMBRA M homozygous rf3 C G ALIXAN M homozygous rf3 C G AMADEUS M homozygous rf3 C G ANAPURNA M homozygous rf3 C G APACHE M homozygous rf3 C G ARKEOS M homozygous rf3 C G ARLEQUIN M homozygous rf3 C G ARTDECO M homozygous rf3 C G ARTURNICK M homozygous rf3 C G ATOMO M homozygous rf3 C G AVENUE M homozygous rf3 C G CEZANNE M homozygous rf3 C G CROISADE M

homozygous rf3 C G FRUCTIDOR M homozygous rf3 C G GAZUL M homozygous rf3 C G
HERMANN M homozygous rf3 C G HORATIO M homozygous rf3 C G KALAHARI M
homozygous rf3 C G

(173) TABLE-US-00023 TABLE 17 Haplotypes at the Rf7 locus for the restorer lines LGWR16-0016 and LGWR16-0026 and for a series of maintainer lines. “—” scores correspond to dominant markers with no amplification in several maintainer lines. R: restorer/ CODE M: maintainer Rf alleles cfn0917304 cfn0919993 cfn0920459 LGWR16-0016 R homozygous Rf1, Rf3, Rf7 T G C LGWR16-0026 R homozygous Rf1, Rf3, Rf7 T G C AIGLE M homozygous rf1, rf3, rf7 G — G AIRBUS M homozygous rf1, rf3, rf7 T G G ALHAMBRA M homozygous rf1, rf3, rf7 G T G ALIXAN M homozygous rf1, rf3, rf7 T T C AMADEUS M homozygous rf1, rf3, rf7 G G C ANAPURNA M homozygous rf1, rf3, rf7 G T G APACHE M homozygous rf1, rf3, rf7 G T C ARKEOS M homozygous rf1, rf3, rf7 G T G ARLEQUIN M homozygous rf1, rf3, rf7 G G C ARTDECO M homozygous rf1, rf3, rf7 G — G ARTURNICK M homozygous rf1, rf3, rf7 T T G ATOMO M homozygous rf1, rf3, rf7 T T G AVENUE M homozygous rf1, rf3, rf7 T T C CEZANNE M homozygous rf1, rf3, rf7 G T G CROISADE M homozygous rf1, rf3, rf7 G T C FRUCTIDOR M homozygous rf1, rf3, rf7 G T G GAZUL M homozygous rf1, rf3, rf7 — G C HERMANN M homozygous rf1, rf3, rf7 T G G HORATIO M homozygous rf1, rf3, rf7 G — G KALAHARI M homozygous rf1, rf3, rf7 G — G

(174) TABLE-US-00024 TABLE 18 Haplotypes at the Rf4 locus for the restorer lines L13 and R113 and maintainer lines. R: restorer/ CODE M: maintainer Rf alleles cfn0393953 cfn0856945 LGWR16-0016 M homozygous rf4 T T LGWR16-0026 M homozygous rf4 T T AIGLE M homozygous rf4 T T AIRBUS M homozygous rf4 T T ALHAMBRA M homozygous rf4 T T ALIXAN M homozygous rf4 T T AMADEUS M homozygous rf4 T T ANAPURNA M homozygous rf4 T T APACHE M homozygous rf4 T T ARKEOS M homozygous rf4 T T ARLEQUIN M homozygous rf4 T T ARTDECO M homozygous rf4 T T ARTURNICK M homozygous rf4 T T ATOMO M homozygous rf4 C T AVENUE M homozygous rf4 T T Cezanne M homozygous rf4 T T CROISADE M homozygous rf4 T T FRUCTIDOR M homozygous rf4 T T GAZUL M homozygous rf4 — T HERMANN M homozygous rf4 T T HORATIO M homozygous rf4 C T KALAHARI M homozygous rf4 T T L13 R homozygous Rf4 C G R113 R homozygous Rf4 C G

Sequences of Snp Markers:

(175) TABLE-US-00025 TABLE 19 Marker sequences and SNP position in sequence. SEQUENCES OF SNP MARKERS Marker ID AlleleX AlleleY Sequence SEQ ID NO:
cfn0523072 C T CAAAGGCTTGACAATGATAATGCCCCCGAATC 3187
TTG[C/T]GATAGACCTCATGCGCTAGAGTTGTT TTCCTCAAT cfn0523109 A C
GACAAAGTTGAGGTGAACAAAACAGGCCTAC 3188
AATC[A/C]GCTAACTTACGTATATCCACATTAG CACACACCAC 276I13_96B22_97797 C T
AAATTCGACAAGTACTATGGCTATGTCTCTGA 3189
ATG[C/T]TTGTTTGGTTTTATTTGTCTATATTGT CGTTGTAT cfn0522096 C G
ATGCAAAGTAGTACTCGTAGAGAGTTAACACA 3190
GAC[C/G]AGTGATTTATTGGGTGGTATTCTACT TGATATTTG cfn0527763 T C
ATAAAGAAAAGTAGAGGAAGCTTATGAATAA 3191
AATGGAAAAGGAATTCAAAATTGCCGATAAA
TATAAACTCATAACAAATCTAGCCACGCAAA
TGCCCG[T/C]GCCGCTCTGCTCGTTTGTACATG
TCTCGGTGGACAAGGAAGAACCCAACAATTGC
ACAGGTCAATCTTATCCAGCAAAACAAGGAA GCAAACCAAACAGG 104A4_105172 TG
CA ATGTTGCCTCTCGCTAGCCGCTGTTCGMACCCA 3192
ATGAATAATGTT[TG/CA]TGGGTTCTGGCTCCG AGAGGATGGCCGGCTYCCC
104A4_105588 A C GTTCCTTGTGACATGTACTCATA[A/C]ACAAGA 3193

GCCATCATACCTCCCTTGCA cfn0373248 T A
GACATAATGTGTAATAACAGCCCATAATGCAA 3194
TAAATATCAATATAAAAAGCATGATGCAAAATG
GACGTATCATTGCCACGRAAAAAATCTCACAA
GATG[T/A]GACCATTTGATCCTCRTAATTGTTG
TTCTAGACCCACTCCTAAGTMTAACATTCTTT
ATGTCTATYCTTCAAATCCCGAAGAGTAATGA AA ACTATCGAA cfn1097828 T C
CCATGAGTACCCGCTACTATCGATCTCCCTCCT 3195
CCCTGTAGGAGGCCTACGAACGATGCCCTCAG
GTCCTGCTTCCTCTCGGTAGCGATGGATCCAC
CTG[T/C]GGTTGCTCTCTCAGGAACCAGTGTTG
GCGGCGGCTCATCCGGGGCGCTGGATCTTGGT
GATGTGCTGGAACA ACTCAACTTGGAAGACGA AGAATTTGAT cfn0527067 A G
GACAATATGATTCACCCTAGATCCTTCACCTT 3196
ACA[A/G]TTCGAAAAAATAAAAGAACAAAAG TAATTTGACA cfn0528390 A G
ACGAAGATGAGGAAGGTCTTCATGTTGGGTTT 3197
ATG[A/G]T TACTAATACTTGCTTGGAATAGATG TTTTGTATC BWS0267 A G
GTTACCCCAATATGCTCCCTCCTTGCACATTTT 3198
CTTCAGCTGCATAAAAAMCAGAATACC[A/G]C
ATCAGTTGCCTGAACCTTAACGCAGGTGCAGA AATAAGGCGACATAATTTYCACTAATC
cfn0527718 C T AGGAAAATAAATTGTTCAACATGGACATGA 3199
GAA[C/T]GGGGCAACCAAAAAGGGAAGAACAT TGGAGGAAAC cfn0524469 G T
TTTGTACTGCACGTAGTAAGTATTGATTTTTTCT 3200
GT[G/T]TGCTCTCTGTGGACTTAGATTTGAAAA TTGGCCTT cfn0524921 A G
ATGCACATTGTTTCCATGTAAAGCTTATATTGT 3201
GC[A/G]TAACTCAAAAGATTGAAATGGAATTA CCAAAGGGC cfn1122326 C T
ACTGACTGTTGGAATCTGATTAAGACGCTGGA 3202
GAA[C/T]CCGAGCCAAGATATGTCACGACTAG GCCATCTGGA cfn1252000 A G
AATCAGATCCTGTTAATGCTGTAGCCATTCTTG 3203
CA[A/G]GCGACACCTTGTCCCAGTCGTCTTATG GGCACCTA IWB14060 A G
GGCAGAGCCGGTCGACGGAGAGGAGCGCCAT 3204
TCGACGCGTCTTCCGCAAT[A/G]TGTTTGCCTG
CTTCGGCCGCGGCCATTTCGGCGAGCTCCCACG CTTCGTCC cfn1249269 A G
CGTTTAAAAGAACACAAATGTGGCCCTAGTGA 3205
TCA[A/G]GTACACATATTTGTCACCTCTTTGAA TCTTACTTA 219K1_166464 C T
CGGGCTGATGAGGCTCTCGACGTGCTGCTTCA 3206
CAGGATGCCTGAGCTGGGCTGCAC[C/T]CCCA
ACGTGGTGGCATATAACCACGGTCATCCACGGC TTCTTTAAGGAAGGC 219K1_158251 G
A GCGCTATCCGGCGTCGTGTTCCCTCTTGGGGG 3207
AATCGTCCTGGAGATGGATCCGGTCA[G/A]AG
GGGCCCCGTGATTTGTGAGGATGTGTGTGTTGT TTCCCGAAAGGCG 219K1_111446 A C
CTTTGACCTTAAATTCTTGTACTAATTTAGCAG 3208
AATCGTTCTTCGAGAAGCACTC[A/C]AAAAAT
GGTTTGTCTTGGGTCTGTATCATATTTTCTCTG AACAAACAGGCGTGA 219K1_110042 T
C GACTTAGCCTCACACGGAATCGAGTCAACCAA 3209
TTCC[T/C]GTCGGTTTTGAGTGGCTCCCTTGAA
GATGCAATCGTTTTTCAGCATGGTCAGATTAAT CAGCGAGCGTGC 219K1_110005 C T
CATGTAGTGGCTGGCGTCTAAGCGCCTTTTCTT 3210
CTTCCAGCATCTA[C/T]GACTTAGCCTCACACG
GAATCGAGTCAACCAATTCCTGTCGGTTTTGA GTGGCTCCCTTGAAGATG

219K1_107461 A C GTCGTATATATGTTGTATTAAAAAGTTGTGT 3211
GTTTTG[A/C]GTCATAATTTTTAAAAATATTATT ATGTCATTTTCAAATTCGCATCAAC
219K1_99688 T C AATCTTCTTGACTTCATCCATCCGCCTTGTTGC 3212
CCTGCGCAAAATCAAAC[T/C]CCCCGTCCTTA
TCATCAAGTCAGGTCCCGCCCTGGGCAGAGAG AG 219K1_37 C T
CGGCAGATATCACAAAGGGCTATCCTGGTGAA 3213
CAA[C/T]AAGATGGGTCAGAATTTGATAATGA AGCCTCAAGCCC cfn1270524 A T
AATAGATGCACGCATCGGCGACCATTTTTTTAG 3214
TACTTTTTGCCTTTTTTTGAAAATTTTGTCTTA
AAAGACAAATGCCTAGTCTATACCTGATAAAC
TAA[A/T]ATCATACATAGAGAAAATGGTCATTT
GGTTGAGTTTCGGTACATGCTGAGATGGTTGC
ACTTCGGTGCATCTGCTTTTGCTTCCATCACATC ATAATGTCT 136H5_3M5_7601 T C
GCTGCTTGTAGCGTCCCCCATGGCACCTG[T/C] 3215
GAAGAGGTTTTTCGGCCACAGAGAAGGGGAAG GCTC cfn1288811 T G
AAAATTACTTTTCACGCGCTTCGTTGGTCTGAC 3216
AGTGCGAGCATAATTTTACTTTTTTCTCAGTTTT
ACTTAATTTGGTTAACCAAATCCTTTTTTGATTT
T[T/G]AACTAGAAAACCGAATGTCAAACATTG
TGCAAATTTGGAAACTGAAACTGAAACCAAA
AACCTAAAAAAATGATTAGTTTGTTTTTTTTGT CTTGTTTCG 136H5_3M5_89176 A G
gtatttCTTAGGATTTTCTCACCGGCATCTCC[A/G] 3217
TTTTTTGAGCAAGAGTATTTAAGGATGGTAGG C 136H5_3M5_89263 C T
AACAAAGATGCTAGTAAGAACATGAACCTAG 3218
TTGCTCATTTTTTAACAACAATTGCCCACCAACC
TGACATGCTCTTCCCATGTTCTTTTTTTTGCTCA
AAA[C/T]AGAGATGCTAGTCCAAATATTTTTCT
AGTTGCTTACATTTTAAACAACAATTGCCTAC
CATCCTTAAATACTCTTGCTCAAAAAACGGAG ATGCCGGTGA 136H5_3M5_138211 T A
AATACAGACTGGGTGCAAAGCCAAGATGAT[T/A] 3219
GTAAAATTGATTGATGGCCGTTGGGAGGT cfn0556874 C T
TGTAAGAAGCTTAACCAGGAAAGCTATCAG 3220
GGCCATAGGGAATGGCTGGTTAGTGACAATTT
TGCCTGCTGGAAATGGGATTTCTTGTTTATTTT
AGTT[C/T]TGCATTGTGTCTGACATGCTCTTTCT
TTTGGGCGCAGGCTGAAGTGAATTACCTTGGA
CAACTATCGCACCCGAATCTTGTAAGCTCGT TGGGTACTGT 136H5_3M5_64154 T C
TGGCGGAGCTGGGGCTGTTCCCTCCTACGCAGG 3221
CGAAACTTCGCCGCGATAAA[T/C]GGA ACTAT CATCAGGTTCCCCGATGATCCATACG
136H5_3M5_68807 A G ACAAGCAACCGAGACAAGTTGCTCTTAATTAT 3222
CTGTGCGT[A/G]CACCTCTAAGTCTTAACCTGA CGTAACCAACCAACCGTGT
136H5_3M5_77916 A G GATGGTTACAAGGCATGCATAGCAAGTAGAGT 3223
TAACTTATCAAGTTATT[A/G]GTATTTTTCTTCT GTGGTACTTAGAGTCTAMAGCTTGAGC
cfn1246088 A C AATGGAAGCTGATGTGCGTTAGCGATAAAGCA 3224
ACAGCGATAACGACGCATGGATCACCATGCTA
CTTGGGGAAGCAGGGACATCTGATGAGCCAG
CATAC[A/C]CCCAGATATGTGTCTTTCCAAATT
CCACGTCCCAACAGATGAGCTATAAATTAATG
CCACCTTCCTCCTACAGCTAAATACTCCATCCG TTTCATAATGT cfn1287194 G A
CAGAGGCATTTCGTGAATTGGGCGAAATCAGA 3225

ACGAAGGACGGATGTTTCAGCGCAGAAGG
CACTGGGAGGGGATTCCAGGGAGGCTGCCCA
CCAGCCC[G/A]CCATCAGATACGGAGGAGGTG
GATCCATGGCCCTACCTGTGTCCTGCGCCGAA
TCTGGACTGTGGTAACTACAGCGCCTGAATCT AGAGGTTTCAGCCTGG cfn1258380 A C
GATCCATCTCCCTTAATAATTTTGCTATTGGTA 3226
TTGGGTATGGACATCTGAAGTGAAGGTTACGG
CCGATTTATAGGAGTGATAGCACCAACAATT
CAT[A/C]AGAGCATCTGCAATAGATGAGTAGA
TGTAAAACTACTTAACTTTTACATCTCCGGGCC
TAAAAACGCATCTGTAATAAGATAATGTAGAT GTAAAGAAAA IWB72107 A G
CGACGACGACGAGGATGCCGAGTTTGATGAC 3227
ATGGAGGATTATATCGACG[A/G]cgcgagactgggacg cgcacatgatgatgatgttcgatgtctgaagga
BS00090770 C T TAGCCGTAGGTCGTAGCACATAGCCGTTTA[C/T] 3228
GTAATGCATAGTTGTCCGAAGGAATGTTTC cfn1239345 A G
TAACCTGGGGCTTCTTTTTTCTCCCTATAATAT 3229
GG[A/G]CTGCCCTTTTAAGAAGGAAGTGCAGC GAGGGTGCA cfn0917304 G T
GACTACGCGTTCCTCCCGGTGGTGGCGCTCTA 3230
CCC[G/T]TTGTGTTGCCTTTCTCCAAGCAGTTGT GCCCTTCG cfn0919993 G T
ATATCTTTACAAGTCATCGACTTACATGCTTCT 3231
TT[G/T]TATTATATGCACCTATGCAGTACTTGTT AATGGGT cfn0920459 C G
GATGATATAACCGTAGCCAAGGAAGCCCAGA 3232
TTTT[C/G]TTCTGTGTATCTATAGGAGCTTAATT AGGAGGAGG cfn0393953 C T
AGTATATAAAAAACAAGTTGTCACCCAGATG 3233
AAT[C/T]CGAAACTATGTCAATGTCGACGGTGA GTGTGGACC cfn0856945 G T
GGACATCGGCACATGCTTTATTACTGATCTGA 3234
TTT[G/T]TTGACTGTTTATTTTAGGTTTGCCTAC ACCACTGA cfn1291249 T G
ATGGTTGAATATGTGACTGCATTTGGACTCAC 3462
TCCTTGTTTCTGCATTTTCATTGAAGATAAGCAT
GGCCTTATCAAGCTTCCCAGATTTAGCATGTG
CAT[T/G]AATCAGTATGTTGAAGATACGACAGT
CAGGTAGAATGCAGTATCTTTCCATTGAATTG
AAGAGATTAATCATATCAACTAAGCATCCTTC GGTGGCATAc cfn0231871 G T
GGAGGCGCCTACTGTATTAAATTAGCTAGTGT 3463
GGC[G/T]TGTTTGAGGATAATGGCACATATACC TTGGCGGTG cfn0867742 G T
TCCAGGCAGGGGGCATCCTCGACAATGATTTTC 3464
ATC[G/T]AAGCTGCATCCCCATTCCATCACGAC GCGGAGGCT cfn0523990 G T
TGTAAGCTAACTATACATGAAGAGTGCAGGCA 3465
CAC[G/T]AAAACGTTTCATCCTGAAGTACAAGA GTTATTTTGG cfn3126082 A C
GGAGAAAGGCGAGATTTTAGCACCTAACGCC 3466
GCAA[A/C]CCAGATCAAATCGCTGTCCCTTT W90K_RAC875_ A G
AACCTTGGAAGCTATCATTGCGCACTTGAAGA 3467 c33564_120
GCA[A/G]TAGTGTGGACATTCCTGTTTATGCTT GGAGCTTAG cfn3407185 A G
CGCAGGCAGCGGGCATGTATCCTCGTCTGACG 3468
GAT[A/G]CCCAGATTATTAACTGTCACCCTGC ACGCCTGCA S100069923 G A
GAGGCTAATCCCGACGTGCCACATTGAGCACG 3469
TGTGTTCTTGCTGTGGCCTGGTCGAAAGACAT
GACGCATGCACGTGCCCCACGCCTCACACGGC
TTGGGTTCTCGCCTGTCCGGTGCTCGACGGAC
CAGTACATATACGCGAGCGCTCCT[G/A]GCCA

CCTCAGTACACTTCACTGCGAGTACAAG
 GCCTCGGCTCTCGGCAGACTCCTCATTGCTGCT
 TCTGCTAGTGAAAAGAGAGATTCTTCAGCGCT
 GCTCCTGAAAGAGATAAGAAATACGATGGCA ACAATGGTCAGAG S3045171 G A
 ACTTACTGCGCGCAGACGTTGCAGCTCTTCCT 3470
 GCAGAAGCCAGGAGCTTCCTTGGTGCCCCACCA
 TATAGTTGGGGTTCTTGGCACACTCCCCA[G/A]
 CGGCAGCCCCTGCGAGCAGAGGACATTCTCG TCCTCGCAGCCGTCACCGGAGCCT
 S3045222 T C AAGCAAGCTACGCGTTGCTCAAAAAAAAAAAAA 3471
 AAGCAAGCAAGCTACGCTGATCAAAGGCTGA
 ATAGTCCAGAGTTACAGGACATGGCTACTCTG
 CAGC[A/G]CCCAGCAAGCTTACTACTTAGGTTG
 GTGGAGAAGCAGCACCCACTCGAGACTCGAC
 AAGCAACCTTGGACGTTCTACTCGCCAGTGCA TTGCTGCTTTACC cfn1087371 A G
 ACCAGAGAAAGAGAGGGGAACCTTTGGGTATA 3472
 CACC[A/G]CATTACCCTAGTGAAAGAAGAAGG GGGTATTATGT cfn0436720 A G
 GAAGGGTCCACTGAGAATTAAGGATGCATTCT 3473
 TTC[A/G]ATTTGGTATATTTGTTGTAAGGAATG AAGAATCGG S100067637 G C
 TTTCGTGGCGGGGGATCTCGTGCCGGTTCGAGG 3474
 AGGTCCACCTCCAGCGAATTCTGCAGCAACCA
 ACACAAACAGGCCCAAATGTCCGAATTCAGA
 GCA[C/G]AGCCCGACCGACCGACCGCGAAATC
 GCGCGGCATGGCGTTGGCGTTTGGCGTTGGCG
 AGAAGGAAAAAGGCACTCTATGCAGACCTTA GCTTGGTTATGGC cfn0554333 C G
 TTGCCAAATTCACACCATCATTGATCTGGGGT 3475
 ATC[C/G]TATGCCTATGTGATGCCTCTCACCTC TTTCTTCCC cfn0238384 C G
 CCGTGAAACCTGTAAAAAGATGTCTGTGTGTC 3476
 TAG[C/G]AAAGCCCTAATTTTAATCACCCCGTA CGCCCCCCT cfn0530841 C T
 GCAGCTTCTTGTTTATATATTCTCTTATCAGAA 3477
 GT[C/T]GGGTAGAACAGCTAACGATGTGCTGCT CATTTCCT cfn1082074 C T
 TGATCTTACTGATAAAATCCGGTTCAAATATA 3478
 TAA[C/T]GGTGAGAAAAAATTAACCAGAGCGA GGCGAGACAT cfn0560679 C T
 GTTCATGTACAATGATGTTTAACATTGGAACG 3479
 GTC[C/T]GGGATCTGTTTGATCTATGCCCCCTT CAACGTCTT cfn0915987 G T
 AAGTCTGCCATCCAGATCATTACCCAACGGCC 3445
 AAT[G/T]GAGCCATGAGGTTTGCCTCGTTGCAC GTTTTGGCT cfn0920253 A C
 GCAACAAAGCTGGTCATCCAAACATTTACATC 3446
 GTT[A/C]GGCAGGCTTTCGCCCCAAACCATGCG GCCGACCTG cfn0448874 C T
 TATGTAAAACCTCTTTGTTTCTAAATAGCTGCG 3447
 GC[C/T]CGCTACCTAAATTTATGTTGAACCTAG AGGCACCC cfn0923814 A C
 GTTCGGCAGAATCCAAGTCGCAAATGTAAGGT 3448
 CAG[A/C]AAATGAATGATGATCATGATAATGA AAATCATAAG cfn0924180 A G
 ACGTATGGAGCTTCCTCTTTTCATCATGCACCA 3449
 TT[A/G]TGATCTCCCTCTTATTTTGTCTGAAGCC ATTCATG cfn0919484 A G
 AGGTCATGAAAATGCAAGTGCGCAATCTTATC 3450
 TCT[A/G]TTATACCATTTGGCAAAACAAAGGCG AGAGTTCTG

Example 19: Test of the Cumulative Effect of Rf Genes

(176) To test the hypothesis of the cumulative effect of the restorer alleles and to identify the combination(s) of Rf genes that can produce a full fertility in the CMS hybrid, the SNPs linked to the 4 major loci mapped are used to create restorer lines combining the different restoring alleles.

The 2 Rf alleles, 3 Rf alleles and 4 Rf alleles combinations (Rf1+Rf3, Rf1+Rf4, Rf1+Rf7, Rf3+Rf4, Rf3+Rf7, Rf4+Rf7, Rf1+Rf3+Rf4, Rf1+Rf3+Rf7, Rf3+Rf4+Rf7 and Rf1+Rf3+Rf4+Rf7) can be created employing different breeding techniques such as pedigree breeding, backcross, single-seed descent or double haploid.

(177) In the example exposed 2 double-haploid lines from the cross TJB155/R204 were created: LGWR16-0016 and LGWR16-0026. Those two lines carry the restorer alleles Rf1, Rf7 (donor R204) and Rf3 (donor TJB155) and are alloplasmic for the cytoplasm from *T. timopheevii* (donor TJB155).

(178) LGWR16-0016 and LGWR16-0026 have winter growth habits and show a normal fertility (LGWR16-0016:2.54 seeds/spikelet, average over 29 individual spikes. LGWR16-0026:2.33 seeds/spikelet, average over 40 individual spikes). LGWR16-0016 and LGWR16-0026 were used as pollinators in a series of crosses onto a series of A-line with elite background (14 A-lines for LGWR16-0016 and 16 A-lines for LGWR16-0026). The F1 plants originating from those crosses were assessed indoor for their fertility.

(179) Fertility Assessment with the Main Tillers

(180) The spikes from the hybrids produced with the restorer lines LGWR16-0016 yielded on average 45.4 grains and did show an average fertility of 2.44 grains/spikelet. The spikes from the hybrids produced with the restorer lines LGWR16-0026 yielded on average 44.7 grains and did show an average fertility of 2.37 grains/spikelet. Neither of these two groups differ significantly in their fertility distribution from the group formed by the elite lines (Table 20. T-test, P-values<0.05). The distribution of the fertility scores are, as a consequence, relatively similar between groups (see Table 20)

(181) TABLE-US-00026 TABLE 20 fertility (expressed as number of seeds/spikelet) of a series of spikes from hybrids produced with the restorers lines LGWR16- 0016 and LGWR16-0026 and from a panel of elite lines. The average number of seeds per spike are indicated in the column "SEEDS". The standard deviation "STD. DEV." are given for each of the three groups. The T-test p-value "P-Value" is given for the two groups of hybrids produced with LGWR16-0016 and LGWR16-0026: it indicates the statistical significance of the difference for the value "Fertility" between the group of hybrids ad the group of elite lines The spikes originate from the tallest tiller(s). STD. P- SPIKES SEEDS FERTILITY DEV. VALUE LGWR16-0016 164 45.4 2.44 0.43 0.090 LGWR16-0026 206 44.7 2.37 0.45 0.692 ELITES inbred 637 44.1 2.36 0.59

Fertility Assessment with all Tillers from F1 Plants

(182) To be considered as complete, the restoration of fertility has to be observed in all the spikes formed by the hybrid plant. For that purpose, the integrality of the spikes of 56 F1 plants produced with LRWG16-0016, of 61 F1 plants produced with LGWR16-0026 and of 52 elite lines were assessed for their fertility and represent, per group, 294, 262 and respectively 288 individual spikes (Table 21). The 56 F1 plants produced with LGWR16-0016 produced on average 41.4 grains/spikes and did show an average fertility of 2.21 grains per spikelet (standard deviation 0.32). The 61 F1 plants produced with LGWR16-0026 produced on average 42.8 grains/spikes and did show an average fertility of 2.27 grains per spikelet (Table 5. Standard deviation 0.32). The fertility distribution of these two groups differ significantly from the fertility distribution of the groups formed by the elite lines (average number of seeds per spike: 38.4. Average fertility: 2.01. Standard deviation 0.44. see Table 21).

(183) TABLE-US-00027 TABLE 21 average spike fertility (expressed as number of seeds/spikelet) of individual plants from hybrids produced with the restorers lines LGWR16-0016 and LGWR16-0026 and from a panel of elite lines. The average number of seeds per spikes are indicated in the column "SEEDS". The standard deviation "STD. DEV." are given for each of the three groups. The T-test p-value "P-Value" is given for the two groups of hybrids produced with LGWR16-0016 and LGWR16-0026: it indicates the significance of the statistical difference for the value "Fertility" between the group of hybrids ad the group of elite lines. On average the F1 plants produced with

the restorer lines LGWR16- 0016 and LGWR16-0026 formed 5.25 and respectively 4.3 spikes and the plants from the elite lines group formed on average 4.35 spikes. STD. P- PLANTS SEED FERTILITY DEV. VALUE LGWR16-0016 56 41.4 2.21 0.32 0.003 LGWR16-0026 61 42.8 2.27 0.32 0.000 ELITES inbred 52 38.4 2.01 0.44

Example 20: Development of Restorer Lines from Different Sources

(184) With the help of the markers developed in the previous examples, new restorer lines comprising Rf1, Rf3 and Rf7 loci were obtained using different sources of restoration locus. Table 22 shows the list of these restorer lines and their characteristics.

(185) LGWR17-0015 is a winter wheat double haploid line produced from the F1 plants from a complex cross: R0934F was first pollinated by Altigo, the created F1 R0934F/Altigo was then crossed with the F1 R197/Apache taken as male and the resulting four ways F1 “R0934F/ALTIGO//R197/APACHE” was then pollinated with the line Altamira. LGWR17-0015 inherited the T-CMS cytoplasm from the restorer line R0934F. LGWR17-0015 has been selected under local environment and is a fully fertile wheat line.

(186) LGWR17-0022 and LGWR17-0157 are winter wheat double haploid lines arising from a complex cross: the F1 formed from the cross between the restorer lines R204 and R213 was pollinated with Aristote, the resulting 3-ways cross “R204/R213//Aristote” was then pollinated with the line NIC07-5520. LGWR17-0022 and LGWR17-0157 inherited the T-CMS cytoplasm from the restorer line R204. They have both been selected under local environment and are fully fertile wheat lines.

(187) LGWR17-0096 and LGWR17-0154 are winter wheat lines developed by conventional pedigree breeding technique from the 3-ways cross R204/R213//ARISTOTE. LGWR17-0096 and LGWR17-0154 inherited the T-CMS cytoplasm from the restorer line R204. They have both been selected under local environment and are fully fertile wheat lines.

(188) All lines have been genotyped with the markers as described above to check for the presence of Rf1, Rf3 and Rf7 haplotypes.

(189) Representative seed samples for each line have been deposited before NCIMB Collection on 25 Sep. 2017. The NCIMB is located at Ferguson Building, Craibstone Estate, Bucksburn, Aberdeen, Scotland, UK, AB21 9YA.

(190) TABLE-US-00028 TABLE 22 Full restorer lines data regarding pedigree, the process of selection, Rf haplotype identified by markers and NCIMB deposit number “HD”: homozygous plant obtained after a Double Haploid process, “F6”: homozygous plant obtained through six generations of self-crosses. NCIMB deposit CODE PEDIGREE SELECTION HAPLOTYPE Rf number LGWR16- TJB155/R204 HD Rf1 + Rf3 + Rf7 NCIMB 0016 42811 LGWR16- TJB155/R204 HD Rf1 + Rf3 + Rf7 NCIMB 0026 42812 LGWR17- R0934F/ALTIGO//R197/ HD Rf1 + Rf3 + Rf7 NCIMB 0015 APACHE///ALTAMIRA 42813 LGWR17- R204/R213//ARISTOTE/// HD Rf1 + Rf3 + Rf7 NCIMB 0022 NIC07-5520 42814 LGWR17- R204/R213//ARISTOTE F6 Rf1 + Rf3 + Rf7 NCIMB 0096 42815 LGWR17- R204/R213//ARISTOTE F6 Rf1 + Rf3 + Rf7 NCIMB 0154 42816 LGWR17- R204/R213//ARISTOTE/// HD Rf1 + Rf3 + Rf7 NCIMB 0157 NIC07-5520 42817

Example 21: Test of the Cumulative Effect of Rf1, Rf3 and Rf7 Genes Outdoor

(191) Fertility from 4 hybrids produced with the restorer lines LGWR17-0022, LGWR17-0153, LGWR17-0154 and LGWR17-0157 was assessed in field as described in example 19 for the fertility assessment of the main tillers. The restorer lines have the pedigree and haplotypes as described in table 23.

(192) The assays are made with agronomically adapted hybrids in three different countries France, Germany and United Kingdom. In total 28 hybrids are tested and compared to 26 control Elite inbreds. The result show that, in the field, the hybrids comprising the combination of the three alleles Rf1, Rf3 and Rf7 restorers performed as well as the elite inbreds (table 24). It is also worth to be noted that, in this combination, the Rf3 weak or Rf3 do not have any impact on the fertility

level.

(193) TABLE-US-00029 TABLE 23 pedigree and haplotype of the restorer lines NCIMB CODE PEDIGREE HAPLOTYPE Rf deposit LGWR17-0022 R204/R213//ARISTOTE///NIC07- Rf1 + Rf3 + Rf7 yes 5520 LGWR17-0153 R204/R213//ARISTOTE Rf1 + Rf3weak + Rf7 no LGWR17-0154 R204/R213//ARISTOTE Rf1 + Rf3weak + Rf7 yes LGWR17-0157 R204/R213//ARISTOTE///NIC07- Rf1 + Rf3 + Rf7 yes 5520

(194) TABLE-US-00030 TABLE 24 fertility (expressed as number of seeds/spikelet) of a series of spikes from hybrids produced with the restorer lines LGWR17-0022, LGWR17-0153, LGWR17-0154 and LGWR17-0157 and from a panel of elite lines. Each spike originates from the tallest tiller. The standard deviation “STD. DEV.” are given for each of the three groups. Fertility SPIKES average STD DEV. Hybrids Total 169 2.71 0.47 Rf1 + Rf3weak + Rf7 43 2.72 0.38 Rf1 + Rf3 + Rf7 126 2.71 0.49 Elite inbreds 226 2.73 0.43

Example 22: Modification of the Endogenous RFL29c Gene Sequence by CRISPR Technologies to Revert a Rf3 Allele to a Rf3 Allele

(195) As shown in example 11, FIG. 5A and FIG. 12, RFL29c nucleotide sequence is characterized by a deletion of 2 nucleotides compared to RFL29a nucleotide sequence creating a frameshift and resulting in an inactive truncated protein. The sequence alignment in FIG. 12 shows that one “T” nucleotide in the RFL29c gene sequence could be removed or 2 nucleotides added in order to reframe the RFL29c gene sequence into a complete and functional RFL29 protein.

(196) Such modification could be achieved in Fielder, which comprises a RFL29c as depicted in SEQ ID NO 3457, by designing, as described in example 15, a suitable guide sequence targeting the frameshift and used it in combination with a base-editing technology such as described in WO2015089406. FIG. 12 shows suitable PAM motif and target sequence for CRISPR cas9 edition.

BIBLIOGRAPHY

(197) Ahmed et al, 2001. QTL analysis of fertility restoration against cytoplasmic male sterility in wheat. *Genes Genet Syst*, 76:33-38. Bahl P N, Maan S S, 1973. Chromosomal location of fertility restoring genes in six lines of common wheat. *Crop Sci* 13:317-320. Bennetzen J L et al, 2012. Reference genome sequence of the model plant *Setaria*. *Nat Biotechnol* 30 (6): 555-+. doi: 10.1038/nbt.2196 Brenchley R, et al, 2012. Analysis of the bread wheat genome using whole-genome shotgun sequencing. *Nature* 491 (7426): 705-710. doi: 10.1038/nature11650 Cannarozzi G, et al, 2014. Genome and transcriptome sequencing identifies breeding targets in the orphan crop tef (*Eragrostis tef*). *Bmc Genomics* 15. doi: Artn 58110.1186/1471-2164-15-581 Chen J F et al, 2013. Whole-genome sequencing of *Oryza brachyantha* reveals mechanisms underlying *Oryza* genome evolution. *Nature Communications* 4. doi: ARTN 159510.1038/ncomms2596 Cheng S F, Gutmann B, Zhong X, Ye Y T, Fisher M F, Bai F Q, Castleden I, Song Y, Song B, Huang J Y, Liu X, Xu X, Lim B L, Bond C S, Yiu S M, Small I (2016) Redefining the structural motifs that determine RNA binding and RNA editing by pentatricopeptide repeat proteins in land plants. *Plant Journal* 85 (4): 532-547. doi: 10.1111/tpj.13121. Christensen A H and Quail P H, 1996. Ubiquitin promoter-based vectors for high-level expression of selectable and/or screenable marker genes in monocotyledonous plants. *Transgenic Res*, May;5 (3): 213-8. Christian et al, 1992. Maize polyubiquitin genes: structure, thermal perturbation of expression and transcript splicing, and promoter activity following transfer to protoplasts by electroporation. *Plant Mol Biol*. 18 (4): 675-89. Curtis and Lukaszewski, 1993. Localization of genes in Rye that restore male fertility to hexaploid wheat with *timopheevii* cytoplasm. *Plant breeding*, 11:106-112. Depigny-This D et al, 1992. The cruciferin gene family in radish. *Plant Molecular Biology*, 20:467-479. Fehr W R et al, 1987. *Principles of Cultivar Development Vol. 1 Theory and Technique*. Macmillan, New York . . . Fujii S, Bond C S, Small I D (2011) Selection patterns on restorer-like genes reveal a conflict between nuclear and mitochondrial genomes throughout angiosperm evolution. *P Natl Acad Sci USA* 108 (4): 1723-1728. doi: DOI 10.1073/pnas. 1007667108. Geyer M et al, 2016. Distribution of the fertility-restoring gene Rf3 in common and spelt wheat determined by an informative SNP

marker. Mol Breeding, 36:167. DOI 10.1007/s11032-016-0592-6. Götz H et al, 2011. Transgene Expression Systems in the Triticeae Cereals. Journal of Plant Physiology 168, no. 1:30-44. doi: 10.1016/j.jplph.2010.07.007. International Brachypodium I (2010) Genome sequencing and analysis of the model grass Brachypodium distachyon. Nature 463 (7282): 763-768. doi: 10.1038/nature08747 Jacquemin J, et al, 2013. The International Oryza Map Alignment Project: development of a genus-wide comparative genomics platform to help solve the 9 billion-people question. Curr Opin Plant Biol 16 (2): 147-156. doi: 10.1016/j.pbi.2013.02.014. Jefferson, R. A., 1987. Assaying chimeric genes in plants: The GUS gene fusion system. Plant Mol. Biol. Report. 5, 387-405. DOI: 10.1007/BF02667740 Jia J, et al, 2013. *Aegilops tauschii* draft genome sequence reveals a gene repertoire for wheat adaptation. Nature 496 (7443): 91-95. doi: 10.1038/nature12028. Jones H D, 2015. Wheat Biotechnology: Current Status and Future Prospects. K. Azhakanandam et al. (eds.), Recent Advancements in Gene Expression and Enabling Technologies in Crop Plants, DOI 10.1007/978-1-4939-2202-4_8. Kay R, et al 1987. Duplication of CaMV 35S promoter sequences creates a strong enhancer for plant genes. Science 236:1299-1302. Kawahara Y et al, 2013. Improvement of the *Oryza sativa* Nipponbare reference genome using next generation sequence and optical map data. Rice 6. doi: Artn 410.1186/1939-8433-6-4. Kihara, 1951, Genome analysis in *Triticum* and *Aegilops* X. Concluding review. Cytologia, 16:101-123. Kojima et al, 1997, High-resolution RFLP mapping of the fertility restoration (Rf3) gene against *Triticum timopheevii* cytoplasm located on chromosome 1BS of common wheat. Genes Genet Syst, 72:353-359. Krasileva K V et al, 2013. Separating homeologs by phasing in the tetraploid wheat transcriptome. Genome Biol 14 (6). doi: ARTN R66 10.1186/gb-2013-14-6-r66. Li et al, 2003. OrthoMCL: Identification of Ortholog Groups for Eukaryotic Genomes. Genome Res. 2003 September; 13 (9): 2178-2189. Li H. and Durbin R, 2010. Fast and accurate long-read alignment with Burrows-Wheeler Transform. Bioinformatics, Epub. [PMID: 20080505] Ling H Q et al, 2013. Draft genome of the wheat A-genome progenitor *Triticum urartu*. Nature 496 (7443): 87-90. doi: 10.1038/nature11997 Longin et al, 2012, Hybrid breeding in autogamous cereals. Theor Appl Genet.: 125:1007-1096. DOI 10.1007/s00122012-1967-7. Ma Z Q and Sorrells M E, 1995, Genetic analysis of fertility restoration in wheat using RFLP. Crop Sci., 35:1137-1143. Mace E S et al, 2013. Whole-genome sequencing reveals untapped genetic potential in Africa's indigenous cereal crop sorghum. Nat Commun 4:2320. doi: 10.1038/ncomms3320. McElroy D et al, 1990. Isolation of an Efficient Actin Promoter for Use in Rice Transformation. The Plant Cell, Vol. 2, 163-171. Martis M M et al, 2013. Reticulate Evolution of the Rye Genome. Plant Cell 25 (10): 3685-3698. doi: 10.1105/tpc.113.114553 Mayer K F X, et al, 2014. A chromosome-based draft sequence of the hexaploid bread wheat (*Triticum aestivum*) Science genome. 345 (6194). doi: ARTN 125178810.1126/science. 1251788 Mayer K F X et al, Conso IBGS, 2012. A physical, genetic and functional sequence assembly of the barley genome. Nature 491 (7426): 711-+. doi: 10.1038/nature 11543. Ouyang S et al, 2007. The TIGR Rice Genome Annotation Resource: Improvements and new features. Nucleic Acids Res 35: D883-D887. doi: 10.1093/nar/gk1976. Pallavi Sinha P et al 2013. Genetic analysis and molecular mapping of a new fertility restorer gene Rf8 for *Triticum timopheevii* cytoplasm in wheat (*Triticum aestivum* L.) using SSR markers. Genetica, 141:131-141. Paterson A H et al, 2009. The *Sorghum bicolor* genome and the diversification of grasses. Nature 457 (7229): 551-556. doi: 10.1038/nature07723. Rathburn and Hedgcoth, 1991. Chimeric open reading frame in the 5' flanking region of coxI mitochondrial DNA from cytoplasmic male-sterile wheat. Plant Mol. Biol., 16:909-912. Rice P et al, 2000. A. EMBOSS: The European molecular biology open software suite. Trends Genet 16, 276-277, 10.1016/S0168-9525 (00) 02024-2. Sakai H, et al, 2013. Rice Annotation Project Database (RAP-DB): An Integrative and Interactive Database for Rice Genomics. Plant Cell Physiol 54 (2): E6-+. doi: 10.1093/pcp/pcs183. Schnable P S, et al, 2009. The B73 Maize Genome: Complexity, Diversity, and Dynamics. Science 326 (5956): 1112-1115. doi: 10.1126/science.1178534. Singh S K et al, 2010. Perspective of hybrid wheat research: a review. Indian J Agric Sci 80:1013-1027.

Song and Hedgcoth, 1994. Influence of nuclear background on transcription of a chimeric gene orf256 and cox1 in fertile and cytoplasmic male sterile wheats. *Genome*, vol. 37 Stojalowski S et al, 2013. The importance of chromosomes from the sixth homeologic group in the restoration of male fertility in winter triticale with *Triticum timopheevii* cytoplasm. *J. Appl. Genetics*, 54:179-184. Verdaguer et al, 1996. Isolation and expression in transgenic tobacco and rice plants, of the cassava vein mosaic virus (CVMV) promoter. *Plant Molecular Biology* 31:1129-1139. Wang M H et al, 2014. The genome sequence of African rice (*Oryza glaberrima*) and evidence for independent domestication. *Nat Genet* 46 (9): 982-+. doi: 10.1038/ng.3044 Wilson J A, Ross W M. 1962. Male sterility interaction of the *Triticum aestivum* nucleus and *Triticum timopheevii* cytoplasm. *Wheat Information Service (Kyoto)* 14:29-30. Wilson J A, Ross W M. 1962. Male sterility interaction of the *Triticum aestivum* nucleus and *Triticum timopheevii* cytoplasm. *Wheat Information Service (Kyoto)* 14, 29-30. Wilson, 1984. Hybrid wheat breeding and commercial seed development. *Plant Breeding Rev.*, 2:303-319. 2 Whitford R et al, 2013. Hybrid breeding in wheat: technologies to improve hybrid wheat seed production. *Journal of Experimental Botany*. doi: 10.1093/jxb/ert333. Zhou et al, 2005. SSR marker associated with fertility restoration genes against *Triticum timopheevii* cytoplasm in *Triticum aestivum*. *Euphytica*, 141:33-40.

Claims

1. A wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm comprising: a Rf1 restorer allele comprising a nucleic acid encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity to SEQ ID NO:361, a Rf3 restorer allele within the Rf3 locus, wherein the Rf3 locus comprises one or more of the following SNP allele(s): TABLE-US-00031

Marker	Restorer	SNP#	Marker Name	SEQ ID	Allele
SNP17	cfn1252000	3203	A	SNP18	IWB14060*
3204	G	SNP19	cfn1249269	3205	G
SNP20	219K1	166464	3206	T	SNP21
219K1	158251	3207	G	SNP22	219K1
111446	3208	A	SNP23	219K1	110042
3209	T	SNP24	219K1	110005	3210
C	SNP25	219K1	107461	3211	A
SNP26	219K1	99688	3212	T	SNP27
219K1	37	3213	C	SNP28	cfn1270524
3214	T	SNP29	136H5	3M5	7601
3215	T	SNP30	cfn1288811	3216	G
SNP31	136H5	3M5	89176	3217	A
SNP32	136H5	3M5	89263	3218	T
SNP33	136H5	3M5	138211	3219	T
SNP34	cfn0556874	3220	C	SNP35	136H5
3M5	64154	3221	C	SNP36	136H5
3M5	68807	3222	G	SNP37	136H5
3M5	77916	3223	A	SNP38	cfn1246088
3224	A	SNP39	cfn1287194	3225	G
SNP40	cfn1258380	3226	A	SNP41	IWB72107*
3227	A	SNP42	BS00090770	3228	T
SNP43	cfn1239345	3229	A	and a Rf7 restorer allele within the Rf7 locus, wherein the Rf7 locus comprises one or more of the following SNP allele(s): TABLE-US-00032	
2. The wheat plant according to claim 1, wherein said Rf3 restorer allele is located within the chromosomal fragment between SNP markers cfn1249269 and BS00090770.
3. The wheat plant according to claim 2, wherein the corresponding amino acid sequence of the Rf3 restorer allele has at least 95% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO: 684.
4. The wheat plant according to claim 1, wherein said Rf3 locus comprises SEQ ID NO:1712, SEQ ID NO:3147, SEQ ID NO:2230, SEQ ID NO:3148 or SEQ ID NO:2238.
5. A method for producing the wheat plant according to claim 1, said method comprising: a. providing a first wheat plant comprising one or two restorer allele selected among Rf1, Rf3 and Rf7 restorer alleles, b. crossing said first wheat plant with a second wheat plant comprising one or two restorer alleles selected among Rf1, Rf3 and Rf7 restorer alleles, wherein Rf1, Rf3 and Rf7 restorer alleles are represented at least once in the panel of restorer alleles provided by the first

plant and the second plant, c. collecting the F1 hybrid seed, and d. obtaining homozygous plants from the F1 plants.

6. A method for producing a wheat hybrid plant, said comprising: a. crossing a sterile female comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant according to claim 1; and b. collecting the hybrid seed.

7. The method according to claim 6, further comprising: detecting the presence of at least three of Rf locus chosen amongst Rf1, Rf3, and Rf7 in the hybrid seeds.

8. A wheat hybrid plant as obtained by the method according to claim 7.

9. A method of identifying a wheat plant according to claim 1, said method comprising: identifying said wheat plant by detecting the presence of the restorer alleles Rf1, Rf3 and Rf7.

10. At least three nucleic acid probes or primers for the specific detection of the restorer alleles Rf1, Rf3 or Rf7 in a wheat plant, wherein the at least three nucleic acid probes or primers are chosen among: SEQ ID NOs: 3254, 3259, 3261-3265, 3269-3275, 3279, 3280, 3282, 3293, 3294, 3309, 3318, 3323, 3325-3329, 3333-3339, 3343, 3344, 3346, 3357, 3358, 3373, 3382, 3387, 3389-3393, 3397-3403, 3407, 3408, 3410, 3421, 3422, and 3437 for the specific detection of restorer SNP marker(s) in locus Rf1, SEQ ID NOs: 3253, 3255, 3260, 3266-3268, 3276-3278, 3281, 3283-3288, 3295-3308, 3310, 3311, 3313-3317, 3319, 3324, 3330-3332, 3340-3342, 3345, 3347-3352, 3359-3372, 3374, 3375, 3377-3381, 3383, 3388, 3394-3396, 3404-3406, 3409, 3411-3416, 3423-3436, 3438, 3439, and 3441-3444 for the specific detection of restorer SNP marker(s) in locus Rf3, and/or SEQ ID NOs: 3256-3258, 3320-3322, and 3384-3386 for the specific detection of restorer SNP marker(s) in locus Rf7.

11. The wheat plant according to claim 3, wherein the corresponding amino acid sequence of the Rf3 restorer allele has at least 98% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO: 684.

12. The wheat plant according to claim 3, wherein the corresponding amino acid sequence of the Rf3 restorer allele has 100% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO: 684.

13. The method according to claim 5, further comprising: detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles in the hybrid seed and/or detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles at each generation.

14. The method according to claim 6, further comprising: detecting hybridity level of the hybrid seeds.
